MILITARY SPECIFICATION

CONTROL, ORDERWIRE C-10716/TRC

This specification is approved for use by the Communications Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of the 'Control, Orderwire C-10716/TRC. This unit is also known as the Orderwire Control Unit Type 1 (OCU-1) and is one of the electronic units that comprise the Digital Group Multiplexer (DGM) family of cable system equipment developed by the Joint Tactical Command, Control and Communications Agency (JTC^{3}A formerly TRI-TAC). See 6.1 for intended use and functional descripition of the unit.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specifi-cation to the extent specified herein. Unless otherwise speci-fied, the issues shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications Electronics Command, ATTN: AMSEL-ED-TO Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Method of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A (Notice 4)	Electromagnetic Emission and Susceptibility Requirements for the Control Of Electromagnetic Interference
MIL-STD-462 (Notice 3)	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781C	Reliability Design Qualification and Production Acceptance Test Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-1332	Definitions of Tactical, Prime, Precise and Utility Terminologies for Classification of the DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-2073-2	Packaging Requirements Codes
MIL-STD-45662	Calibration Systems Requirements

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments
CR-CX-0178-001	TEMPEST Provisions for the Family of Orderwire Unit Equipments

DRAWINGS

Communications-Electronics Command

SM-D-884567	Circuit Card Assembly (CCA) Multi Vdc
SM-D-884571	CCA, 52 Vdc
SM-D-884579	CCA, AC Input
DL-SM-B-959043	Control Orderwire C-10716()/TRC
SM-D-959060	CCA, Control Panel Assembly (Remote)
SM-D-959064	CCA, Control Panel Logic
SM-D-959069	CCA, Analog Xmtr Amplifier
SM-D-959072	CCA, Analog Signaling
SM-D-959075	CCA, Analog Bridge
SM-D-959084	CCA, Telemetry Combiner/Generator
SM-D-959093	CCA, VINSON Buffer/Ring Detector
SM-D-959102	CCA, Digital Bridge/Matrix
SM-D-959105	CCA, BITE 3
SM-D-959108	CCA, Alarm Interface
SM-D-959286	CCA, Lamp Driver/Ring Generator
SM-D-959290	CCA, Timing Source
SM-A-984722	Filter,Analog
A3051665	Error Inserter
A3051666	Timing Source
A3051669	DVOW/DOW Simulator

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article testing units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with the requirements of this specification and of DL-SM-B-959043, including the requirements for parts, materials and processes therein.

3.2.2 Printed wiring boards. The printed wiring boards shall meet the requirements of MIL-STD-454, Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454, Requirements 30 and 64.

3.3. Parts, materials, and processes general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and DL-SM-B-959043.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. A unique serial number is required on each nomenclatured item, and CCA. The number shall be consecutively assigned.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled C-10716/TRC shall be electrically preconditioned for a

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minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.14 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Balanced equipment-side signal characteristics. (See 4.7.1) The equipment shall accept and deliver balanced data and timing at 2 and 16 kilobits per second (kb/s). Timing signals also shall be delivered at rates of 2, 16, 32, 128, 144, 256, 288, 512, 576, 1024, 1152, 2304, or 4608 kHz. A timing input at rates of 16, 128, 144, 256, 288, 512, 576, 1024, 1152, 1536, 2048, 2304,4096, or 4608 kHz shall be accepted.

3.7.1.1 Line-to-line voltage levels. (See 4.7.1.1) The line-to-line voltage shall be between +0.1 and +4.0 volts for a digital 1 and between -0.1 and -4.0 for a digital 0, measured into a 100 ohm dummy load. For the 2 kb/s data and timing outputs, the voltages shall be between +1.2 and +4.0 volts or -1.2 and -4.0 volts respectively, as measured across a 100 ohm load center tapped to ground.

3.7.1.2 Line-to-ground voltage. (See 4.7.1.2) The magnitude of the line-to-ground voltage shall not exceed 3.0 volts except for the 2 kb/s data and timing outputs, where the limit shall be 3.25 volts.

3.7.1.3 Clock/data phasing. (See 4.7.1.3) For those data signals having a related clock signal, the time difference between the upward transition of the clock signal and either the upward or downward transition of the data signal shall be no more than 1500 nanoseconds (ns) for 2 and 16 kb/s signals. Times shall be measured at the zero voltage point of the signals. The negative to positive transition of the TX clk input shall be followed by TXDPT data transition within ±125 microseconds.

3.7.1.4 Offset. (See 4.7.1.4) The magnitude of the offset voltage measured between the center tap of 100 ohm dummy load and ground shall not be greater than 3 volts.

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3.7.1.5 Signal balance. (See 4.7.1.5) The magnitude of the difference between the magnitudes of the digital 1 level and the digital 0 level shall be no greater than 10% of the larger of the two.

3.7.1.6 Slew rate. (See 4.7.1.6) When 100 feet of RG-108A/U cable terminated in a 78 ohm dummy load is connected to the driver output, the transition between ±40 MV (either direction) shall take no more than the following times:

Signal	Maximum Slew Time
16 to 4608 kHz group timing source outputs	10 ns
2 kb/s DOW data and timing outputs	100 ns
16 kb/s DVOW data & timing outputs	100 ns
2 or 16 kb/s test data out	100 ns

3.7.1.7 Receiver dynamic range. (See 4.7.1.7) The receiver shall deliver the proper digital output when a line-to-line voltage of +50 mV to +4V for a digital 1 or -50 mV to -4V for a digital 0 is applied to the receiver input terminals.

3.7.1.8 Input load impedance. (See 4.7.1.8) The input impedance shall be 100 ohms ± 10% except that the 2 kb/s input impedance shall be 2000 ohms minimum.

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3.7.1.9 Clock asymmetry. (Se 4.7.1.9) The positive half of the clock waveform shall be equal to half the clock period within a tolerance of ± 1500 ns for clock rates from 2 to 32 kHz, ±120 ns for clock rates from 128 to 576 kHz, and ± 15 ns for clock rates from 1024 to 4608 kHz.

3.7.1.10 Unloaded output. (See 4.7.1.10) The 2 kb/s data and timing voltage magnitudes, line to line shall not exceed 6 volts measured into an open circuit.

3.7.2 DLED resync interface, (See 4.7.2)

3.7.2.1 Driver output voltage. (See 4.7.2.1) The voltage for a digital 1 shall be +5 ±1 volts, and the voltage for a diagital 0 shall be -5 ± 1 volt. These voltage levels shall be met when measured into an open circuit and into a 20 kilohm load.

3.7.2.2 Transmitter output impedance. (See 4.7.2.2) Transmitter output impedance shall be no greater than 150 ohms.

3.7.2.3 Rise and fall times. (See 4.7.2.3) Rise and fall times shall be no greater than 3 microseconds (3 us).

3.7.2.4 Sync signal. (See 4.7.2.4) The sync signal shall be a positive going pulse, having a width at zero crossing of 30 to 200 us.

3.7.3 Unbalanced VINSON interface. (See 4.7.3)

3.7.3.1 Driver output voltage. (See 4.7.3.1) The voltage for a digital 1 shall be between +3 and +6 volts and the voltage for a digital zero shall be between -3 and -6 volts. These voltage levels shall be met when measured into a 20 kilohm load.

3.7.3.2 Receiver operating range. (See 4.7.3.2) The receiver shall deliver a digital 1 for an input voltage of +0.6 to +7.0 volts. The receiver shall deliver a digital 0 for an input voltage of -7.0 to -0.6 volts.

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3.7.3.3 Rise and fall times. (See 4.7.3.3) Rise and fall times shall be 3 to 15 us.

3.7.4. Analog voice orderwire (AVOW) channel interface. (See 4.7.4)

3.7.4.1 Input and output impedance. (See 4.7.4.1) Input and output impedance shall be 600 ohms ±10%.

3.7.4.2 Output level. (See 4.7.4.2) With an input test tone level of -4 dBm, the output level shall be -4 dBm ± 1 dB.

3.7.4.3 Frequency response. (See 4.7.4.3) Frequency response shall be ±0.5 dB from 300 to 3400 Hz, referred to 1000 Hz.

3.7.4.4 Signal generator frequency. (See 4.7.4.4) Signal output frequency shall be 1600 Hz ±2%.

3.7.4.5 Signal detector operating range. (See 4.7.4.5) Signal detector operating range shall be -16 to -2 dBm for frequencies of 1600 Hz ±2%.

3.7.4.6 Signal generator output. (See 4.7.4.6) Signal generator output shall be -4 dBm ±1 dB.

3.7.5 AVOW handset interface. (See 4.7.5)

3.7.5.1 Output. (See 4.7.5.1) With a -4 dBm test tone level applied at the AVOW channel input, the output shall be +2 to +5 dBm into 1000 ohms.

3.7.5.2 Input impedance. (See 4.7.5.2) Input impedance shall be 136 to 165 ohms at 1 kHz.

3.7.5.3 Input level. (See 4.7.5.3) With a 1.4 mV rms input applied, the output at the AVOW channel shall be -4 dBm ±1 dB.

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3.7.5.4 Frequency response. (See 4.7.5.4) Frequency response shall be ±2 dB from 300 to 3000 Hz, referenced to 1 kHz.

3.7.6 Alarm interface. (See 4.7.6)

3.7.6.1 Relay contact alarms. (See 4.7.6.1) A closure shall cause the appropriate alarm. Forty eight (48) alarms shall be provided as shown in table I.

3.7.6.2 Logic driver alarms. (See 4.7.6.2) An input voltage of +2 to +6 volts shall cause the appropriate alarm. Seven (7) alarms shall be provided as shown in table II. An input voltage of -6 to +0.8 volts shall not cause an alarm. The input impedance shall be no less than 6 kilohms.

3.7.7 Bit error rate (BER) interface. (See 4.7.7) The signal characteristics of the strobe and BCD data lines are balanced as described in 3.7.1 with the additional characteristics of 3.7.7.1 through 3.7.7.3. Offset requirement of 3.7.1.4 does not apply.

3.7.7.1 Strobe pulsewidth. (See 4.7.7.1) Strobe pulsewidth shall be 25 to 70 usec.

3.7.7.2 Slew time. (See 4.7.7.2) When the strobe is connected to a 100 ohm dummy load, the transition between ±40 mV (either direction) shall take no more than 100 ns.

3.7.7.3 Strobe voltage. (See 4.7.7.3) Strobe voltage shall be:

Idle: -4.0 to - 0.1 volts

Active: +0.1 to +4.0 volts

Both idle and active strobe voltages shall be measured line-to-line into a load of 50 ohms from both lines to ground.

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Table I. Alarm messages.

ALARM NO.	Message	ALARM NO.	Message	ALARM NO.	Message	ALARM NO.	Message
1	RCVSA1	13	GMSA3	25	LCDSA3	37	RCDSA1
2	RCVSA2	14	GMSA4	26	LCDLS1	38	RCDSA2
3	RCVSA3	15	GMLS1	27	LCDLS2	39	LGM1
4	RCVRS1	16	GMLS2	28	LCDLS3	40	LGM2
5	RCVRS2	17	GMLS3	29	TGM1	41	RCDLS1
6	RCVRS3	18	GMLS4	30	TGM2	42	RCDLS2
7	MGMSA1	19	HCDSA1	31	GMSA5	43	28V
8	MGMSA2	20	HCDSA2	32	GMSA6	44	
9	MGMFS1	21	HCDLS1	33	GMLS5	45	XMTR1
10	MGMFS2	22	HCDLS2	34	GMLS6	46	XMTR2
11	GMSA1	23	LCDSA1	35	CNVSA1	47	XMTR3
12	GMSA2	24	LCDSA2	36	CNVSA2	48	OCU

Table II Logic driver alarms.

ALARM	ALARM WORD
49	DLED
50	DLED
51	TED1
52	TED2
53	TED3
54	TED4
55	VINSON

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3.7.8 Digital voice orderwire (DVOW) performance. (See 4.7.8)

3.7.8.1 Signaling code generation. (See 4.7.8.1) Upon operator selection, the equipment shall be able to generate any one of the 16 codes listed in table III. A single operation of the signal button shall generate a transmission of two to three seconds duration.

Table III Ring codes.

Dial Position	Party	Required Code	Dial Position	Party	Required Code
1	1	10000001	9	9	10110001
2	2	10000010	10	10	10101001
3	3	10000100	11	11	10110100
4	4	10001000	12	12	10110010
5	5	10000111	13	13	10011111
6	6	10001110	14	14	10101111
7	7	10011100	15	15	10110111
8	8	10111000	16	16	10111011

3.7.8.2 Signaling detection. (See 4.7.8.2) The equipment shall be able to detect upon selection by internal switches any one of the 16 ring codes listed in table III.

3.7.8.2.1 Detection with errors. (See 4.7.8.2.1) Ring detection of the selected signaling code shall be accomplished with a 90% probability or greater in an error environment of 20% BER random bursts interspersed with 0.1% BER random error intervals. The time between successive bursts shall be random between 50 ms and one second, and the burst length shall be five percent of the time between the start of the burst and the start of the next burst.

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3.7.8.2.2 False ring detection. (See 4.7.8.2.2) The equipment shall have a probability of false ring detection of 10^{-5} or less when in the presence of an idle DVOW channel (all ONES) in the burst BER of 3.7.8.2.1. The mean time between false ring detection with random digital data on all incoming lines shall be 24 hours or more.

3.7.8.3 Operator access and bridging. (See 4.7.8.3) The equipment shall provide for a half duplex connection between any channel and the operator in isolation from the bridge. The equipment shall provide for half duplex connections between any number of channels and the operator through the bridge. The equipment shall provide for full duplex connection between any pair of channels up to three pairs.

3.7.8.4 Bridge seizure and release. (See 4.7.8.4) The bridge shall be seized by channel activity within 50 ms. The bridge shall be released by cessation of channel activity within 20 ms.

3.7.8.5 VINSON buffer. (See 4.7.8.5) The VINSON buffer shall provide for at least 2.2 minutes of error free operation with a frequency difference between the data source and the equipment timing distributor of ±30 parts per million (ppm).

3.7.9 Timing source. (See 4.7.9) The timing source shall have a frequency accuracy in the master mode of ±10 ppm. It shall be capable of operating in the slave mode with external sources having frequency deviations up to ±10 ppm from nominal. Frequencies accepted from external sources shall include 16, 128, 144, 256, 288, 512, 576, 1024, 1152, 1536, 2048, 2304, 4096, and 4608 kHz. The following output shall be supplied:

Two group outputs independently selectable from these frequencies: 128, 144, 256, 288, 512, 576, 1024, 1152, 2304, and 4608 kHz.

Two outputs at 2 kHz.

Three outputs at 16 kHz.

Three outputs at 32 kHz.

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3.7.10 AVOW performance. (See 4.7.10)

3.7.10.1 Ring generation. (See 4.7.10.1) A single operation of the ring pushbutton shall generate a ring signal transmission of two to three seconds duration.

3.7.10.2 Ring detection. (See 4.7.10.2) Under the following conditions, the equipment shall detect a tone burst no less than 90% of the time:

a. Tone level from -16 to -2 dBm.

b. Tone to random noise level ratio of 15 dB or better.

c. Tone frequency between 1568 and 1632 Hz.

d. Tone burst two seconds minimum.

3.7.10.3 False ring detection. (See 4.7.10.3) For a period of one hour, the equipment shall not detect a tone with Gaussian noise in the band 300 to 3400 Hz at -4 dBm.

3.7.10.4 Operator access and bridging. (See 4.7.10.4) The equipment shall provide a full duplex connection between any channel and the operator in isolation from the bridge. The equipment shall provide a full duplex connection between any number of channels and the operator through the bridge.

3.7.11 Data orderwire (DOW) performance. (See 4.7.11)

3.7.11.1 Format. (See 4.7.11.1) The equipment shall generate a 20 character message block as shown in table IV. The equipment shall provide 13 channels, any one of which can be used for the message block. For one or two alarms, one message block shall be transmitted. For three or four alarms, two message blocks shall be transmitted. For five or six alarms, three message blocks shall be transmitted. For seven or eight alarms, four message blocks shall be transmitted. For more than eight alarms, one message block shall be transmitted with the message "MAJOR MAJOR".

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Table IV. Character message block.

CH1 CHARACTER	SYMBOL	CHARACTER	SYMBOL	CHARACTER	SYMBOL
1	STX	7	↑	13	↑
2	A	8	↑	14	↑
3	O	9	↑	15	↑
4	O	10	↑	16	↑
5	P	11	↑	17	↑
6		12	↑	18	↑
				19	ETX
				20	ETB

3.7.11.2 Telemetry decombiner frame acquisition and maintenance. (See 4.7.11.2) With a steady bit error rate of 10^{-3}, the decombiner shall lock-in within 0.4 seconds for no less than 90% of the trials. Once framed-in, the decombiner shall declare out of frame within 825 ms of removal of data for no less than 90% of the trials. Once framed, and with 10^{-3} bit error rate, the decombiner shall not declare loss of frame more often than once per 24 hours on the average.

3.7.11.3 Resync command pulse rate. (See 4.7.11.3) When a decombiner number 1 is out of frame, it shall issue a resync command pulse to the DLED every 1.6 to 5.0 seconds.

3.7.11.4 DOW BCI. (See 4.7.11.4) When the decombiner timing is offset ±1 ppm with respect to the combiner timing, loss of bit count integrity (BCI) on the channels shall occur no more than once in 15 minutes (100 ppm = 9.0 seconds).

3.7.12 Prime power variation. (See 4.7.12) The ac primary power source shall be 115V ±10% at frequiences of 50 Hz, 60 Hz, and 400 Hz ±5% single phase. The power input to the unit shall not exceed 80 watts. The ac primary power shall conform to MIL-STD-1332 for Type 1, Class 1 and 2, Mode I, II, and III generator sets.

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3.7.13 BITE and alarms. (See 4.7.13) The built-in test equipment (BITE) shall monitor signals in and out of the unit and internal circuits and power supply voltages. The BITE shall operate without interrupting traffic or interfering with normal system operation. Alarm conditions are removed automatically when proper operation is restored. Failures detected by the BITE circuitry shall cause the following:

a. S alarm. The S alarm shall indicate any failure detected including loss of input power. This alarm shall provide a visible indication on the front panel except for input power failure.

b. SA alarm. The SA alarm shall indicate that the malfunction is within the unit or the unit has lost input power. If the power switch is set to OFF, there shall be no SA alarm. The SA alarm shall cause the status light to go red and the alarm word OCU to be transmitted.

3.7.13.1 Alarm operation. A check of the operation of the alarms shall not interrupt the operation of the equipment.

3.7.14 Operational check. (See 4.7.14)

3.7.14.1 Alarm monitor check. (See 4.7.14.1) There shall be no alarm lights illuminated.

3.7.14.2 DVOW error rate test. (See 4.7.14.2) There shall be no errors for a period of 30 seconds.

3.7.14.3 DVOW ring test. (See 4.7.14.3) The DVOW shall be able to cause itself to ring when looped back.

3.7.14.4 AVOW levels. (See 4.7.14.4) With a channel input of -4 dBm at 1000 Hz, the channel output shall be between -5 and -3 dBm; the handset output shall be +2 to +5 dBm into 1000 ohms.

3.7.14.5 AVOW self-ring test. (See 4.7.14.5) The AVOW shall be able to ring itself when looped back and the self-ring switch is activated.

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3.7.14.6 Timing source outputs. (See 4.7.14.6) All timing source outputs shall be checked for a frequency accuracy of ±10 ppm.

3.8 Service conditions. (See 4.8) The operational test of 3.7.14 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +63°C (+145°F), and shall withstand exposure (non-operating) to ambient air temperatures as high as +71°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient air temperatures as low as -31.7°C (-25°F), and shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall withstand, in both operating and non-operating conditions, the effects of dust particles that may accumulate within the shelter assembly or vehicle.

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

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3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Vibration and shock. (See 4.8.7). The equipment shall withstand vibration and shock induced during vehicular transport as part of a mobile assemblage, over all types of roads and cross-country terrain, and shocks encountered during servicing.

3.9 Electromagnetic interference (EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4.

CE02		CS02	RE02 RS02
CE03	CS06	RE02.1	RS03
CE04			
CE05			

NOTES:

a. RE02.1 is applicable over the frequency range of 14 kHz to 1000 MHz.

b. RS03 is applicable over the frequency range of 10 kHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to dc, 60 Hz and 400 Hz power lines.

3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The C-10716/TRC shall have no less than 3500 hours mean-time-between-failures (MTBF), 0_0, rate when operating in the normal modes of operation.

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3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement drawings, as referenced in the following subassembly drawings:

CCA	DRAWING
Multi Vdc	SM-D-884567
52 Vdc	SM-D-884571
AC Input	SM-D-884579
Control Panel Logic	SM-D-959064
Analog Signaling	SM-D-959072
Analog Bridge	SM-D-959075
Telemetry Combiner/Generator	SM-D-959084*
VINSON Buffer/RingDetector	SM-D-959093*
Analog Xmtr Amplifier	SM-D-959096
Digital Bridge/Matrix	SM-D-959102*
BITE 3	SM-D-959105*
Alarm Interface	SM-D-959108*
Lamp Driver/Ring Generator	SM-D-959286*
Timing Source	SM-D-959290

* This drawing contains automatic test equipment (ATE) acceptance procedures and programs.

3.13 Operational tests. (See 4.7.14) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.14 after all other inspections and tests are completed to determine that the unit is operating prior to packaging. The operational test of Group A satisfies this requirement for those units for which that is the last inspection or test.

3.14 Systems safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, and 8 shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with the requirement has been achieved shall be through a visual examination and inspection.

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3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Nuclear survivability. (See 4.16) The equipment shall comply with the nuclear survivability requirements stated in section 3 of CR-CX-0173-001 with the exceptions of paragraphs 3.3.1.1 (Blast) and 3.3.1.2 (Thermal).

3.16 TEMPEST. (See 4.17) The requirements of section 3 of CR-CX-0178-001 shall be met.

3.17 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirements 1,5 and 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (Harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections .

(1) Quality conformance inspections of equipment before packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish a minimum of 15 first article units of the complete Control Unit, Orderwire C-10716/TRC. The first article quantity shall be as specified in the contractor or purchase order.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table V and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in-process group A inspection prior to burn-in.

4.3.3 First article inspection data. The first article inspection test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE V. First article inspection

Inspection	Rqmt Para	Test Para	UNIT 1	2	3	4	5	6	7	8-15
1. Inspections by Subsidiary Documents	N/A	4.4	Inspections to be performed on all units.							
2. Burn-in	3.6	4.6	Inspections to be performed on all units.							
3. Group A Inspection	(See Table II)		Inspections to be performed on all units.							
4. Group B Inspection	(See Table III)		Inspections to be performed on 15 units.							
5. Group C Inspection										
Subgroup 1										
High Temperature	3.8.1.1	4.8.1.1	1							
Low Temperature	3.8.1.2	4.8.1.2	2							
Subgroup 2										
EMI	3.9	4.9		1						
TEMPEST	3.16	4.17		2						
Vibration/Shock	3.8.7	4.8.7			1					
Subgroup 3										
Humidity (Note c)	3.8.2	4.8.2				1				
Subgroup 4										
Low Pressure (Altitude)	3.8.3	4.8.3					1			
Dust (Note c)	3.8.4	4.8.4					2			
Salt Fog (Note c)	3.8.5	4.8.5					3			
Fungus (Note c)	3.8.6	4.8.6						1		
Subgroup 5										
Line-to-Ground Voltage Level	3.7.1.2	4.7.1.2							1	**
Clock/Data Phasing	3.7.1.3	4.7.1.3							2	
Offset	3.7.1.4	4.7.1.4							3	**
Signal Balance	3.7.1.5	4.7.1.5							4	

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TABLE V. First article inspection -continued.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8-15
Slew Rate	3.7.1.6	4.7.1.6							5	
Input Load Impedance	3.7.1.8	4.7.1.8							6	
Clock Asymmetry	3.7.1.9	4.7.1.9							7	
Unloaded Output	3.7.1.10	4.7.1.10							8	
Transmitter Output Impedance	3.7.2.2	4.7.2.2							9	
Rise and Fall Times	3.7.2.3	4.7.2.3							10	
Sync Signal	3.7.2.4	4.7.2.4							11	
Rise and Fall Times	3.7.3.3	4.7.3.3							12	
AVOW Input and Output Impedance	3.7.4.1	4.7.4.1							13	
Frequency Response	3.7.4.3	4.7.4.3							14	**
Input Impedance	3.7.5.2	4.7.5.2							15	
Frequency Response	3.7.5.4	4.7.5.4							16	**
Strobe Pulse Width	3.7.7.1	4.7.7.1							17	
Slew Time	3.7.7.2	4.7.7.2							18	
Signaling Code Generation	3.7.8.1	4.7.8.1							19	**
Signaling Detection	3.7.8.2	4.7.8.2							20	
Detection with Errors	3.7.8.2.1	4.7.8.2.1							21	
False Ring Detection	3.7.8.2.2	4.7.8.2.2							22	**
Bridge Seizure and Release	3.7.8.4	4.7.8.4							23	
VINSON Buffer	3.7.8.5	4.7.8.5							24	
Timing Source	3.7.9	4.7.9							25	
Ring Detection	3.7.10.2	4.7.10.2							26	
False Ring Detection	3.7.10.3	4.7.10.3							27	**
Format	3.7.11.1	4.7.11.1							28	
Telemetry Decombiner Framing	3.7.11.2	4.7.11.2							29	
(Frame Maintenance only)		4.7.11.2.1							30	**
Resync Command Pulse Rate	3.7.11.3	4.7.11.3							31	
DOW BCI	3.7.11.4	4.7.11.4							32	* **
Prime Power Variation	3.7.12	4.7.12							33	
Dimensional Interchange-ability	3.11	4.11							34	
6. Group D inspection										
Reliability	3.10	4.10								8 Units

** One time engineering test to be covered by a separate engineering report. Not part of first article test report.

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NOTES for table V:

a. The table V inspections 1 through 4, in the order shown, shall be performed on all first article inspection units before subjecting these units to any other inspection requirements specified in the table.

b. The number in the unit columns in the table covered by Inspections 5 and 6 specify the order of inspection for the indicated unit. An asterisk (*) indicates the use of an auxillary C-10716()/TRC unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article inspection units shall be subjected to group A and group B inspections.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where required
Printed wiring boards	(3.2.2)
Semiconductor devices and integrated circuits	(3.2.3)
Parts, materials, and processes	(3.3)
Finish	(3.4)
Marking	(3.5)

4.5 Quality conformance inspections of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in table VI shall be inspected for conformance to the inspections specified in table VI.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in the order listed in table VI. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table VI shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table VI.

b. Only for spares that are not installed into end equipment, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table VI (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipment which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and a 100% end equipment visual/mechanical and electrical inspection in accordance with table VI in that order.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table VI. Group A electrical inspection shall be performed in the order shown in table VI.

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TABLE VI. Group A inspection.

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
CCA Analog Xmtr Amplifier	3.17	4.13	2.5%	4.0%
CCA Multi V DC	3.17	4.13	2.5%	4.0%
CCA 52 V DC	3.17	4.13	2.5%	4.0%
CCA AC Input	3.17	4.13	2.5%	4.0%
Control Panel Assembly (Remote)	3.17	4.13	2.5%	4.0%
CCA Control Panel Logic	3.17	4.13	2.5%	4.0%
CCA Analog Signaling	3.17	4.13	2.5%	4.0%
CCA Analog Bridge	3.17	4.13	2.5%	4.0%
CCA Telemetry Combiner/ Generator	3.17	4.13	2.5%	4.0%
CCA VINSON Buffer/Ring Detector	3.17	4.13	2.5%	4.0%
CCA Digital Bridge/Matrix	3.17	4.13	2.5%	4.0%
CCA BITE 3	3.17	4.13	2.5%	4.0%
CCA Alarm Interface	3.17	4.13	2.5%	4.0%
CCA Lamp Driver/Ring Generator	3.17	4.13	2.5%	4.0%
CCA Timing Source	3.17	4.13	2.5%	4.0%
Case and Covers	3.17	4.13	2.5%	10.0%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
CCA Analog Xmtr Amplifier	3.12	4.12	1.0%	*
CCA Multi V DC	3.12	4.12	1.0%	*
CCA 52 V DC	3.12	4.12	1.0%	*
CCA AC Input	3.12	4.12	1.0%	*
Control Panel Assembley (Remote)	3.12	4.12	1.0%	*
CCA Control Panel Logic	3.12	4.12	1.0%	*
CCA Analog Signaling	3.12	4.12	1.0%	*
CCA Analog Bridge	3.12	4.12	1.0%	*
CCA Telemetry Combiner/ Generator	3.12	4.12	1.0%	*
CCA VINSON Buffer/Ring Detector	3.12	4.12	1.0%	*

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TABLE VI. Group A inspection-continued.

Inspection	Reqt Para	Insp Para	ALQ Major	ALQ Minor
CCA Digital Bridge/Matrix	3.12	4.12	1.0%	*
CCA BITE 3	3.12	4.12	1.0%	*
CCA Alarm Interface	3.12	4.12	1.0%	*
CCA Lamp Driver/Ring Generator	3.12	4.12	1.0%	*
CCA Timing Source	3.12	4.12	1.0%	*
ELECTRICAL				
BITE and Alarms	3.7.13	4.7.13	1.0%	*
Operational Check	3.7.14	4.7.14	1.0%	*
DVOW Operator Access and Bridging	3.7.8.3	4.7.8.3	1.0%	*
AVOW Operator Access and Bridging	3.7.10.4	4.7.10.4	1.0%	*

*All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table VII and to the special inspection levels of table 1 of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B AQL shall be as shown in table VII and the inspection level shall be S-4.

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in the order listed in table VII.

TABLE VII. GROUP B inspection.

Inspection	Reqt Para	Insp Para	AQL
Line-to-line Voltage Levels	3.7.1.1	4.7.1.1	6.5%
Receiver Dynamic Range	3.7.1.7	4.7.1.7	6.5%
Driver Output Voltage	3.7.2.1	4.7.2.1	6.5%
Receiver Operating Range	3.7.3.2	4.7.3.2	6.5%

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TABLE VII Group B inspections - continued.

Inspection	Reqt Para	Insp Para	AQL
Output Level	3.7.4.2	4.7.4.2	6.5%
Signaling Frequency	3.7.4.4	4.7.4.4	6.5%
Signal Detector Operating Range	3.7.4.5	4.7.4.5	6.5%
Signal Generator Output	3.7.4.6	4.7.4.6	6.5%
Output	3.7.5.1	4.7.5.1	6.5%
Input Level	3.7.5.3	4.7.5.3	6.5%
Relay Contact Alarms	3.7.6.1	4.7.6.1	6.5%
Logic Driver Alarms	3.7.6.2	4.7.6.2	6.5%
Strobe Voltage	3.7.7.3	4.7.7.3	6.5%
Ring Generation	3.7.10.1	4.7.10.1	6.5%
Driver Output Voltage	3.7.3.1	4.7.3.1	6.5%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table VIII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1. Sampling for group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to vibration and shock testing with the unit subjected to testing in that order. The unit selected every eighth month shall be subjected to EMI and TEMPEST testing.

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4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.'

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to Altitude, Dust and Salt Fog testing and one unit shall be subjected to Fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical test in the order shown in table VIII.

TABLE VIII. Group C inspections.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High Temperature	3.8.1.1	4.8.1.1
Low Temperature	3.8.1.2	4.8.1.2
Subgroup 2		
EMI	3.9	4.9
TEMPEST	3.16	4.17
Vibration/Shock	3.8.7	4.8.7
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt Fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)

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TABLE VIII. Group C inspections - continued.

Inspection	Reqt Para	Insp Para
Subgroup 5		
Clock/Data Phasing	3.7.1.3	4.7.1.3
Signal Balance	3.7.1.5	4.7.1.5
Slew Rate	3.7.1.6	4.7.1.6
Input Load Impedance	3.7.1.8	4.7.1.8
Clock Asymmetry	3.7.1.9	4.7.1.9
Unloaded Output	3.7.1.10	4.7.1.10
Transmitter Output Impedance	3.7.2.2	4.7.2.2
Rise and Fall Times	3.7.2.3	4.7.2.3
Sync Signal	3.7.2.4	4.7.2.4
Rise and Fall Times	3.7.3.3	4.7.3.3
AVOW Input and Output Impedance	3.7.4.1	4.7.4.1
Input Impedance	3.7.5.2	4.7.5.2
Strobe Pulse Width	3.7.7.1	4.7.7.1
Slew Time	3.7.7.2	4.7.7.2
Signaling Detection	3.7.8.2	4.7.8.2
Detection with Errors	3.7.8.2.1	4.7.8.2.1
Bridge Seizure and Release	3.7.8.4	4.7.8.4
VINSON Buffer	3.7.8.5	4.7.8.5
Timing Source	3.7.9	4.7.9
Ring Detection	3.7.10.2	4.7.10.2
Format	3.7.11.1	4.7.11.1
Telemetry Decombiner Framing	3.7.11.2	4.7.11.2
Resync Command Pulse Rate	3.7.11.3	4.7.11.3
Prime Power Variation	3.7.12	4.7.12
Dimensional Inter-changeability	3.11	4.11

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried and refurbished if necessary, before proceeding with subsequent tests.

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4.5.3.2 Order of inspection within group C. Group C inspections shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample unts which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in table IX and shall be performed on units from lots which have been subjected to and have met group A and group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE IX. Group D inspection.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for group D inspection. Eight units shall be selected from the first month's production and subjected to the test of 4.10. For subsequent group D inspections, a maximum of eight units shall be selected at random from every fourth month's production.

4.5.5.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming group D samples units. Unless otherwise specified, samples units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass group A inspection only.

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4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected per figure 1. Test conditions of 4.10.3 shall apply except that duty cycle shall be 100%. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, and the preconditioning cycle reinstituted. The requirements of 3.6 shall be met.

a. Set control panel switches per table X. Set CCA switches per table X.

b. Test unit power ON. Run the unit continuously for 48 hours, using BITE to indicate failure. If a failure is detected, it is to be repaired and recorded. The last 24 hours of burn-in must be failure free as indicated by BITE.

c. Provisions shall be made to latch all S and SA alarms.

d. At the conclusion of the 48 hour burn-in, perform the operational test procedures of 4.7.14.

4.7 Electrical. (See 3.7) Unless otherwise specified, all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in table XI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (See table XII and figure 2) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test listed in table XIII. Table XIV identifies the external interface connections with the test figures. Unless otherwise noted all resistors in the test figures shall be .5 watts.

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TABLE X . Control panel and CCA switch positions for burn-in.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	2	3	4	5	6	1

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	ON
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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TABLE XI. Standard test equipment.

Description	Mfg/Model No. (or equivalent)
DC Power Supply	Power Designs 510A
Oscilloscope	Tektronix 7904
Attenuator	H.P. 350D
AC Voltmeter	H.P. 403B
Audio Oscillator	H.P. 200CD
Distortion Analyzer	H.P.333A
RMS Voltmeter	H.P. 400EL
Selective Voltmeter	H.P. 302A
Counter	H.P. 5345A
Frequency Synthesizer	H.P.3320B
Frequency Synthesizer	H.P. 3335A
Oscilloscope	Tekronix 453
Digital Volt-Ohmmeter	H.P. 3465A
Serial Data Analyzer	Spectron D-502
Time Interval Counter	H.P. 5233L
Logic Analyzer	H.P.1600A
Wattmeter	Weston 432
AC Power Source	Invertron 1501A
HD Comparator	Tekronix 7413
Dual Time Base	Tekronix 7853A
Counter	H.P. 5360A
Counter Input Module	H.P.5365A
Counter Time Internal Plug-In	H.P. 5379A

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TABLE XII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Timing Source	A3051666	The Timing Source provides four independent group outputs, switch-selectable to positions between 72 and 4608 kb/s. Also provided are four independent Station Clock/Telemetry Clock outputs, switch-selectable to 2, 16, or 32 kHz.
Timing Source	(Alternate)	An alternate Timing Source is shown in figure 2 that may be used in lieu of the above.
DVOW/DOW Simulator	A3051669	The DVOW/DOW Simulator provides a pseudo-random data generator and correlator at 2, 16, 32 kbps with a balanced signal interface. A TTL error output signal is provided for counting errors externally.
Error Inserter	A3051665	The Error Inserter provides a means of injecting a 0.1% continuous BER with 20% random bursts.

TABLE XIII. Government furnished test equipment.

Equipment Name	Description
Handset H-250( )/U	The handset utilizes a 1000 ohm earphone element a 150 ohm dynamic noise canceling microphone, a push-to-talk switch and 5 pin audio connector.

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TABLE XIV. C-10716( )/TRC interface connections.

Connector	Function	Hi	Lo	Gnd	Signal Type
DIGITAL VOICE ORDERWIRE	CH1 DVOW IN	C	B	A,s,t	NRZ
	TIM IN	d	c		SQWV
	DVOW OUT	R	P		NRZ
	CH2 DVOW IN	E	D		NRZ
	TIM IN	f	e		SQWV
	DVOW OUT	T	S		NRZ
	CH3 DVOW IN	G	F		NRZ
	TIM IN	h	g		SQWV
	DVOW OUT	V	U		NRZ
	CH4 DVOW IN	J	H		NRZ
	TIM IN	j	i		SQWV
	DVOW OUT	X	W		NRZ
	CH5 DVOW IN	L	K		NRZ
	TIM IN	m	k		SQWV
	DVOW OUT	Z	Y		NRZ
	CH6 DVOW IN	N	M		NRZ
	TIM IN	p	n		SQWV
	DVOW OUT	b	a		NRZ
TIMING	Group 1 INPUT	C	B	A,s,t	SQWV
	OUTPUT	E	D		SQWV
	Group 2 OUTPUT	G	F		SQWV
	MOW DATA IN	J	H		NRZ
	REC TIM IN	L	K		SQWV
	DATA OUT	N	M		NRZ
	XMT TIM IN	R	P		SQWV
	16 Kc OUT 1	T	S		SQWV
	2	V	U		SQWV
	3	X	W		SQWV
	32 Kc OUT 1	Z	Y		SQWV
	2	b	a		SQWV
	3	d	c		SQWV
	2 Kc OUT 1	f	e		SQWV
	2	h	g		SQWV
	TEST DATA IN	j	i		NRZ
	OUT	m	k		NRZ

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TABLE XIV. C-10716( )/TRC interface connections-continued.

CONNECTOR	FUNCTION	HI	LO	GND	SIGNAL TYPE
VINSON BLACK and TD 2,3	DATA IN	A		D	NRZ
	DATA OUT	B			NRZ
	PTT IN	C			RELAY CLOSURE
	TD2 DATA IN	F	E	T,S,U	NRZ
	TIM IN	H	G		SQWV
	TD3 DATA IN	K	J		NRZ
	TIM IN	M	L		SQWV
ANALOG VOICE ORDERWIRE	CH1 IN	B	A	h,j,T	ANALOG
	OUT	P	N		ANALOG
	CH2 IN	D	C		ANALOG
	OUT	S	R		ANALOG
	CH3 IN	F	E		ANALOG
	OUT	V	U		ANALOG
	CH4 IN	H	G		ANALOG
	OUT	X	W		ANALOG
	CH5 IN	K	J		ANALOG
	OUT	Z	Y		ANALOG
	CH6 IN	M	L		ANALOG
	OUT	b	a		ANALOG
	MOW IN	d	c		ANALOG
	OUT	f	e		ANALOG
DLED RED	RXDPT	B	A		NRZ
	RSCLK	D	C		SQWV (To DELED 1)
	SYNC CMD TX	E			PULSE
	TXDPT	K	J		SQWV (To DELED 2)
	TXCLK	M	L		
	SIGNAL GROUND			G	
	SHIELD GROUND			T,S,U	
ALARMS 1-27	ALARM 1	A			CONTACT CLOSE
	2	B			CONTACT CLOSE
	3	C			CONTACT CLOSE
	4	D			CONTACT CLOSE
	5	E			CONTACT CLOSE
	6	F			CONTACT CLOSE
	7	G			CONTACT CLPSE
	8	H			CONTACT CLOSE
	9	J			CONTACT CLOSE
	10	K			CONTACT CLOSE
	11	L			CONTACT CLOSE
	12	M			CONTACT CLOSE

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TABLE XIV. C-10716( )/TRC interface connections-continued.

CONNECTOR	FUNCTION	HI	LO	GND	SIGNAL TYPE
ALARMS 1-27 (continued)	ALARM 13	N			CONTACT CLOSE
	14	P			CONTACT CLOSE
	15	R			CONTACT CLOSE
	16	S			CONTACT CLOSE
	17	U			CONTACT CLOSE
	18	V			CONTACT CLOSE
	19	W			CONTACT CLOSE
	20	X			CONTACT CLOSE
	21	Y			CONTACT CLOSE
	22	Z			CONTACT CLOSE
	23	a			CONTACT CLOSE
	24	b			CONTACT CLOSE
	25	c			CONTACT CLOSE
	26	d			CONTACT CLOSE
	27	e			CONTACT CLOSE
	SIGNAL GROUND			f,g,h,j	
	SHIELD GROUND			T	
ALARMS 28-55	ALARM 28	A			
	29	B			
	30	C			CONTACT CLOSE
	31	D			CONTACT CLOSE
	32	E			CONTACT CLOSE
	33	F			CONTACT CLOSE
	34	G			CONTACT CLOSE
	35	H			CONTACT CLOSE
	36	J			CONTACT CLOSE
	37	K			CONTACT CLOSE
	38	L			CONTACT CLOSE
	39	M			CONTACT CLOSE
	40	N			CONTACT CLOSE
	41	P			CONTACT CLOSE
	42	R			CONTACT CLOSE
	43	S			CONTACT CLOSE
	44	U			CONTACT CLOSE
	45	V			CONTACT CLOSE
	46	W			CONTACT CLOSE
	47	X			CONTACT CLOSE
	48	Y		connected	internally
	49	Z			LOGIC DRIVER
	50	a			LOGIC DRIVER
	51	b			LOGIC DRIVER

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TABLE XIV. C-10716( )/TRC interface connections-continued.

CONNECTOR	FUNCTION	HI	LO	GND	SIGNAL TYPE
ALARMS 28-55 (continued)	52 53 54 55 SIGNAL GROUND SIGNAL GROUND	c d e f		 g,h,j T	LOGIC DRIVER LOGIC DRIVER LOGIC DRIVER LOGIC DRIVER LOGIC DRIVER
BIT ERROR RATE	BER 1 STROBE BER 2 STROBE BER 3 STROBE BER BIT 1(LSB) BER BIT 2 BER BIT 3 BER BIT 4	K M P B D F H	J L N A C E G	R,S,U,T	
POWER		A	c	B	
HANDSET (CONT. PANEL)	HANDSET XMT HANDSET RCV PTT	D B C		A	ANALOG ANALOG
VINSON RED (CONT.PANEL)	AUDIO IN AUDIO OUT PTT OUT	B C E		D(SIG) A,F,G(SQ)	ANALOG ANALOG

REMOTE CABLE CONNECTIONS

REMOTE CABLE 1 On the C-10716 main unit connects to REMOTE CABLE 1 on the control panel. Pins at one end of the cable connect to pins of the same letter at the other end. The following pins are used: R, B, C, D, E, F, G, J, K, L, M, N, P, HH, S, T, U, V, W, X, Y, Z, a, b, c, d, e, f, g, h, i, j, k, m, n, p, q, r, s, t, u, v, w, x, y, z, AA, BB, CC, H, and GG. Pin A is the cable shield at both ends.

REMOTE CABLE 2 On the C-10716 main unit connects to REMOTE CABLE 2 on the control panel. Pins at one end of the cable connect to pins of the same letter at the other end. The following pins are used: d, e, f, g, h, j, k, m, n, p, i, R, S, T, U, V, W, X, Y, E, F, G, H, J, K, L, M, N, P, Z, a, b, c, q, r, s, and t. Pin A is the cable shield at both ends. The maximum voltage between any wire and the shield is 56 volts. A maximum cable length of 50 feet can be used if the maximum wire gauge is #30.

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4.7.1 Balanced equipment side signal characteristics. (See 3.7.1)

4.7.1.1 Line-to-line voltage levels. (See 3.7.1.1).

a. Connect the equipment as shown in figure 3 using oscilloscope setup A.

b. Set the UUT as shown in table XV.

c. For each of the signals listed in table XVI measure the positive and negative voltage levels line-to-line. Measure Channel 2 through 6 DVOW DATA OUT signals with the DVOW/DOW input at channel 1. Measure Channel 1 OUT with the DVOW/DOW input at Channel 2.

d. For an all "1's" pattern, an all "0's" pattern and a random pattern verify that the line-to-line voltage level requirements of 3.7.1.1 are met.

4.7.1.2 Line-to-ground voltage. (See 3.7.1.2) Measure the line-to-line ground voltage for each of the signals listed in table XVI using the test configuration shown in figure 3 using oscilloscope setup B. Verify that the requirements of 3.7.1.2 are met.

4.7.1.3 Clock/data phasing. (See 3.7.1.3)

a. Connect the equipment as shown in figure 4.

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TABLE XV Switch positions for the line-to-line voltage levels.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	B	B	B	B	B	B

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	ON
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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TABLE XVI Line-to-line signals.

DVOW DATA OUT CH 1	16 kHz CLOCK OUT 1
DVOW DATA OUT CH 2	16 kHz CLOCK OUT 2
DVOW DATA OUT CH 3	16 kHz CLOCK OUT 3
DVOW DATA OUT CH 4	STROBE OUT 1
DVOW DATA OUT CH 5	STROBE OUT 2
DVOW DATA OUT CH 6	STROBE OUT 3
MOW DATA OUT	2 kHz CLOCK OUT 1
TXDPT	2 kHz CLOCK OUT 2
TEST DATA OUT	GROUP 1 OUT
32 kHz CLOCK OUT 1	GROUP 2 OUT
32 kHz CLOCK OUT 2	
32 kHz CLOCK OUT 3	

b. Set the UUT circuit board switches and the control panel switches as shown in table XVII.

c. Push the DVOW Channel 1 pushbutton and close the PTT switch.

d. For each of the following pairs of signals measure the delay between the zero voltage points on both the data rise and fall time slopes to the zero voltage points of the timing rise slopes as indicated in figure 5.

(1) DVOW DATA CH 1 OUT/16 kHz CLK 1 OUT

(2) DVOW DATA CH 1 OUT/16 kHz CLK 2 OUT

(3) DVOW DATA CH 1 OUT/16 kHz CLK 3 OUT

(4) TX CLK IN/TX DPT OUT

e. Verify that the delays meet the requirements of 3.7.1.3.

4.7.1.4 Offset. (See 3.7.1.4)

a. Connect the scope connections as shown in figure 6(A).

b. For each of the output signals listed in table XVI (except the strobe outputs) measure the dc off-set between the high and low pins.

c. Verify that the dc offset voltages satisfy the requirements of 3.7.1.4.

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TABLE XVII Switch positions for the clock/data phasing test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	T	T	T	T	T	T

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG	GROUP 1	14
SCE	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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4.7.1.5 Signal balance. (See 3.7.1.5)

a. Connect the equipment as shown in figure 3.

b. Set the DVOW/DOW simulator switches as shown in table XV.

c. For each of the signals listed in table XVI, measure the positive and negative voltage levels line-to-line.

NOTE: Measure Channel 2 through 6 DVOW DATA OUT signals with the DVOW/DOW input at Channel 1. Measure Channel 1 OUT with the DVOW/DOW input at Channel 2.

d. Compare the value of the magnitudes of the positive and negative voltage levels. Determine the larger of the two values. Then, compare the difference between the positive and negative values to 10% of the larger value.

e. Verify that the requirements of 3.7.1.5 are met.

4.7.1.6 Slew rate. (See 3.7.1.6)

a. The equipment setup is the same as in 4.7.1.1 except change the scope connections as shown in figure 6 (B). All measurements shall be made through 100 feet of RG-108 cable.

b. For each of the signals listed in table XVI, measure the time between the +40 mV and -40 mV points around the zero voltage levels of the rise and fall slopes as shown in figure 7.

c. Verify that the requirements of 3.7.1.6 are met.

4.7.1.7 Receiver dynamic range. (See 3.7.1.7)

a. Connect the equipment as shown in figure 8. Set the balanced attenuators (A) to give ±50mV input into the unit, the unbalanced attenuator (B) to give ±0.6 volt input into the unit ohm load, and the balanced attenuators (C) to give ±200 mV input into the unit. Measure the attenuator output levels using an oscilloscope set up in the diffential mode referenced to ground.

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b. Set the control panel switches and CCA switches as shown in table XVIII.

c. Measure the DVOW Channel 1 through 6 BER as follows:

(1) Set the STATION TIMING thumbwheel switch to the selected channel.

(2) Connect the DVOW/DOW Simulator data and timing output to the selected channel inputs. Connect the DVOW/DOW Simulator data input to the selected channel output, close the PTT switch.

(3) Monitor the DVOW/DOW Simulator for a period of 30 seconds. Verify that no errors are indicated.

d. Verify that there are no S or LOI alarms.

e. Set the timing switch to G.

f. Connect the DVOW/DOW Simulator data input and output to channel 1.

g. Verify that there are no alarms.

h. Set the BER Code Switch to position 7 on the BER number generator. Verify that the Summary Status Light on the Control Monitor is AMBER.

i. Set the BER Code Switch to position 8 on the BER number generator. Verify that the Summary Status Light is GREEN.

j. Connect TXDPT to RXDPT (TD1) and 2 kHz TIMING to RXCLK and TXCLK. Set the TD1 switch to ON. Verify that TD OOF and LOI LED's are off, then set TD1 to off.

k. Connect TXDPT to TD2 DATA IN and 2 kHz TIMING to TD2 TIM IN. Set the TCG TD2 switch to ON. Verify that the TD OOF and LOI LED's are off. Then set TD2 to off.

l. Repeat k. substituting TD3 for TD2.

m. Push AVOW 1 button. Push and hold the SIG A SELF RING switch. Verify that Channel 1 flashes.

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TABLE XVIII. Switch position for receiver sensitivity and operating range test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	B	B	B	B	B	B

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION		AS REQUIRED						

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	FRG	ON
TMG	GROUP 1	14
SCE	GROUP 2	14
VERC	VERC RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMS 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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4.7.1.8 Input load impedance. (See 3.7.1.8)

a. Connect the equipment as shown in figure 10.

b. Connect the test setup to the inputs listed in table XIX (100, 2000 and 6000 ohm inputs).

c. For the 100 ohm inputs, use R = 100 ohms ±1%, adjust the audio oscillator at a frequency of 1.0 kHz to produce an output voltage at V1 of 1.0 volts. Then measure the V2 voltage. Record the V2 voltage.

d. For the 6K ohm inputs, use R = 6000 ohms ±1% and 10 Hz, proceed as in c.

e. For the 2K ohm inputs, use R = 2000 ohms ±1% and 1.0 kHz, proceed as in c.

f. Calculate all the input impedances using the formulas below:

$$Z = (R)\frac{V2}{V1} \quad \text{ohms}$$

NOTE: Use R = 100 ohms, R = 6000 ohms R = 2000 ohms as required.

g. Verify that the requirements of 3.7.1.8 are met.

TABLE XIX. Signals for the input load impedance test.

100 Ohms Input	
Test Data In	BER Data In 3
Group TIM In	BER Data In 4
MOW Data In	DVOW CH 1 DATA IN
MOW RCV TIM IN	DVOW CH 3 DATA IN
MOW XMT Tim In	DVOW CH 4 DATA IN
BER Data In 1	DVOW CH 5 DATA IN
BER Data In 2	DVOW CH 6 DATA IN

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TABLE XIX. Signals for the input load impedance test (continued).

100 Ohm Input
DVOW CH 1 TIM IN
DVOW CH 2 TIM IN
DVOW CH 3 TIM IN
DVOW CH 4 TIM IN
DVOW CH 5 TIM IN
DVOW CH 6 TIM IN

6000 Ohms Input
Logic Alarm 49
50
51
52
53
54
55

2000 Ohms Input
TD DATA IN 2
TD DATA IN 3
RXCLK
RXDPT
TXCLK
TD TIM IN 2
TD TIM IN 3

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4.7.1.9 Clock asymmetry. (See 3.7.1.9)

a. The equipment test setup is as specified in 4.7.1.3.

b. For each of the signals listed below measure the time (A) between the rise and fall slopes at the 50% points. Then measure the time (B) between two successive rise slopes at the 50% points as indicated in figure 11.

c. For the following signals verify that the requirements of 3.7.1.9 are met.

32 kHz CLOCK	1 OUT
	2 OUT
	3 OUT
16 kHz CLK	1 OUT
	2 OUT
	3 OUT
2 kHz CLK	1 OUT
	2 OUT

d. GROUP 1 CLK OUT @ 4608

e. GROUP 2 CLK OUT @ 4608

4.7.1.10 Unloaded output. (See 3.7.1.10)

a. The equipment connections are as specified in 4.7.1.1, except that the 50 ohm terminations are removed.

b. Measure the line-to-line voltage levels of the TXDPT OUT and the 2 kHz timing 1 and 2 outputs.

c. Verify that the requirements of 3.7.1.10 are met.

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4.7.2 DLED resync interface. (See 3.7.2)

4.7.2.1 Driver output voltage DLED and VINSON interface. (See 3.7.2.1)

a. Connect the equipment as shown in figure 12.

b. Set the UUT circuit board switches and the control panel switches as shown in table XX. Push the DVOW Channel 1 pushbutton on the control panel.

c. Using the oscilloscope measure the SYNC CMD TX Voltage Levels both positive and negative into an open circuit and into a 20 k ohm load.

d. Using the oscilloscope terminated in 20K ohms, measure the VINSON DATA OUT voltage levels both positive and negative.

e. Verify that the SYNC CMD TX and VINSON DATA OUT requirements of 3.7.2.1 and 3.7.3.1 are met.

4.7.2.2 Transmitter output impedance. (See 3.7.2.2)

a. Connect the equipment as shown in figure 13. The UUT circuit board and Control panel switches are as shown in table XX.

b. Using the oscilloscope, measure the peak-to-peak voltage into a 6000 ohm load on the output of the SYNC CMD TX.

c. Repeat b using the 600- ohm load.

d. Calculate the output impedance using the following formula:

$$Z = \frac{6000\ (V_{6K} - V_{600})}{(10V_{600} - V_{6K})}$$

e. Verify that the requirements of 3.7.2.2 are met.

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TABLE XX Switch positions for driver output voltage DLED and VINSON interface test, and transmitter output impedance test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	B	B	B	B	B	B

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMP 1	ON
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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4.7.2.3 Rise and fall times. (See 3.7.2.3)

a. The equipment setup and switch positions are the same as in 4.7.2.2. Push the DVOW Channel 1 button.

b. Using the oscilloscope terminated in 6K ohms, measure the SYNC CMD TX and the VISION DATA OUT rise and fall times at the 10% and 90% points.

c. Verify that the rise and fall times meet the requirements 3.7.2.3.and 3.7.3.3 respectively.:

4.7.2.4 Sync signal. (See 3.7.2.4)

a. The equipment setup and UUT switch positions are the same as 4.7.2.3.

b. Measure the pulse width of the SYNC CMD TX signal at zero volts. Verify that the pulse width is within the required limits of 3.7.2.3.

4.7.3 Unbalanced VINSON interface. (See 3.7.3)

4.7.3.1 Driver output voltage. (See 3.7.3.1) The driver output voltage requirement of 3.7.3.1 is demonstrated as part of the Driver Output Voltage DLED and VINSON Interface Tests of 4.7.2.1.

4.7.3.2 Receiver operating range. (See 3.7.3.2) The receiver operating range requirement of 3.7.3.2 is demonstrated as part of the Receiver Sensitivity Tests of 4.7.1.7.

4.7.3.3 Rise and fall times. (See 3.7.3.3) The rise and fall time requirement of 3.7.3.3 is demonstrated as part of the Rise and Fall Times Tests of 4.7.2.3.

4.7.4 Analog voice orderwire (AVOW) channel interface. (See 3.7.4).

4.7.4.1 AVOW input and output impedance. (See 3.7.4.1)

a. Connect the equipment as shown in figure 14.

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b. Set the control panel AVOW toggle switches 1 through 6 to terminate.

c. Connect the oscillator in turn to the signals listed below at a frequency of 1 kHz with an output level of 1.0 volt as measured on the V1 voltmeter. Measure and record the V2 voltage.

CH 1 AVOW IN	CH 1 AVOW OUT
CH 2 AVOW IN	CH 2 AVOW OUT
CH 3 AVOW IN	CH 3 AVOW OUT
CH 4 AVOW IN	CH 4 AVOW OUT
CH 5 AVOW IN	CH 5 AVOW OUT
CH 6 AVOW IN	CH 6 AVOW OUT
MOW IN	MOW OUT

d. Repeat c. at 300 Hz and 3400 Hz.

e. Calculate the input and output impedance using the formula:

$$Z = 600 \frac{V_2}{V_1} \text{ Ohms}$$

f. Verify that the specified requirements are met.

4.7.4.2 Output levels. (See 3.7.4.2) The output test tone level requirement of 3.7.4.2 is demonstrated as part of the AVOW Operator Access and Bridging (Gain) Tests of 4.7.10.4.

4.7.4.3 Frequency response. (See 3.7.4.3)

a. Connect the equipment as shown in figure 15 (S1 closed) and put all AVOW channels on the bridge.

b. For each of the AVOW channel outputs 1-6 measure and record the frequency response of each channel at 300, 400, 1000, 3000, and 3400 Hz.

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NOTE: Measure AVOW Channels 2 through 6 with the oscillator set to 977mV connected to Channel 1, then measure Channel 1 with the oscillator connected to Channel 2. This level will produce -4 dBm into the 600 ohms.

c. Verify that the specified requirements of 3.7.4.3 are met.

4.7.4.4 Signal generator frequency and output level. (See 3.7.4.4)

a. Connect the equipment as specified in figure 16 with S1 open.

b. Push the analog Channel 1 operator control switch on the control panel.

c. Push the SELF RING button and measure and record the AVOW CH 1 Output frequency and voltage level. For Channels 2 through 6 measure the output voltage level.

d. Verify that the requirements of 3.7.4.4 and 3.7.4.6 are met.

4.7.4.5 Signal detector operating range. (See 3.7.4.5)

a. Connect the equipment as shown in figure 16 (S1 closed).

b. Set the oscillator frequency to 1568 Hz.

c. Set the oscillator output level to 245 mV. (This will produce -16 dBm into a 600 ohm load).

d. Open S1 the input to AVOW CH 1 IN, reset the CH 1 indicator to OFF by pushing the CH 1 selector switch (wait approximately 3 seconds).

e. Monitor the Channel 1 pushbutton selector switch then close S1 the input to AVOW CH 1 IN, the CH 1 selector shall flash within 2 seconds of connection.

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f. Repeat d. and e. for channels 2 through 6.

g. Repeat d. through f. with an oscillator level of 1.23 Volts (this will produce -2 dBm into a 600 ohm load).

h. Repeat d. through g. at a frequency of 1632 Hz.

i. Verify that the requirements of 3.7.4.5 are met.

4.7.4.6 Signal generator output. (See 3.7.4.6) The signal generator output requirement of 3.7.4.6 is demonstrated as part of signal frequency tests of 4.7.4.4.

4.7.5 AVOW handset interface. (See 3.7.5)

4.7.5.1 Output. (See 3.7.5.1) The output requirement of 3.7.5.1 is demonstrated as part of the AVOW Operator Access and Bridging (Gain) Tests of 4.7.10.4.

4.7.5.2 Input impedance. (See 3.7.5.2)

a. Connect the equipment as shown in figure 17. Push CH 1 AVOW button.

b. Adjust the oscillator at a frequency of 1000 Hz to produce a V1 voltage level of 1.0 volt, then measure and record the V2 voltage.

c. Calculate the impedance using the formula:

$$Z = 150\frac{V_2}{V_1} \text{ ohms}$$

d. Verify that the input impedance requirements of 3.7.5.2 are met.

4.7.5.3 Input level. (See 3.7.5.3) The input level requirement of 3.7.5.3 is demonstrated as part of the AVOW Operator Access and Bridging (Gain) Tests of 4.7.10.4.

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4.7.5.4 AVOW handset frequency response. (See 3.7.5.4)

a. Connect the equipment as shown in figure 18A. Push CH 1 AVOW button.

b. With the oscillator set to 1.4 mVrms at 1000 Hz record the output in dBm.

c. Repeat b. with the oscillator frequency set to 300, 400, 1600, and 3000 Hz. Measure and record the output at each frequency in dBm.

d. Verify that the frequency response is within ± 2 dB maximum from the reference value at 1000 Hz.

e. Connect the equipment as shown in figure 18B. Push CH 1 AVOW button.

f. With the oscillator set to 977 mV (-4 dBm into the unit) at 1000 Hz. Record the output in dBm.

g. Repeat f. with the oscillator frequency set to 300, 400, 1600,and 3000 Hz. Measure and record the output at each frequency in dBm.

h. Verify that the frequency response is within ±2 dB maximum from the reference value at 1000 Hz.

4.7.6 Alarm interface. (See 3.7.6)

4.7.6.1 Relay contact alarms. (See 3.7.6.1)

a. Connect the equipment as shown in figure 19. Set the control panel switches as shown in table XXI.

b. For each alarm shown in table I, short the appropriate alarm signal pin (see table XIV) to ground to activate the alarm message.

c. Verify that as each alarm is activated the RED STATUS light comes ON and the GREEN STATUS light goes OFF and the UNIT ALARM sounds and the corresponding error message is generated.

NOTE: To initiate Alarm 48, Push Lamp Test Button on Front Panel.

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TABLE XXI. Switch positions for relay contract alarm tests.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION						(AS REQUIRED)										
DIGITAL	1	2	3	4	5	6										
POSITION	T	T	T	T	T	T										

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X
ANALOG	1	2	3	4	5	6		
BRIDGE								
TERMINATE	X	X	X	X	X	X		

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG	GROUP 1	14
SCE	GROUP 2	14
VBRD	VBRD RING CODE	AS ROD
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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4.7.6.2 Logic driver alarms. (See 3.7.6.2)

a. The equipment setup is the same as in 4.7.6.1.

b. Apply +2.0 Vdc to Alarm 49. Verify that the RED STATUS light comes ON and the GREEN STATUS light goes OFF. The UNIT ALARM sounds and the proper ALARM word as indicated in table II is generated.

c. Repeat step b for +6.0 Vdc input.

d. Apply +0.8 Vdc to Alarm 49. Verify that the control panel summary status indicator is GREEN and no alarm sounds.

e. Repeat step d for -6.0 Vdc input.

f. Repeat b through e for the remaining logic alarms

4.7.7 Bit error rate (BER) interface. (See 3.7.7)

4.7.7.1 Strobe pulse width. (See 3.7.7.1)

a. Connect the equipment as shown in figure 3 with the oscilloscope connected to strobe 1.

b. Measure the strobe pulse width (A) at zero output as indicated in figure 20.

c. Verify that the strobe pulse widths are within the specified requirements of 3.7.7.1.

d. Repeat b and c, for strobes 2 and 3.

4.7.7.2 Slew time. (See 3.7.7.2) The slew time requirement of 3.7.7.2 is demonstrated as part of the slew rate tests of 4.7.1.6.

4.7.7.3 Strobe voltage. (See 3.7.7.3) The strobe voltage levels requirement of 3.7.7.3 is demonstrated as part of the balanced equipment side signal characteristics line-to-line voltage level tests of 4.7.1.1.

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4.7.8 Digital voice orderwire (DVOW) performance. (See 3.7.8)

4.7.8.1 Signaling code generation. (See 3.7.8.1)

a. Connect the equipment as shown in figure 3 with the oscilloscope connected to channel 1.

b. Set the UUT and control panel switches as shown in table XXI.

c. Set the RING CODE switch on the control panel to position 1.

d. Push the Channel 1 digital pushbutton, then push the RING button.

e. Verify that the ring code (1) is as shown in table III.

f. Repeat c. through e. for each remaining ring code.

g. Using the oscilloscope, measure the time between the first and last bits of Code 16 verify that the ring code duration is within the limits specified in 3.7.8.1.

4.7.8.2 Signaling detection. (See 3.7.8.2)

a. Set the UUT switch positions and control panel switches as shown in table XXI.

b. Set the control panel and VBRD ring codes to Position 1.

c. Connect DVOW CH1 output to DVOW CH1 input and the 16 KHz CLK 1 OUT to CH1 TIM IN. Push the CH1 button, wait three (3) seconds then push the ring button on the control panel. Verify that only channel 1 flashes. Then, push the DVOW 1 push button.

d. Repeat c. substituting ring codes 2 through 16 for CH1 (one at a time). Verify that only the selected channel flashes.

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e. With the input at channel 6, and the VBRD ring code set to Position 1, set the control panel ring code to Code 2 through 16, ring at each setting and verify that no channel flashes.

f. Set the VBRD ring code to 2 then ring with codes 1 through 16. Verify that CH6 flashes only on Code 2.

g. Repeat f. for VBRD ring codes 3 through 16.

4.7.8.2.1 Detection with errors. (See 3.7.8.2.1)

a. Connect the equipment as shown in figure 21.

b. The UUT and control panel switch positions are the same as in 4.7.8.2. Set the control panel and VBRD ring codes to 1.

c. Push DIGITAL DVOW 1 selector pushbutton on the control panel.

d. Set the Error Inserter to give a 10^{-3} error rate with bursts of 20 percent errors at a duty cycle of 5 percent.

e. Ring 20 times. Channel 1 shall respond at least 18 times.

NOTE: Wait four (4) seconds after pushing channel button before ringing.

4.7.8.2.2 False ring detection. (See 3.7.8.2.2)

a. Connect the equipment as shown in figure 22.

b. The UUT switches and control panel switches are shown in table XXI except, set the VBRD RING Code to 14.

c. Run the unit for 24 hours. There shall be no ring detections.

4.7.8.3 DVOW operator access and bridging. (See 3.7.8.3)

a. Connect the equipment as shown in figure 23.

b. Set the unit switches and control panel switches as specified in table XXII. Close PTT.

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TABLE XXII. Switch positions for DVOW operator access and bridging test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
	B	B	B	B	B	B

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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c. Push the digital OW B pushbutton on the control panel.

d. Connect the DVOW/DOW Simulator to each of the combinations of channels marked C shown in table XXIII.

TABLE XXIII. DVOW operator access and bridging channel combinations.

		DVOW/DOW #2						
		1	2	3	4	5	6	OP
DVOW/DOW #1	1	0	C	C	C	C	C	C
	2	-	0	C	C	C	C	C
	3	-	-	0	C	C	C	C
	4	-	-	-	0	C	C	C
	5	-	-	-	-	0	C	C
	6	-	-	-	-	-	0	C
	OP	-	-	-	-	-	-	0

e. Connect the simulator #1 Data Input to Channel 1 Data Output and simulator #2 Data Input to Channel 2 Data Output. Connect the simulator #1 Data Output to Channel 1 Data Input and verify error free transmission to simulator #2 for at least 30 seconds.

f. Leaving these connections in place connect the Data Output from simulator #2 to Channel 2 Data Input. Verify that error free transmission is maintained at simulator #2.

g. Disconnect the Data Output from simulator #1 and verify error free transmission to simulator #1 for at least 30 seconds.

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h. Repeat e. through g. for all combinations in the table which are marked C.

i. Disconnect simulator #2. Reconnect simulator #1 Data Input and Output to Channel 1 Data Output and Input.

j. Verify that no activity is received by simulator #1.

k. Repeat i, and j. for all channels marked 0 in table XXIII.

l. Push the DVOW 1 button. Connect simulator #1 DATA IN and OUT to CH1 and simulator #2 DATA IN and OUT to operator. Verify error free transmission in both directions.

m. Repeat 1. using Channels 2 through 6 to the operator. Verify error free transmission for each selected channel.

n. Insure that all AVOW and DVOW channel lights are OFF. Select the channels specified in table XXIV and demonstrate error free operation.

TABLE XXIV. DVOW operator access and bridging channel matrix.

CH	Switch Setting	CH	Switch Setting	CH	Switch Setting	CH	Switch Setting	CH	Switch Setting
1	2	2	3	3	4	4	5	5	6
1	3	2	4	3	5	4	6	-	-
1	4	2	5	3	6	-	-	-	-
1	5	2	6	-	-	-	-	-	-
1	6	-	-	-	-	-	-	-	-

NOTE: When testing combination 1-2, set CH 1 thumbwheel SW to Pos 2 and Channel 2 to Position 1. Repeat for all other combinations.

o. Verify that the specified requirements of 3.7.8.3 are met.

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4.7.8.4 Bridge seizure and release. (See 3.7.8.4)

a. Connect the equipment as shown in figure 25. Set the UUT and control panel switches as shown in table XXV.

b. Start the counter upon application of data (START = negative transition). Then measure the time from the start signal to the appearance of data (bridge seizure).

c. Repeat b. for a total of 20 times.

d. Start the counter upon squelch of the data (START = negative transition). Then measure the time from the start signal to the appearance of all ones (bridge release).

NOTE: When data is squelched the input to channel 1 must be all zeroes.

e. Repeat d. for a total of 20 times.

f. Verify that the specified requirements of 3.7.8.4 are met.

4.7.8.5 VINSON buffer. (See 3.7.8.5)

a. Connect the equipment as shown in figure 27.

b. Step 2 Set the UUT and control panel switches as shown in table XXVI.

c. Connect the frequency synthesizer to DVOW/DOW Simulator Channel 2 at a frequency of 16,000.48 Hz (+30 ppm).

d. Push DVOW 1 selector on the control panel and close the PTT switch.

e. Measure and record the time between BCI losses.

f. Repeat c. through e. with the synthesizer set to 15,999.52 Hz (-30 ppm).

g. Open the PTT switch. Verify that Channel 1 output is all ones.

h. Verify that the specified requirements are met.

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TABLE XXV. Switch position for bridge seizure and release test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	B	B	T	T	T	T

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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TABLE XXVI. Switch position for VINSON buffer test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	T	T	T	T	T	T

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION	X							

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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4.7.9 Timing source. (See 3.7.9)

a. Connect the equipment as shown in figure 28.

b. Connect a frequency counter to the Group 1 output.

c. Set the Group 1 rate switch on the Timing Source CCA to 2 and the front panel timing switch to M.

d. Measure the Group 1 frequency.

e. Repeat c. and d. for Group 1 rate switch settings of 3, 4, 5, 6, 7, 8, 9, 12, and 14.

f. Verify that the specified requirements in table XXVII are met.

g. Set the Group 2 rate switch on the Timing Source CCA to 14. Set the Timing Switch, on the front panel to G.

h. Connect the frequency counter to the Group 2 output.

i. Connect a timing source with frequency synthesizer to the Group 1 input.

j. Set the Group 1 rate switch to 2.

k. Set the synthesizer to 128001.3 Hz.

l. Measure the Group 2 output rate.

m. Verify that it is 4,608,046.1 Hz.

n. Set the synthesizer to 127998.7 Hz.

o. Measure the Group 2 output rate.

p. Verify that it is 4,607,953.9 kHz.

q. Set the synthesizer to 144 kHz (nominal frequency).

r. Set Timing Source and Group 1 Rate switch to Rate 3.

s. Verify that the Group 2 output Rate is 4608 kHz $\pm$1 ppm.

t. Repeat q. through s. for Group 1 rate switch settings of 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14. Verify that the requirements of 4608 kHz $\pm$ 1 ppm are met.

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u. Set the synthesizer to 16 kHz.

v. Connect timing source with frequency synthesizer to DVOW Channel 1 timing input.

w. Set UUT timing switch to 1.

x. Verify that the group 2 output rate is 4608 kHz ±1 ppm.

y. Repeat v. through x. using DVOW Channels 2, 3, 4, 5 and 6.

TABLE XXVII. Timing source master mode requirements.

GROUP 1 OUTPUT	REQUIREMENT
128	127998.72 to 128001.28
144	143998.56 to 144001.44
256	255997.44 to 256002.56
288	287997.22 to 288002.88
512	511994.88 to 512005.12
576	579994.24 to 576005.76
1024	1023989.76 to 1024010.24
1152	1151988.48 to 1152011.52
2304	2303976.96 to 2304023.04
4608	4607953.92 to 4608046.08

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4.7.10 AVOW performance. (See 3.7.10)

4.7.10.1 Ring generation. (See 3.7.10.1)

a. Connect the equipment as shown in figure 29.

b. Push the AVOW 1 pushbutton on the control panel, and the RING button.

c. Measure the duration of the ring signal.

d. Verify that the requirements of 3.7.10.1 are met.

4.7.10.2 Ring detection. (See 3.7.10.2)

a. Connect the equipment as shown in figure 30.

b. Set all channel switches to terminate.

c. Set the oscillator to 1632 Hz at a level of -2 dBm (noise off).

d. Turn the audio oscillator off and set the noise generator to -17 dBm (true rms reading).

e. Key the audio oscillator on for 2 seconds a minimum of 20 times and measure and record the number of detections.

f. Repeat c. through e. with the audio oscillator set to -16 dBm and a noise level of -31 dBm.

g. Repeat c. through f. with an audio oscillator frequency of 1568 Hz.

h. Verify that the requirements of 3.7.10.2 are met.

4.7.10.3 False ring detection. (See 3.7.10.3)

a. Connect the equipment as shown in figure 30.

b. Turn off the audio oscillator and set the noise level to -4 dBm. Monitor for detections a minimum of 1 hour.

c. Verify that the requirements of 3.7.10.3 are met.

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4.7.10.4 Operator access and bridging (gain), input/output test tone levels.

a. Connect the equipment as shown in figure 32 (See 3.7.4.2, 3.7.5.1, 3.7.5.3 and 3.7.10.4).

b. Set all the AVOW toggle switches on the control panel to bridge and push the AVOW B button.

c. Connect the oscillator and voltmeters in turn to the channel shown in table XXVIII .

d. The oscillator shall be set as follows: Set the oscillator output through 600 ohms to 1 kHz at +2 dBm (equivalent to -4 dBm into the unit) for channel inputs and 1.4 mV rms for handset inputs.

e. At each combination measure and record the output voltages.

f. Verify that the requirements of 3.7.4.2, 3.7.5.1, 3.7.5.3 and 3.7.10.4 are met.

4.7.11 Data orderwire (DOW) performance. (See 3.7.11)

4.7.11.1 Format. (See 3.7.11.1)

a. Connect the equipment as shown in figure 33. Turn the strobe select switch on the BER generator to OFF.

b. Set the control panel and CCA switches as shown in table XXIX.

c. Verify that the status indicator on the control panel is green, and that the combined format is as shown in figure 34 using the Serial Data Analyzer. Verify that the idle character state is all 1's.

d. Verify that the single status block for Channel 1 is as shown in table IV.

e. Substitute in turn Channels 2 through 13.

NOTE: It will be necessary as each channel is selected to set the S4 DIP switch positions 1, 2, 3 and 4 to the corresponding code for selected channel. (See table XXX).

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TABLE XXVIII. AVOW Operator access and bridging.

MEASUREMENT	REQUIREMENT INPUT CH1	REQUIREMENT INPUT CH2	REQUIREMENT INPUT CH 3	REQUIREMENT INPUT CH4	REQUIREMENT INPUT CH5	REQUIREMENT INPUT CH6	REQUIREMENT INPUT OPER.
CH1	-25 dBm max	-5 to -3 dBm	-5 to -3 dBm	-5 to -3dBm	-5 to -3 dBm	-5 to -3 dBm	-5 to -3 dBm
CH2	-5 to -3 dBm	-25 dBm max					
CH3		-5 to -3 dBm	-25 dBm max				
CH4			-5 to -3 dBm	-25 dBm max			
CH5				-5 to -3 dBm	-25 dBm max		
CH6					-5 to -3 dBm	-25 dBm max	
QP	1.26 to 1.78 Volts	1.26 to 1.78 Volts	1.26 to 1.78 Volts	1.26 to 1.78 Volts	1.26 to 1.78 Volts	1.26 to 1.78 Volts	NA

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TABLE XXIX. Switch positions for DOW performance test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	T	T	T	T	T	T

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VERD	VERD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	OFF
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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TABLE XXX. DOW format switch settings.

STATION ID				
S1 SWITCH SETTINGS	FIRST CHARACTER			
	1	2	3	4
A	ON	ON	ON	ON
B	ON	ON	ON	OFF
C	ON	ON	OFF	ON
D	ON	ON	OFF	OFF
E	ON	OFF	ON	ON
F	ON	OFF	ON	OFF
G	ON	OFF	OFF	ON
H	ON	OFF	OFF	OFF
L	OFF	ON	ON	ON
M	OFF	ON	ON	OFF
P	OFF	ON	OFF	ON
R	OFF	ON	OFF	OFF
S	OFF	OFF	ON	ON
T	OFF	OFF	ON	OFF
X	OFF	OFF	OFF	ON
Z	OFF	OFF	OFF	OFF

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TABLE XXX DOW format switch settings. (Continued)

S2 SWITCH SETTINGS	SECOND CHARACTER			
	1	2	3	4
0	ON	ON	ON	ON
1	ON	ON	ON	OFF
2	ON	ON	OFF	ON
3	ON	ON	OFF	OFF
4	ON	OFF	ON	ON
5	ON	OFF	ON	OFF
6	ON	OFF	OFF	ON
7	ON	OFF	OFF	OFF
8	OFF	ON	ON	ON
9	OFF	ON	ON	OFF
S3 SWITCH SETTINGS	**THIRD CHARACTER**			
0	ON	ON	ON	ON
1	ON	ON	ON	OFF
2	ON	ON	OFF	ON
3	ON	ON	OFF	OFF
4	ON	OFF	ON	ON
5	ON	OFF	ON	OFF
6	ON	OFF	OFF	ON
7	ON	OFF	OFF	OFF
8	OFF	ON	ON	ON
9	OFF	ON	ON	OFF

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TABLE XXX. DOW format switch settings. (Continued)

S4 SWITCH SETTINGS	FOURTH CHARACTER			
	1	2	3	4
1	ON	ON	ON	OFF
2	ON	ON	OFF	ON
3	ON	ON	OFF	OFF
4	ON	OFF	ON	ON
5	ON	OFF	ON	OFF
6	ON	OFF	OFF	ON
7	ON	OFF	OFF	OFF
8	OFF	ON	ON	ON
9	OFF	ON	ON	OFF
10	OFF	ON	OFF	ON
11	OFF	ON	OFF	OFF
12	OFF	OFF	ON	ON
13	OFF	OFF	ON	OFF

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f. Verify that activity is present in the channel selected and all other channels are 1.

g. Reset the channel select settings for Channel 1.

h. Verify that channel character 2 is A.

i. Select in turn each of the station ID alpha positions; Set S1 (CCA A15) positions 1, 2, 3 and 4 to the corresponding positions indicated in table XXX. As each position is selected verify that the CH1 character 2 matches the selected number.

j. Select in turn each of the station numerics 0 through 9. Set S2 (TCG CCA) positions 1, 2, 3 and 4 to the corresponding numerics indicated in table XXX. As each numeric is selected, verify that character 3 matches the selected number.

k. Repeat j. for S3 and character 4 .

l. Return the station ID to A00. Apply any three alarms simultaneously by either grounding the appropriate line for contact closure alarms as shown in table I or applying +5 volts to the appropriate line for logic driver alarms as shown in table II.

m. Verify that the two block status message for Channel 1 is in accordance with the requirements specified in table XXXI and that the status error messages selected appear in the appropriate locations.

n. Apply any five alarms simultaneously as described in 1.

o. Verify that the three block status message for Channel 1 is in accordance with the requirements specified in table XXXII and that the status error messages selected appear in the appropriate locations.

p. Apply any seven alarms simultaneously as described in 1.

q. Verify that the four block status message for Channel 1 is in accordance with the requirements specified in table XXXIII and that the status error messages selected appear in the appropriate locations.

r. Apply any nine alarms simultaneously as described in 1.

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TABLE XXXI. Two block status message.

MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL	MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL
1	STX	21	FF
2	A	22	A
3	0	23	0
4	0	24	0
5	E	25	E
6	↑	26	↑
7		27	
8	First *	28	Third *
9	Alarm	29	Alarm
10	Word	30	Word
11		31	
12		32	
13		33	↑
14	Second *	34	↑
15	Alarm	35	↑
16	Word	36	↑
17		37	↑
18		38	↑
19	↑	39	EXT
20	ETB	40	ETB

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TABLE XXXII. Three block status message.

MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL	MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL
1	STX	41	FF
2	A	42	A
3	0	43	0
4	0	44	0
5	E	45	E
6	↑	46	↑
7		47	
8	First *	48	Fifth*
9	Alarm	49	Alarm
10	Word	50	Word
11		51	
12		52	
13		53	↑
14	Second *	54	↑
15	Alarm	55	↑
16	Word	56	↑
17		57	↑
18		58	↑
19	↑	59	ETX
20	ETB	60	ETB
21	FF		
22	A		
23	0		
24	0		
25	E		
26	↑		
27			
28	Third *		
29	Alarm		
30	Word		
31			
32			
33			
34	Fourth*		
35	Alarm		
36	Word		
37			
38			
39	↑		
40	ETB		

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TABLE XXXIII. Four block status message.

MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL	MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL
1	STX	41	FF
2	A	42	A
3	0	43	0
4	0	44	0
5	E	45	E
6	↑	46	↑
7		47	
8	First *	48	Fifth *
9	Alarm	49	Alarm
10	Word	50	Word
11		51	
12		52	
13		53	
14	Second *	54	Sixth *
15	Alarm	55	Alarm
16	Word	56	Word
17		57	
18		58	
19	↑	59	↑
20	ETB	60	ETB
21	FF	61	FF
22	A	62	A
23	0	63	0
24	0	64	0
25	E	65	E
26	↑	66	↑
27		67	
28	Third *	68	Seventh *
29	Alarm	69	Alarm
30	Word	70	Word
31		71	
32		72	
33		73	↑
34	Fourth *	74	↑
35	Alarm	75	↑
36	Word	76	↑
37		77	↑
38		78	↑
39	↑	79	EXT
40	ETB	80	ETB

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s. Verify that the single block status message for channel 1 is in accordance with the requirements specified in table XXXIV A.

t. Set the BER Code Switch to 0 and the Strobe Select Switch to BER Strobe 1 as shown in table XXXIV B.

u. Verify that the single block status message for channel 1 is of the form shown in table XXXIV B with m=1 and n=0.

v. Set the BER Code Switch to each position as shown in table XXXIV B and verify that the n position in the status message changes correspondingly.

w. Set the Strobe Select Switch to BER Strobe 2 and verify that the m position in the status message is a 2.

x. Repeat w. with the Strobe Select Switch set to BER Strobe 3.

y. Remove all alarms, verify that the status indicator is green.

4.7.11.2. Telemetry decombiner frame acquisition and maintenance. (See 3.7.11.2)

a. Connect the equipment as shown in figure 35.

b. Set the UUT and control panel switches as shown in table XXXV.

c. Connect the stop channel of the interval counter to I/O pin 89 on the TCG CCA (OOF signal) (rising edge).

d. Connect the start channel of the interval counter to trigger upon application of data (rising edge).

e. Set the BER on the error inserter to 10^{-3}, measure and record the lock in time on the counter 20 times.

f. Connect the stop channel of the counter to I/O pin 89 on the TCG CCA.

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TABLE XXXIV. Single block status message.

A.

MEASUREMENT CH 1 CHARACTER	REQUIREMENT SYMBOL
1	STX
2	A
3	0
4	0
5	E
6	
7	M
8	A
9	J
10	0
11	R
12	
13	M
14	A
15	J
16	0
17	R
18	
19	ETX
20	ETB

B.

STROBE SELECT SWITCH	BER CODE SWITCH					STATUS MESSAGE
m	n	8	4	2	1	Rmn
1	0					R10
1	1				x	R11
1	2			x		R12
1	3			x	x	R13
1	4		x			R14
1	5		x		x	R15
1	6		x	x		R16
1	7		x	x	x	R17
1	8	x				R18
1	9	x			x	R19

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TABLE XXXV. Switch position for frame acquisition and maintenance test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	T	T	T	T	T	T

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	OFF
TMG	GROUP 1	14
SCE	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	ON
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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g. Set the start channel to trigger upon cessation of the data (falling edge).

h. Measure and record the drop-out time on the counter 20 times.

i. Verify that the requirements of 3.7.11.2 are met.

4.7.11.2.1 Frame maintenance. (See 3.7.11.2)

a. Connect the equipment as shown in figure 35. The UUT and switch positions are shown in table XXXV.

b. Set the BER to 10^{-3} and gate the data on.

c. Connect the oscilloscope to LED OOF, monitor the LED for 24 hours. There shall be no more than 1 indication during the 24 hour period.

4.7.11.3 Resync command pulse rate. (See 3.7.11.3)

a. The equipment test setup is the same as in 4.7.11.2. and connect the oscilloscope to the resync command signal.

b. Gate the DATA OFF; using the oscilloscope, measure the time between resync commands. Verify that the time requirements of 3.7.11.3 are met.

4.7.11.4 DOW BCI. (See 3.7.11.4)

a. Connect the equipment as shown in figure 36 such that timing source #1 and #2 are operating with a frequency offset determined by the frequency synthesizer.

b. Set the UUT and control panel switches as shown in table XXIX.

c. With the frequency set to 2000.2 Hz and 1999.8 Hz, measure and record the time between reset indications (this is 100 times the specified offset). Locate the 16 bit word for FIFO Reset on the Logic Analyzer. Then measure the time between consecutive FIFO reset indications.

NOTE: FIFO RESET = 01A1

PROGRAM ADDRESS

0000/0001/1010/0001

d. Verify that the FIFO resets no more often than 1 time in 9 seconds.

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4.7.12 Power variation test. (See 3.7.12)

a. Connect the equipment as shown in figure 37.

b. Set the UUT and control panel switches as shown in table XXXVI. Adjust the input power to 115 Vac at 60 Hz.

c. With the oscillator connected to AVOW CH1 IN, adjust the oscillator output to -4 dBm at 1600 Hz. Verify that the AVOW CH1 pushbutton flashes and there are no alarms or BITE indications.

NOTE: Alarms may come on during remainder of test.

d. Set the oscillator frequency to 1000 Hz, and push the AVOW CH1 pushbutton twice. Verify that the AVOW CH1 pushbutton light goes out.

e. With the oscillator connected to AVOW CH1, measure the AVOW 2 through 6 output level.

f. Connect the frequency counter to AVOW CH1 OUT, push CH1 button and push SELF RING button, measure the output frequency.

g. Push the DVOW CH3 pushbutton, set the ring code to position 1 and push the RING button. Verify that DVOW Channel 3 button flashes.

h. Remove the Channel 3 data and timing inputs and reset DVOW CH3 pushbutton.

i. Measure the DVOW Error Rate at Channel 2 over a 30-second period.

j. Reverse the connections for Channel 1 and Channel 2. Measure the DVOW Error Rate at Channel 1 over a 30 second period.

k. Repeat i. and j. for Channels 3 and 4 and Channels 5 and 6.

l. Open the connection between TEST DATA IN and TEST DATA OUT (S1). Verify that the ERROR LED is lit. Close (S1) and verify that the ERROR LED goes out.

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TABLE XXXVI. Switch positions for power variation test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	B	B	B	B	B	B

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION								X

ANALOG	1	2	3	4	5	6
BRIDGE	X	X	X	X	X	X
TERMINATE						

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 kb/s
	PRG	ON
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	ON
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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m. Open the connection between RXDPT and TXDPT (S2). Verify that the LOI is illuminated. Close (S2) and verify that the LOI and OOF are off.

n. Measure the input power (line voltage, frequency and current).

o. Repeat b. through n. for each of the combinations of voltage and frequency:

60 Hz	47.5 Hz	420 Hz	47.5 Hz	420 Hz
115Vac	103.5Vac	103.5Vac	126.5Vac	126.5Vac

p. Apply 156 Vac at 60 Hz for 6 seconds maximum. Verify that there is no power supply out of tolerance indication and that the pilot light remains ON.

q. Apply 115 Vac at 60 Hz and verify that there is no power supply out of tolerance indication and that the pilot lamp is illuminated.

4.7.13 BITE and alarms. (See 3.7.13)

a. Connect the equipment as shown in figure 38.

b. Set the UUT switches and Control Panel switches as shown in table XXXVII.

c. Open S1, verify that the ERROR LED on the BITE CCA is OFF.

d. Turn ON the PRG switch on the BITE 3 CCA.

e. Verify that the ERROR LED goes OFF when S1 is closed and ON when S1 is open.

f. Repeat d and e with BIT RATE SELECT switch on the BITE 3 CCA set to 2.

g. With S1 Closed, S2 set to TXDPT, and S3 Closed, verify the following requirements.

MEASUREMENT	REQUIREMENT
No Fault Operation	All LEDs are off Summary status is GREEN Summary status RED and AMBER are off

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TABLE XXXVII. Switch position for BITE and alarms test.

CONTROL PANEL SWITCH

RING CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
POSITION	X															

DIGITAL	1	2	3	4	5	6
POSITION	B	B	B	B	B	B

TIMING SELECT	G	1	2	3	4	5	6	M
POSITION	X							

ANALOG	1	2	3	4	5	6
BRIDGE						
TERMINATE	X	X	X	X	X	X

CCA SWITCH

CCA	SWITCH	POSITION
BITE 3	RATE SELECT	16 Kb/s
	PRG	OFF
TMG SCE	GROUP 1	14
	GROUP 2	14
VBRD	VBRD RING CODE	1
AMPLA	MOW	NORM

CCA	SWITCH	POSITION
TCG	TLMY DECOMB 1	ON
	2	OFF
	3	OFF
	STA ID ALPHA	A
	STA ID NUMERIC(1)	0
	STD ID NUMERIC(2)	0
	CHANNEL SELECT	1
	BER THRESHOLD	8

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h. Set S2 to the DOW position. Verify the following requirements. Then set S2 back to TXDPT.

MEASUREMENT	REQUIREMENT
Summary Alarm and Out-of-Frame	Summary FAULT on OOF LED on TCG is on Status indicator remains GREEN

i. Verify that the Timing Input is set to G, set S2 to TXDPT and Open S3. Verify the following requirements. Close S3.

MEASUREMENT	REQUIREMENT
LOI	Summary FAULT is on Summary status GREEN LOI is on

j. Press and hold the Alarm Test button on the power supply. Verify that the following requirements are met.

MEASUREMENT	REQUIREMENT
Alarm Test	Summary FAULT is on Status lamps; RED, GREEN and Amber are on Audible alarm sounds LOI is ON, POT is on

4.7.14 Operational test. (See 3.7.14)

a. Connect the equipment as shown in figure 39.

b. Set the UUT CCA switches and control panel switches as shown in table XXXVII, except that the timing input switch shall be set to master.

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4.7.14.1 Alarm monitor check. (See 3.7.14.1) Proper operation is monitored by verification that no BITE or fault lamps are illuminated. The GREEN status light shall be on.

a. Observe that there are no alarms or LED fault indications.

b. Open S2 (TXDPT/RXDPT).

c. Verify that Summary Fault and LOI lights go on, when S2 is open and, Summary Fault and LOI lights are extinguished when S2 is closed.

d. Close S3. Verify that the RED Status Light comes on and the GREEN light goes off. Then open S3.

e. Close S4. Verify that the RED Status Light comes on and the GREEN light goes off. Then open S4.

f. Verify that the following requirements are met.

STATUS	REQUIREMENT
No Fault Operation	All LEDs are off Summary Status is GREEN Summary Status RED and AMBER are off
S2 Open	Summary Fault LOI are on
S3 Closed	RED Status Light Comes on GREEN Light Goes off
S4 Closed	RED Status Light Comes on GREEN Light Goes off

4.7.14.2 DVOW Error rate test. (See 3.7.14.2)

a. Connect the DVOW/DOW Simulator data output to DVOW Channel 1 input and the DVOW/DOW Simulator input to DVOW Channel 6 output.

b. Monitor the DVOW Error Detector counter for a period of 30 seconds. Verify that there are no errors generated.

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c. Connect the DVOW/DOW Simulator data output consecutively to DVOW Channels 2 through 5. Measure the error rate as in b. at each chananel. (Note move 16 kHz CLK2 out to each channel being checked.

d. Connect the DVOW/DOW Simulator input to DVOW Channel 1 output and Simulator output to Channel 6 input. Measure the error rate as in b.

e. Connect the DVOW/DOW Simulator Data output to DVOW Channel 1 input and the DVOW/DOW Simulator input to Channel 1 output.

f. Push the Channel 1 DVOW button, close the PTT switch, and measure the Error Rate as in b.

4.7.14.3 DVOW Ring Test. (See 3.7.14.3)

a. Connect the output of DVOW Channel 2 to the input of DVOW Channel 1.

b. Push the DVOW Channel 2 pushbutton.

c. Push the RING button on the control panel. Verify that the DVOW Channel 1 light flashes.

d. Push the DVOW Channel 1 button. Verify that the DVOW Channel 1 light is on stedily.

e. Push the DVOW Channel 1 button. Verify that the DVOW Channel 1 light goes out.

4.7.14.4 AVOW Level (Gain) (See 3.7.14.4)

a. Apply oscillator at +2 dBm and 820 Hz through a 600 ohm resistor to the MOW IN at the AVOW connector.

b. Measure the output into 600 ohms of the MOW output at the AVOW connector.

c. Verify that the MOW output requirement of 3.7.14.4 is met.

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4.7.14.5 AVOW self-ring test. (See 3.7.14.5)

a. Loop Channel 1 Output to Channel 1 Input.

b. Press the AVOW Channel 1 pushbutton.

c. Push and hold the SELF RING switch on the SIG A CCA until the Channel 1 light flashes and the buzzer sounds. Push the CH 1 button and verify that the light goes off and the buzzer no longer sounds.

4.7.14.6 Timing source outputs. (See 3.7.14.6)

a. Measure the frequency of each of the following signals;

32 kHz CLK #1, #2 and #3
Group 1 @ 4608 kHz
Group 2 @ 4608 kHz

b. Verify that the measured frequencies are within 10 ppm of the nominal.

4.8 Service conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2 Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +63°C (+145°F). Maintain internal chamber temperature for 6 hours at +63°C (+145°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +63°C (+145°F) within a time period of one hour. Repeat the above steps two additional times for a total of three 12-hours cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +63°C (+145°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

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4.8.1.2 Low temperature. (See 3.8.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C(-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation. (Para. II- 3.2 of Method 502.2). The chamber temperature in step 1 of Procedure II shall be adjusted to -31.7°C(-25°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III.

4.8.3 Low pressure (altitude). (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (20.6 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.4) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. In step 2 and step 5, the air velocity shall be 200 ±100 feet per minute. In step 5, the air temperature shall be adjusted to +63°(+145°F). The equipment shall be reoriented periodically to expose all its sides to the dust stream. The equipment shall be operated before step 1, during step 8 and during step 11.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C(+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall be no clogging or binding of moving mechanical parts.

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4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies, with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs, etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Vibration and shock. (See 3.8.7)

4.8.7.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3, Procedure I, Category 8 (Ground Mobile). The vibration level shall be in accordance with figures 514.3-7 through 514.3-9 of MIL-STD-810D. The axes designated on the schedules are the vehicle axes. The orientation of the equipment relative to the vehicle, therefore, must be considered for proper direction of vibration input. Test time shall be 4 hours in each of three mutually perpendicular directions for a total of 12 hours. Prior to and following these tests, the equipment shall be operated in accordance with MIL-STD-810D general test performance guidance paragraph 4.5. The equipment shall not have any degradation in performance and shall not suffer any mechanical damage.

4.8.7.2 Shock.

4.8.7.2.1 Functional. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure I. The shock pulse shall be 20g peak for a nominal duration of 11 msec in accordance with figure 516.3-4.

4.8.7.2.2 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI.

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4.9 Electromagnetic interface tests(EMI). (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government-approved test plan. During susceptibility testing, the integral test facility plug-in subassembly shall be used to monitor operation of the unit.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Reliability testing. (See 3.10)

4.10.1 First article inspection reliability test. First article inspection reliability testing shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subjected to the following conditions:

a. Duty cycle. Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle.

b. Chamber temperature shall 40 $\pm$5°C (95°F to 113°F).

c. Vibration shall be 2.2g $\pm$10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliability test method. The detailed test procedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor-prepared, government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.14.

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4.10.6 Failure reporting and analysis. A failure reporting and anlaysis system shall be implemented per MIL-STD-781C. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such demensions affect insertion of items.

c. Location of switches, sockets, and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings.

4.13 Visual and mechanical inspection. (See 3.14 and 3.17) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.17. An inspection shall be performed to verify compliance with those portions of 3.14 (Systems safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System Safety Engineering.)

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4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Material inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table XXXVIII, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and AQL of 4.0% defective.

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TABLE XXXVIII Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Visual
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect or illegible	Visual

4.15 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item and circuit card assembly.

4.16 Nuclear tests. (See 3.15) Tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

4.17 TEMPEST tests. (See 3.16) TEMPEST tests shall be performed on the equipment in accordance with CR-CX-0178-001.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, tables I through VII, coded as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED

5.1.2 Level B. 2M-1-00-00-NR-X-ED

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The equipment is intended for use as an orderwire termination for JTC^{3}A equipments. It is capable of handling six digital voice orderwires, six analog voice orderwires and three data orderwires.

6.1.1 Functional description. The C-10706/TRC is a shelter-mounted unit which provides an equipment assemblage operator access to, supervision of, and bridging facilities for analog and digital voice orderwires from other assemblages via radio or cable. The C-10716 also monitors status signals of equipment in an assemblage, formats this status information and forwards the information to the Communications Nodal Control Element (CNCE) via data orderwire.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment there to.

b. Level A or level B preservation and packing. (See section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article testing units required.

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e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirements.

i. MIL-Q-9858.

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipment. (See 4.5.3.3) Initial shipments of end items will be contingent upon successful completion of all first article tests.

6.5 Location of operational inspection. (See 4.7.14) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system quality history and the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in 4.7.

6.9 Changes from previous issue. Asterisks (*) (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject terms (key-words) list.

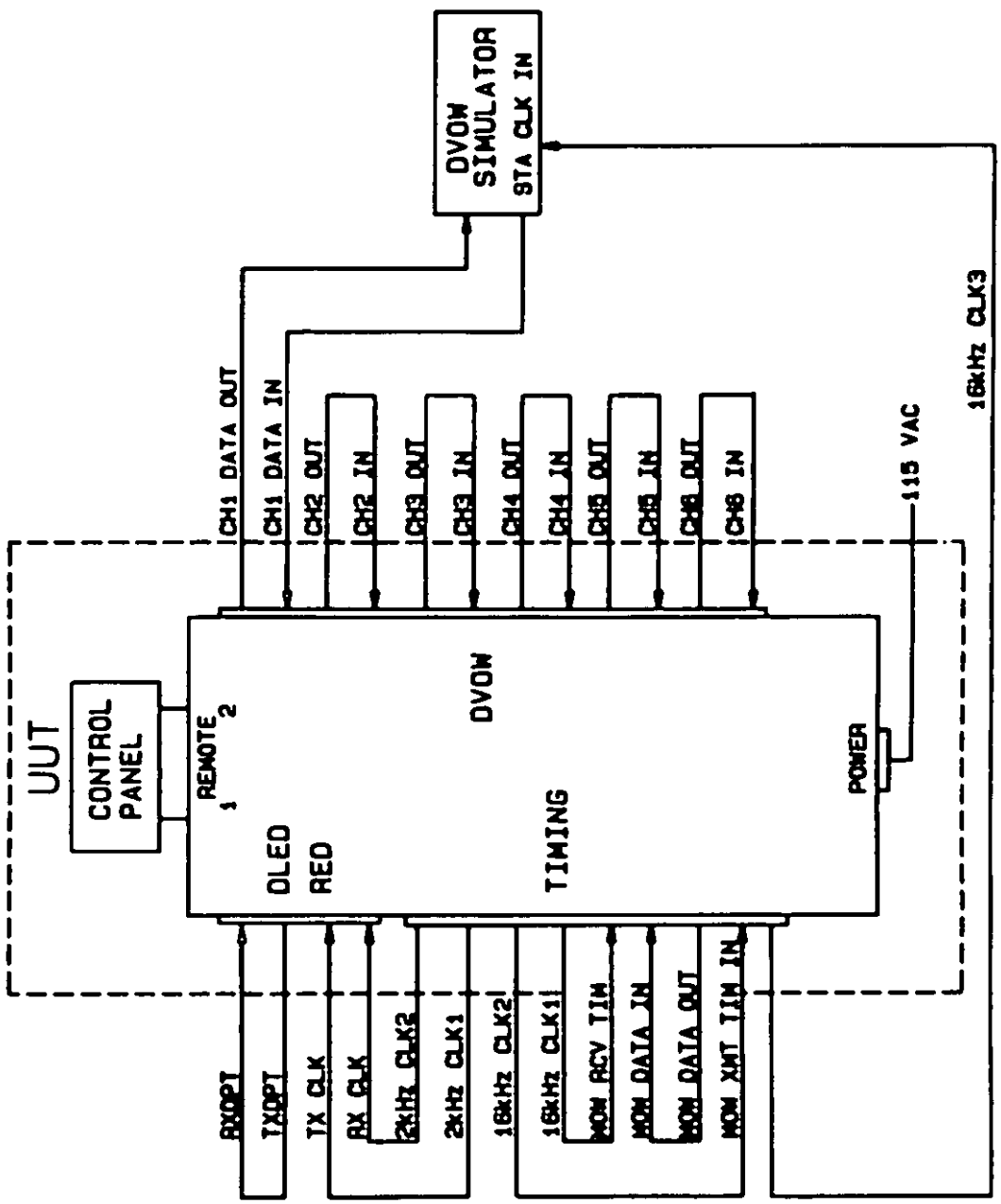
Alarm Interface
Analog Voice Orderwire
AVOW
Balanced Equipment-side Signal Control
Control, Orderwire
Control, Orderwire, C-10716/TRC
Data Orderwire
Digital Group Multiplex
Digital Voice Orderwire
DLED
DOW
Format
OCU-I
Orderwire Control Unit (OCU)
Telemetry
Timing Source
Unbalanced VINSON Interface

CUSTODIAN:

ARMY-CR

PREPARING ACTIVITY
ARMY-CR
Project Nr. 5820-A835

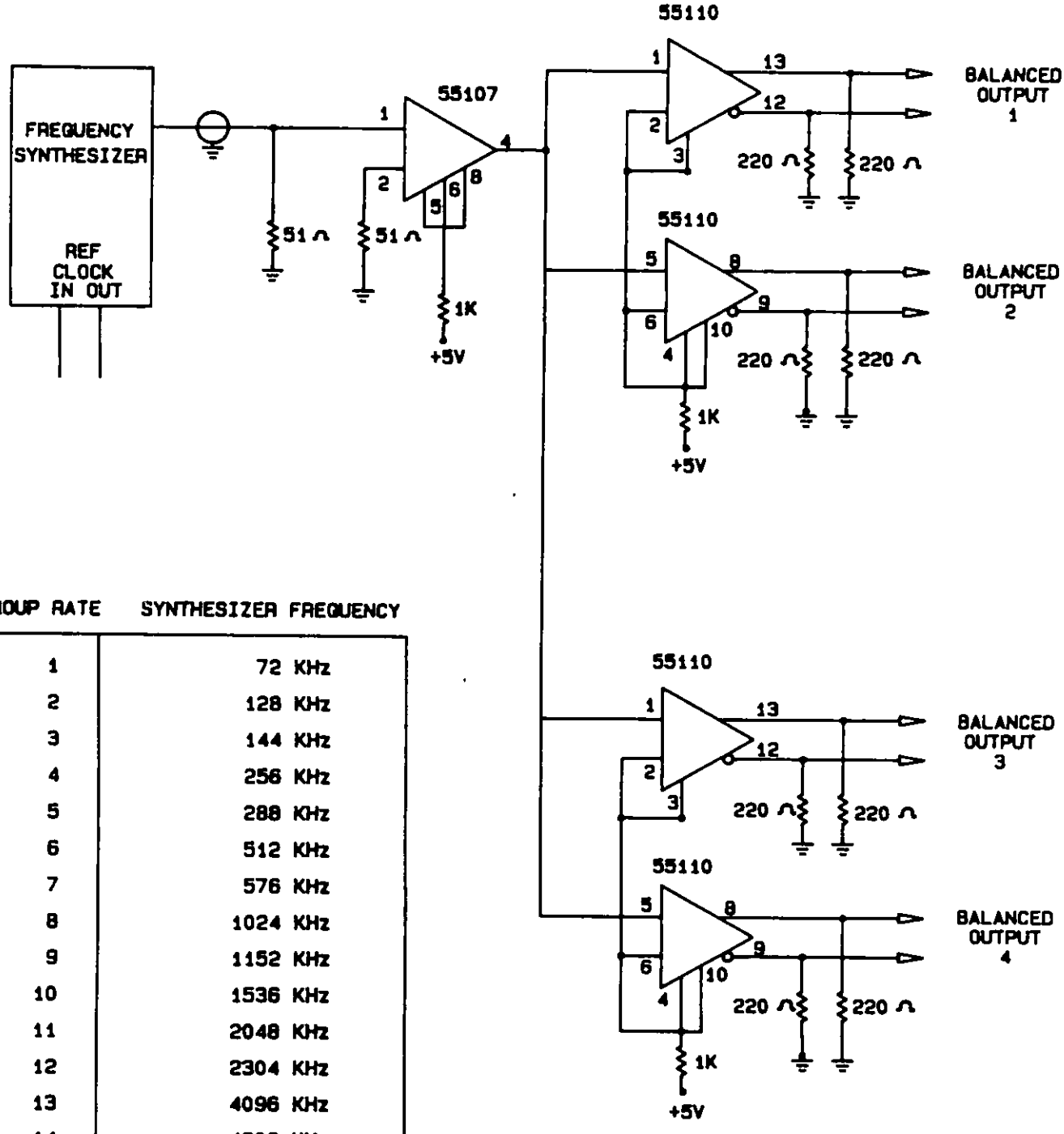
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Set the UUT for MASTER MODE, TD2 & 3 OFF, BRIDGE MATRIX 6, 1, 2, 3, 4 & 5

FIGURE 1.. Burn-in.

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GROUP RATE	SYNTHESIZER FREQUENCY
1	72 KHz
2	128 KHz
3	144 KHz
4	256 KHz
5	288 KHz
6	512 KHz
7	576 KHz
8	1024 KHz
9	1152 KHz
10	1536 KHz
11	2048 KHz
12	2304 KHz
13	4096 KHz
14	4608 KHz
15	4915.2 KHz

FIGURE 2. Alternate timing source.

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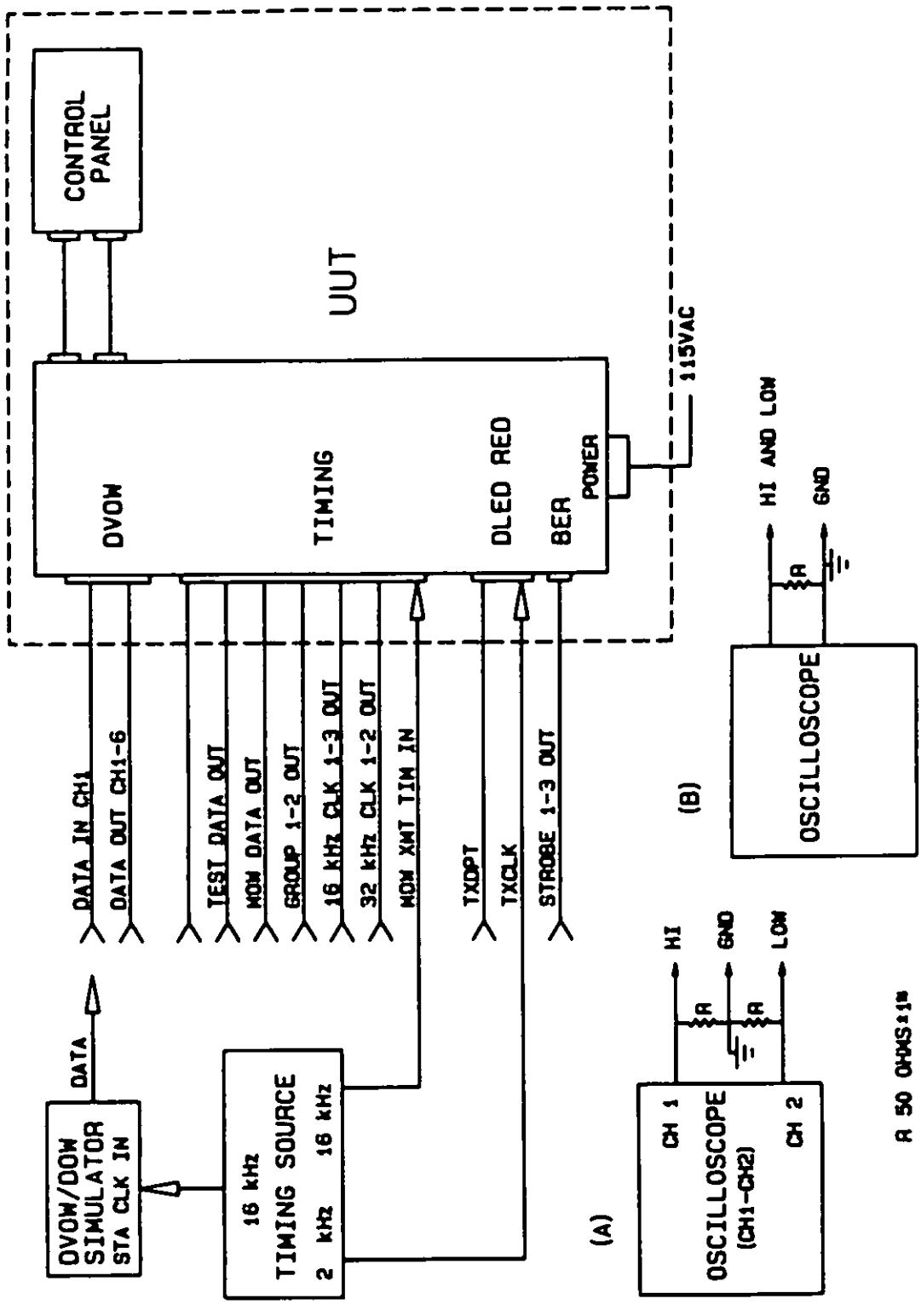


FIGURE 3. Line-to-line voltage levels.

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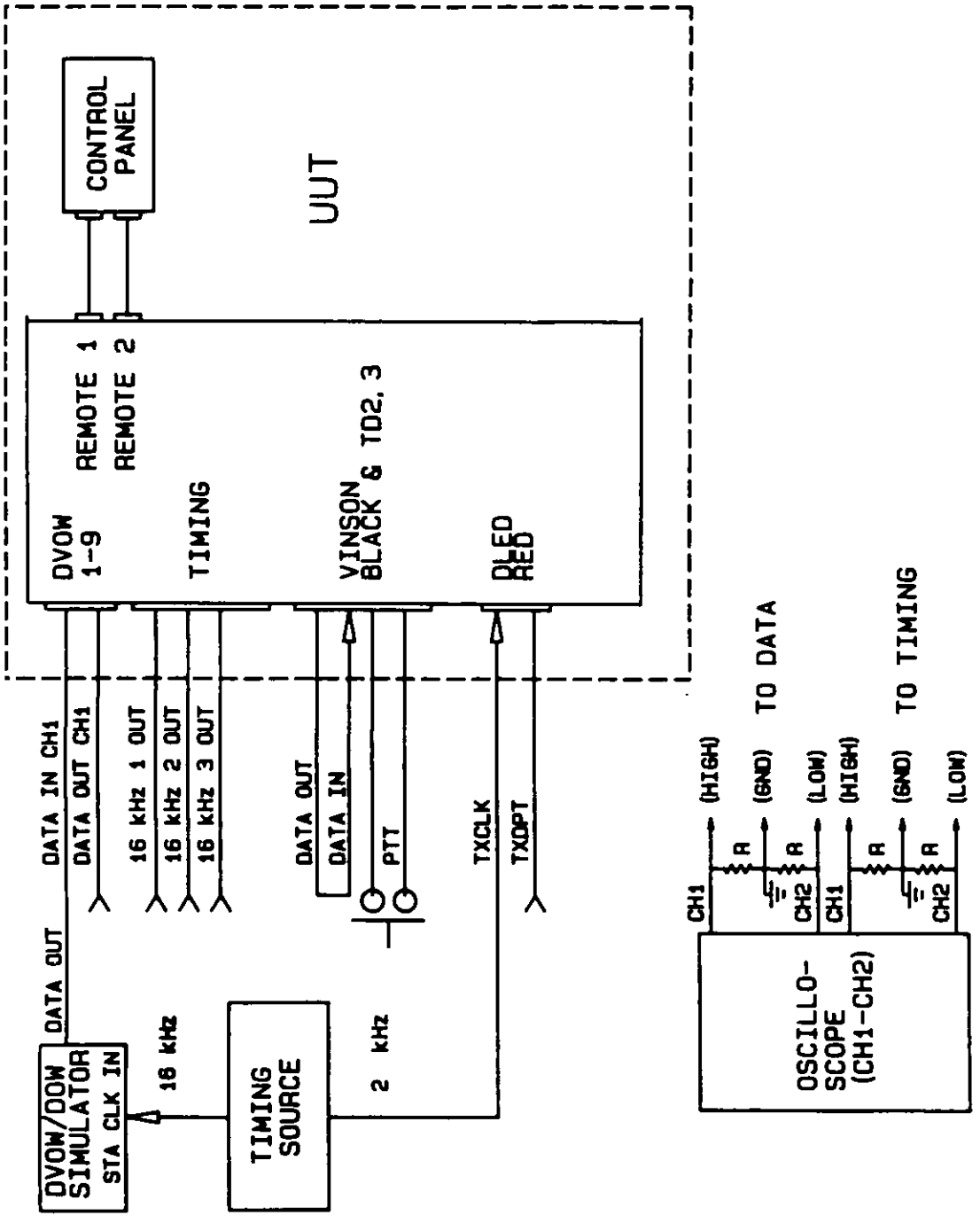


FIGURE 4. Clock data phasing.

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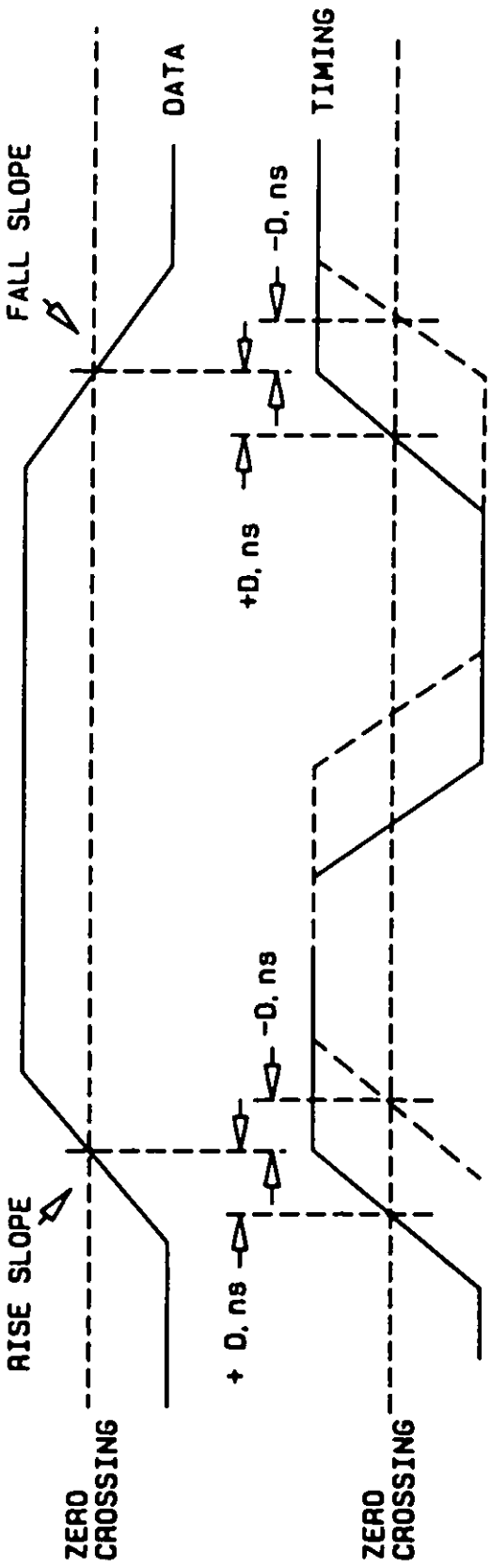
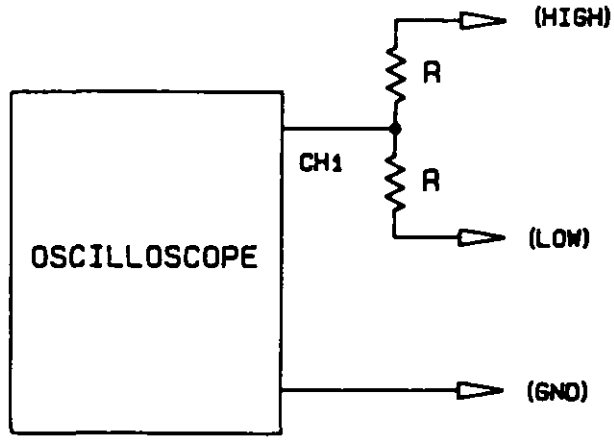


FIGURE 5. Delay measurement (clock/data phasing).

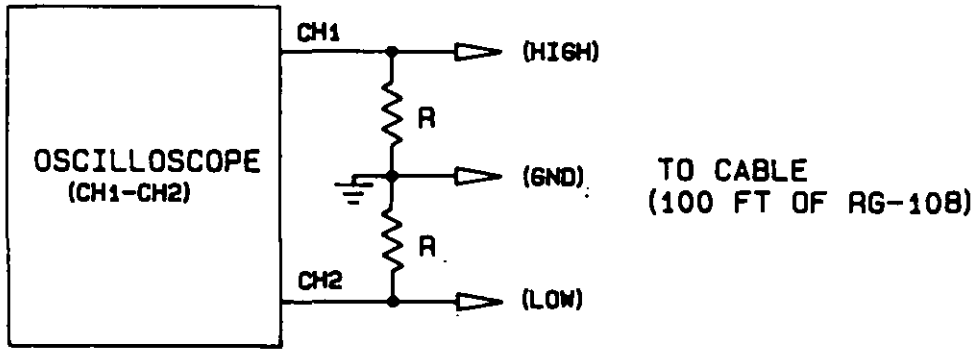
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(A) OFFSET



R 50 OHMS ± 1%

(B) SLEW RATE



R 39 OHMS ± 1%

FIGURE 6. Oscilloscope connections.

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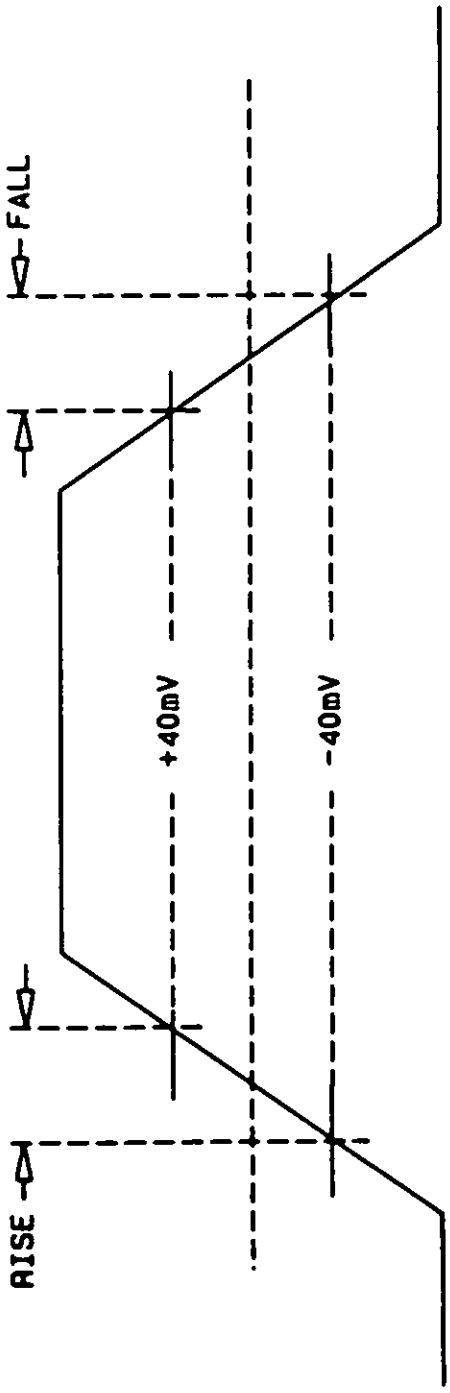


FIGURE 7. Rise and fall slopes (slew rate).

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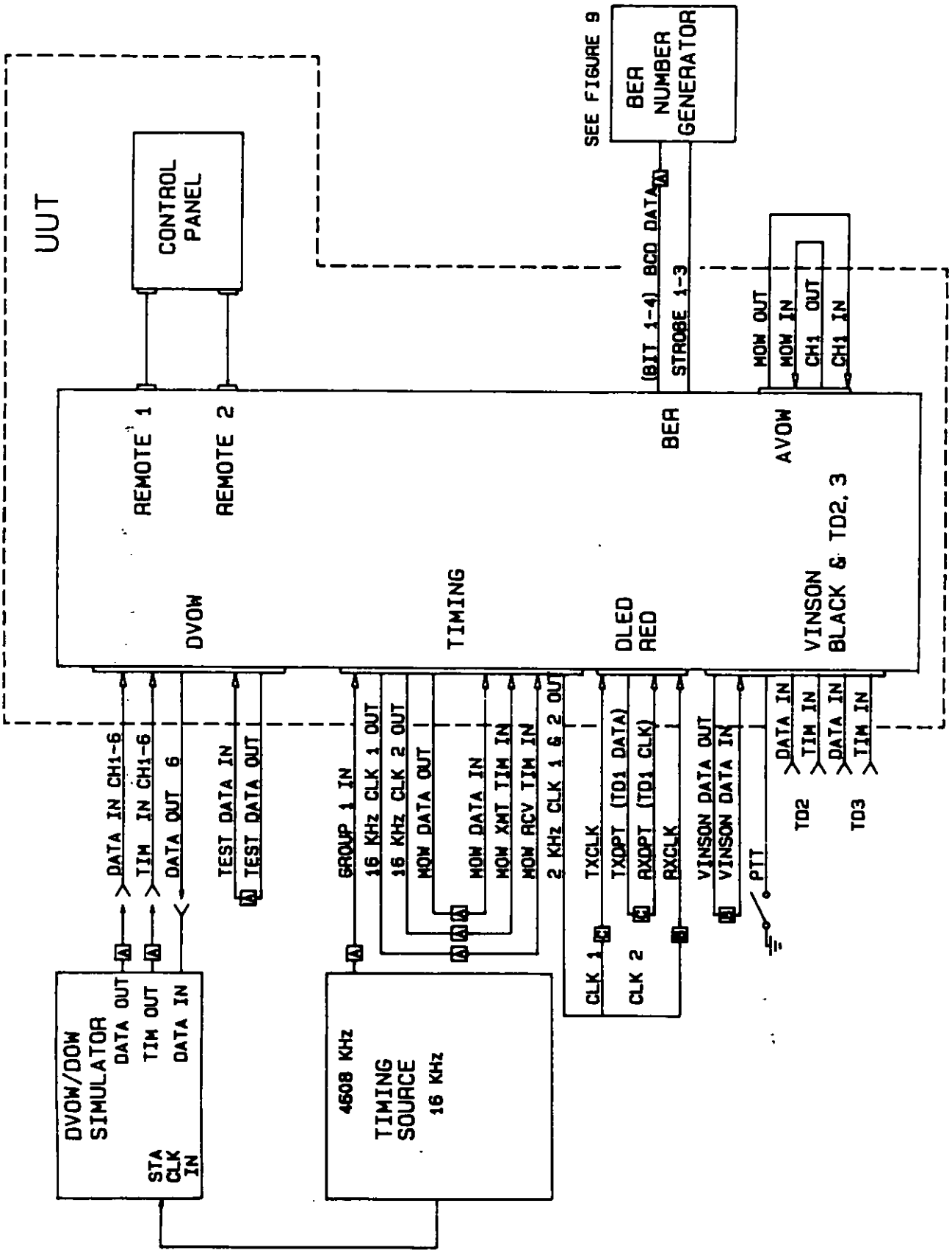


FIGURE 8. Receiver sensitivity.

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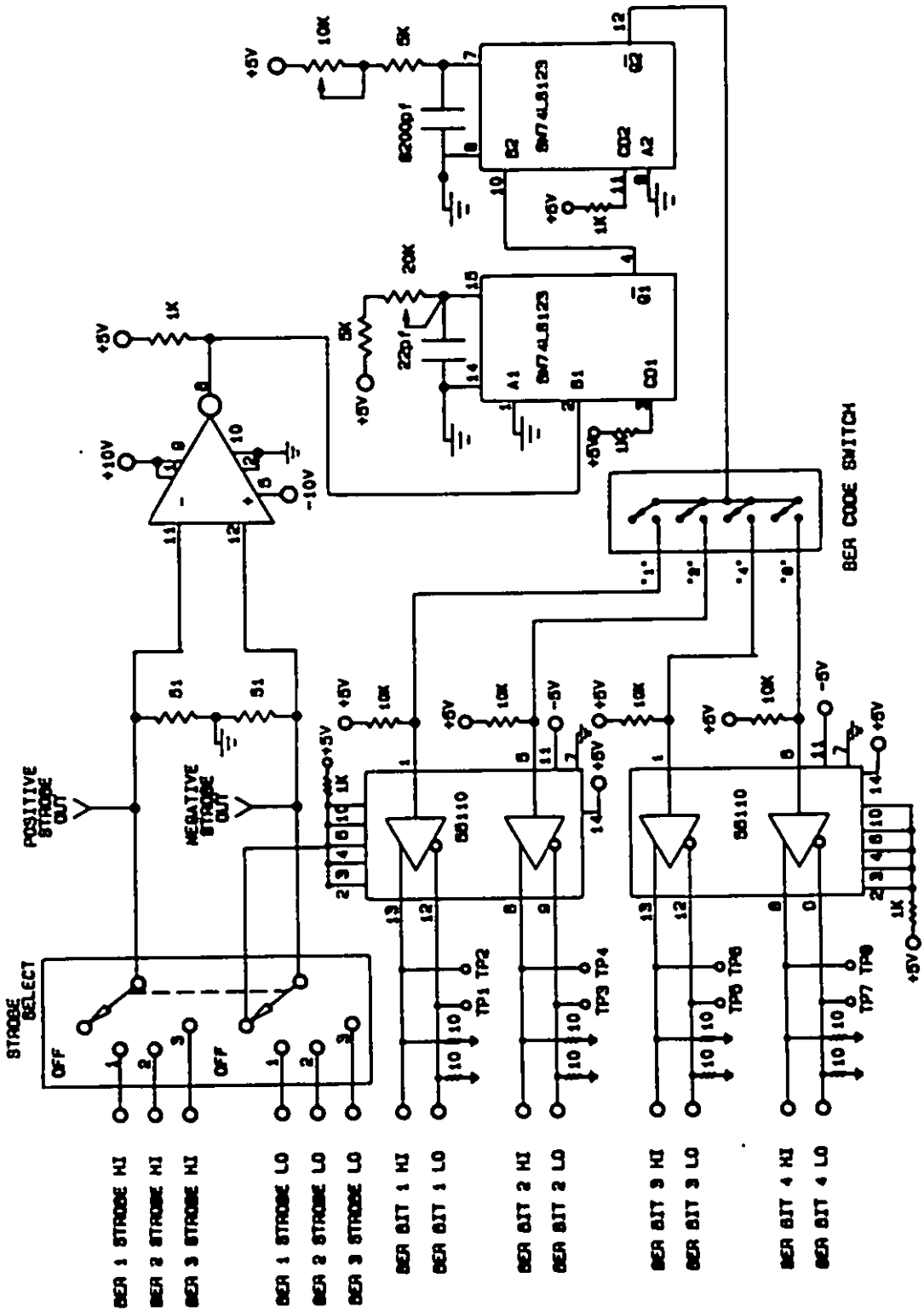
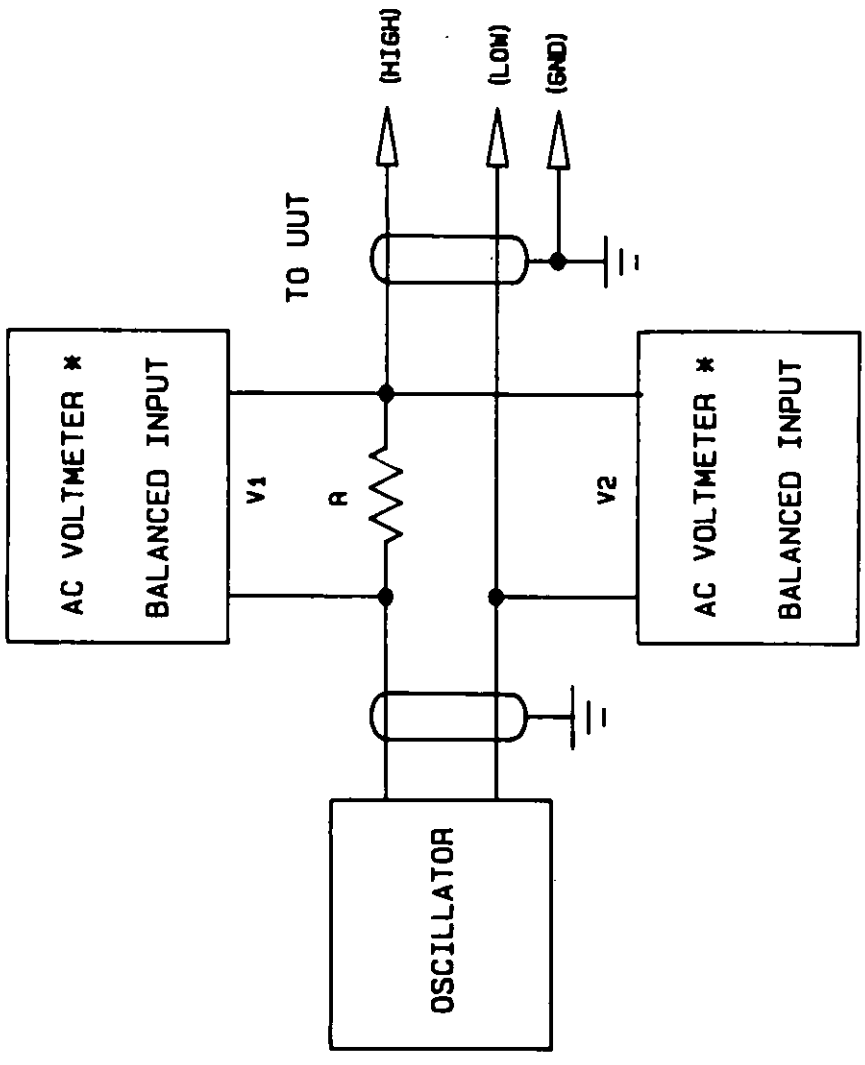


FIGURE 9. BER number generator.

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R 100 OHMS ±1%, 6000 OHMS ±1% OR 2000 OHMS ±1% (SEE TEXT).
* Use battery powered voltmeters.

FIGURE 10. Input load impedance.

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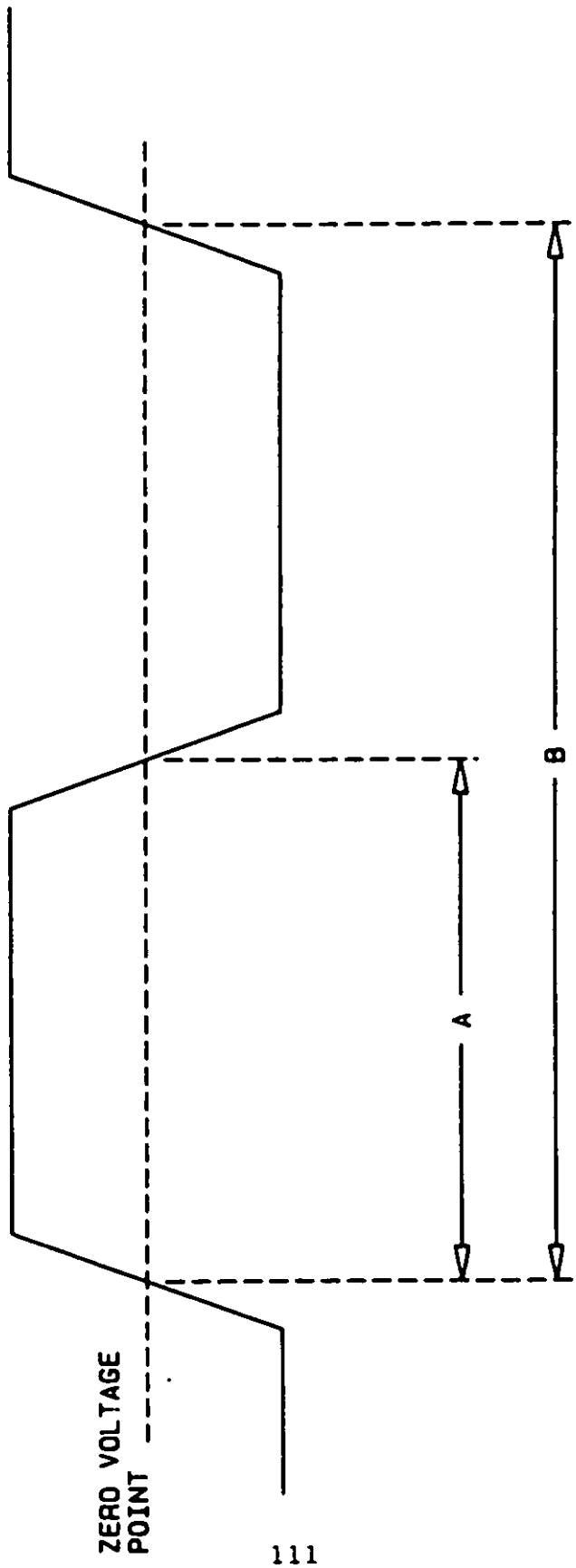


FIGURE 11. Clock asymmetry.

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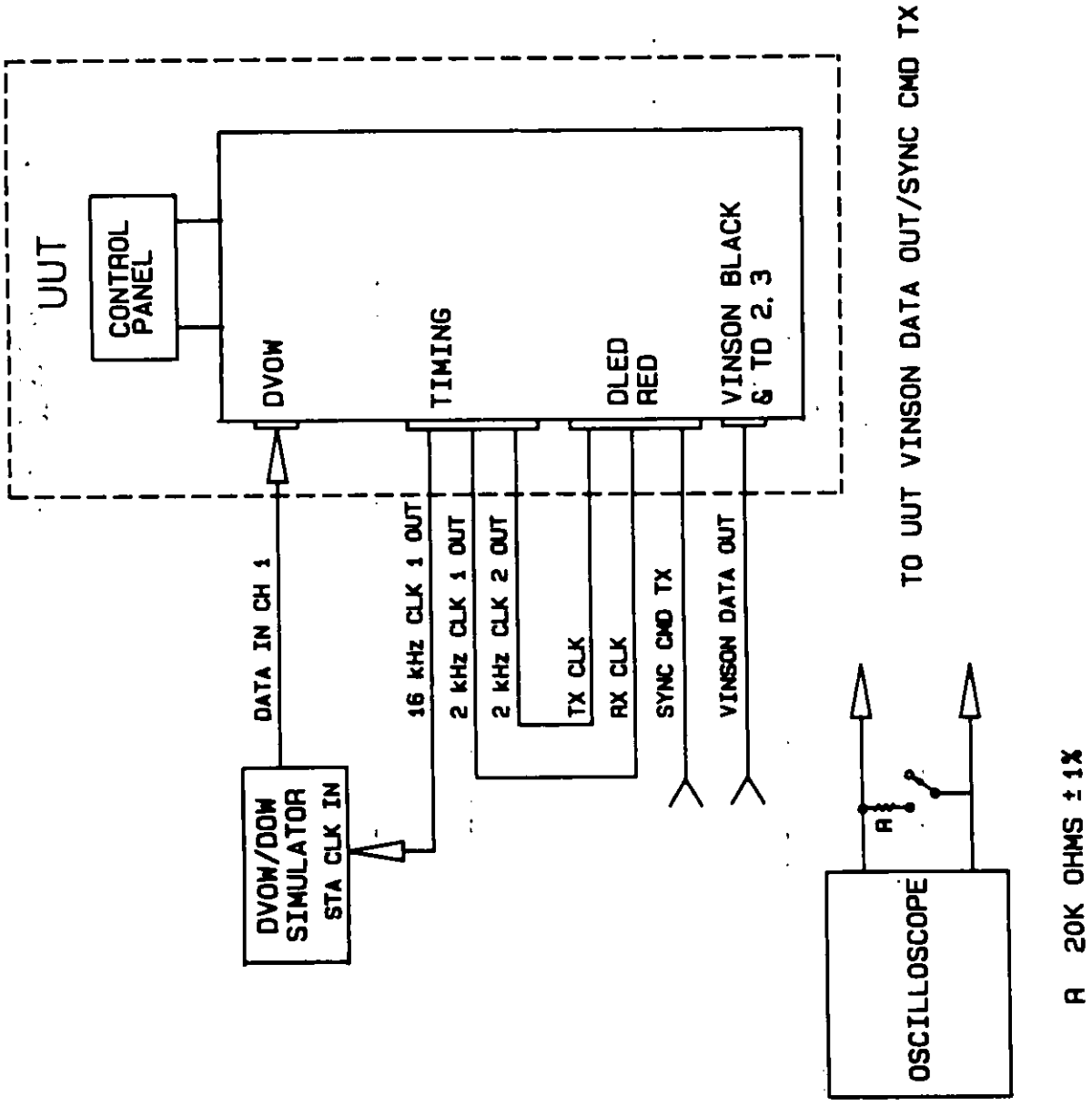


FIGURE 12. Driver output voltage.

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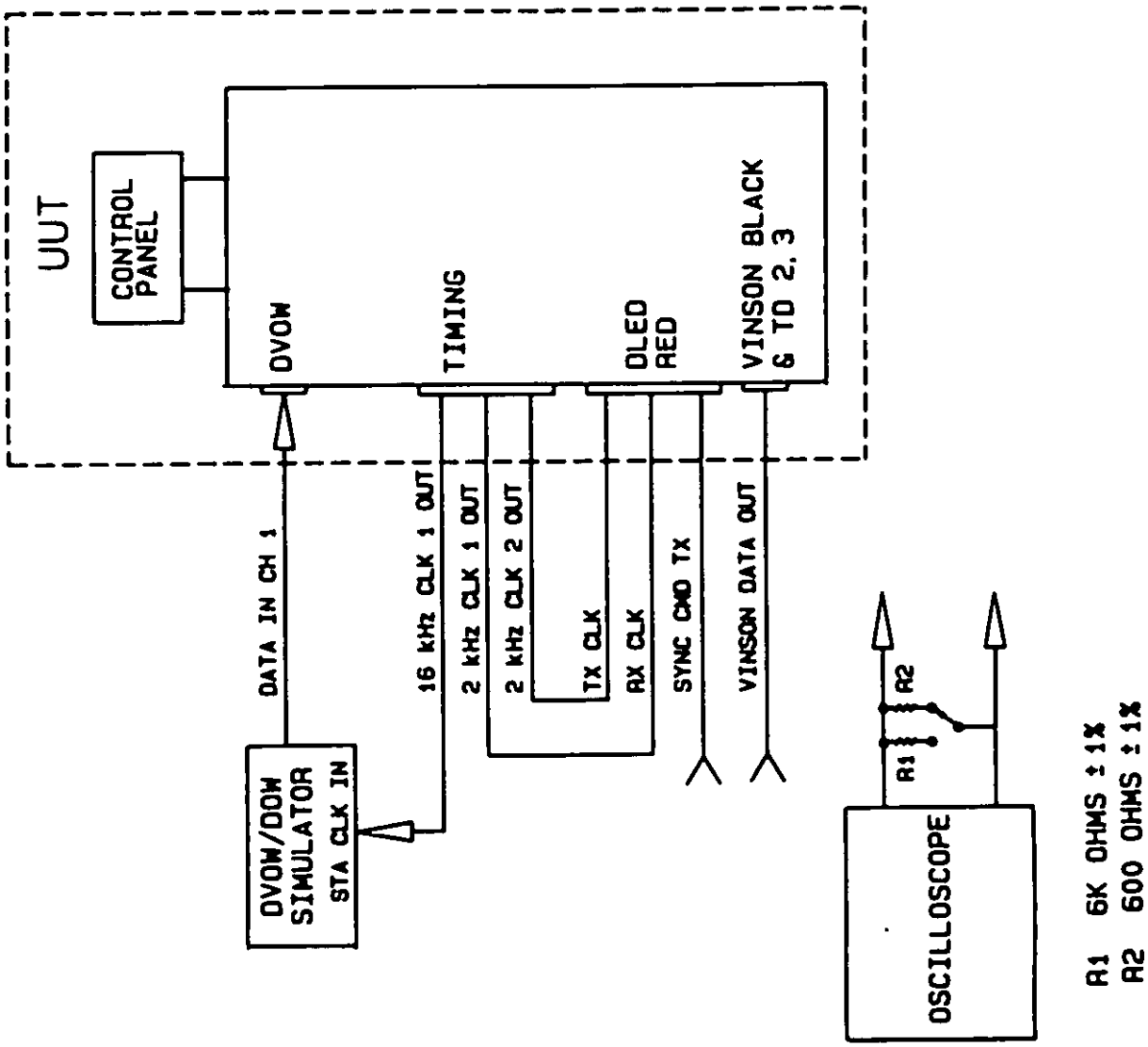


FIGURE 13. Transmitter output impedance.

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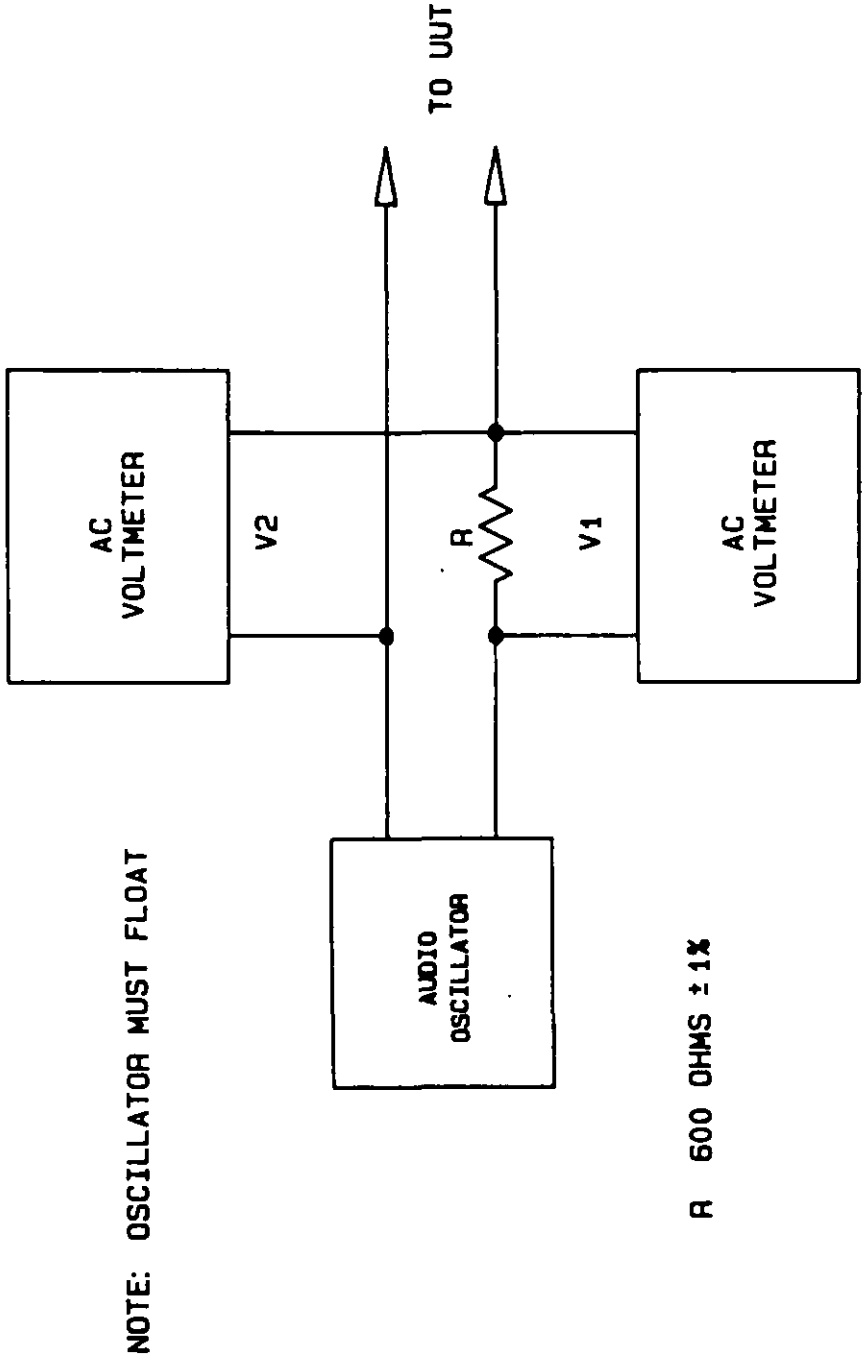


FIGURE 14. AVOM input/output impedance.

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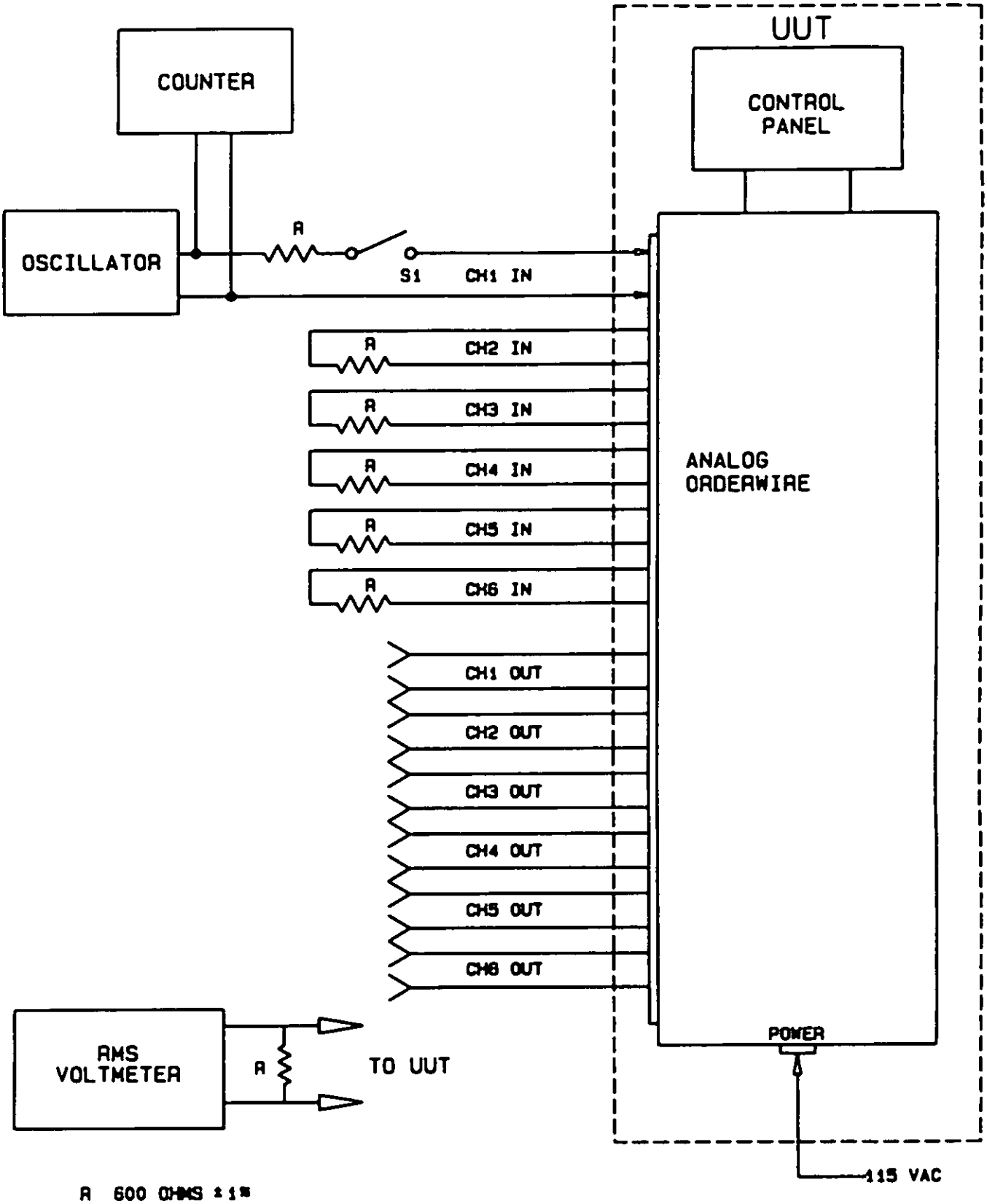
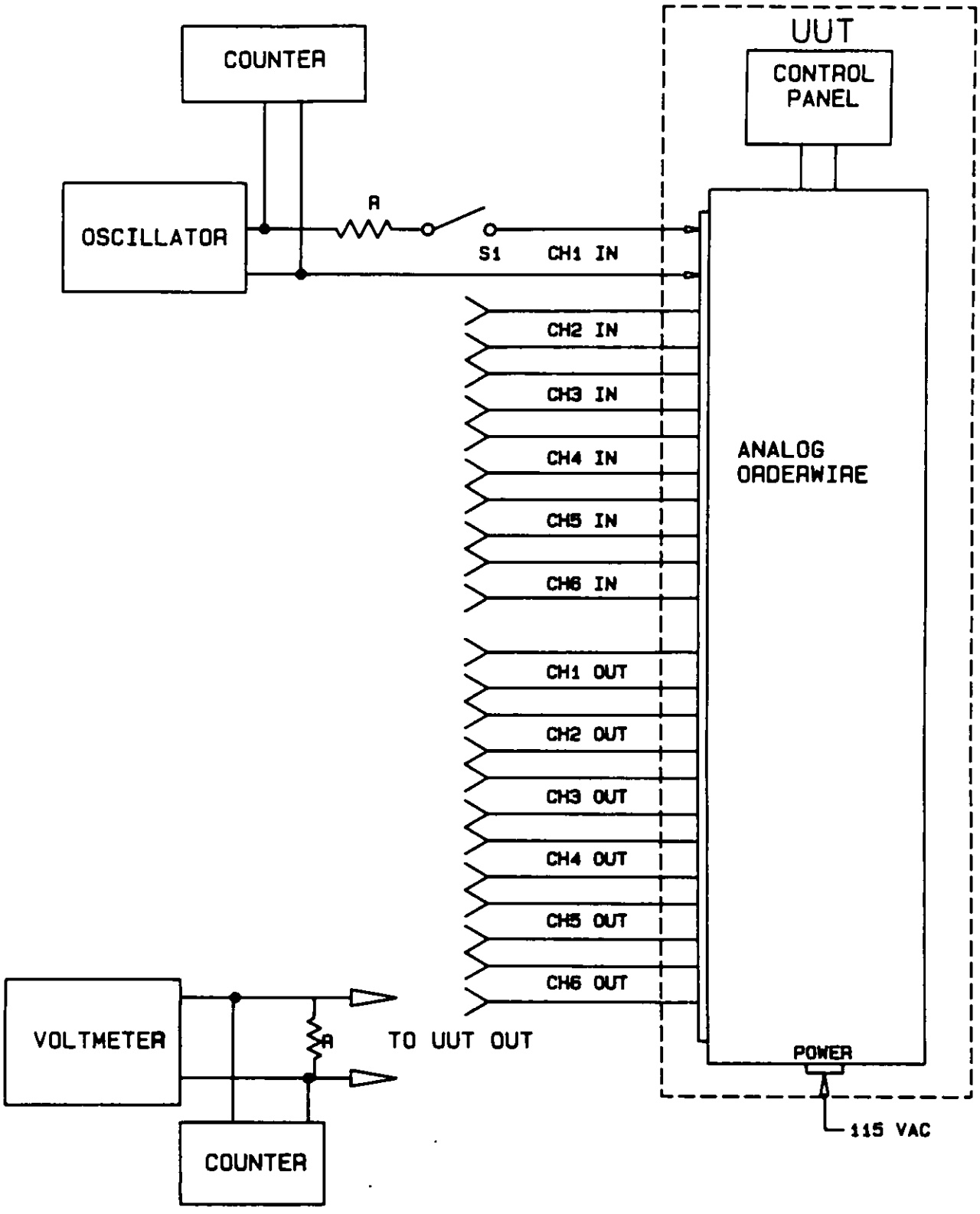


FIGURE 15. Frequency response.

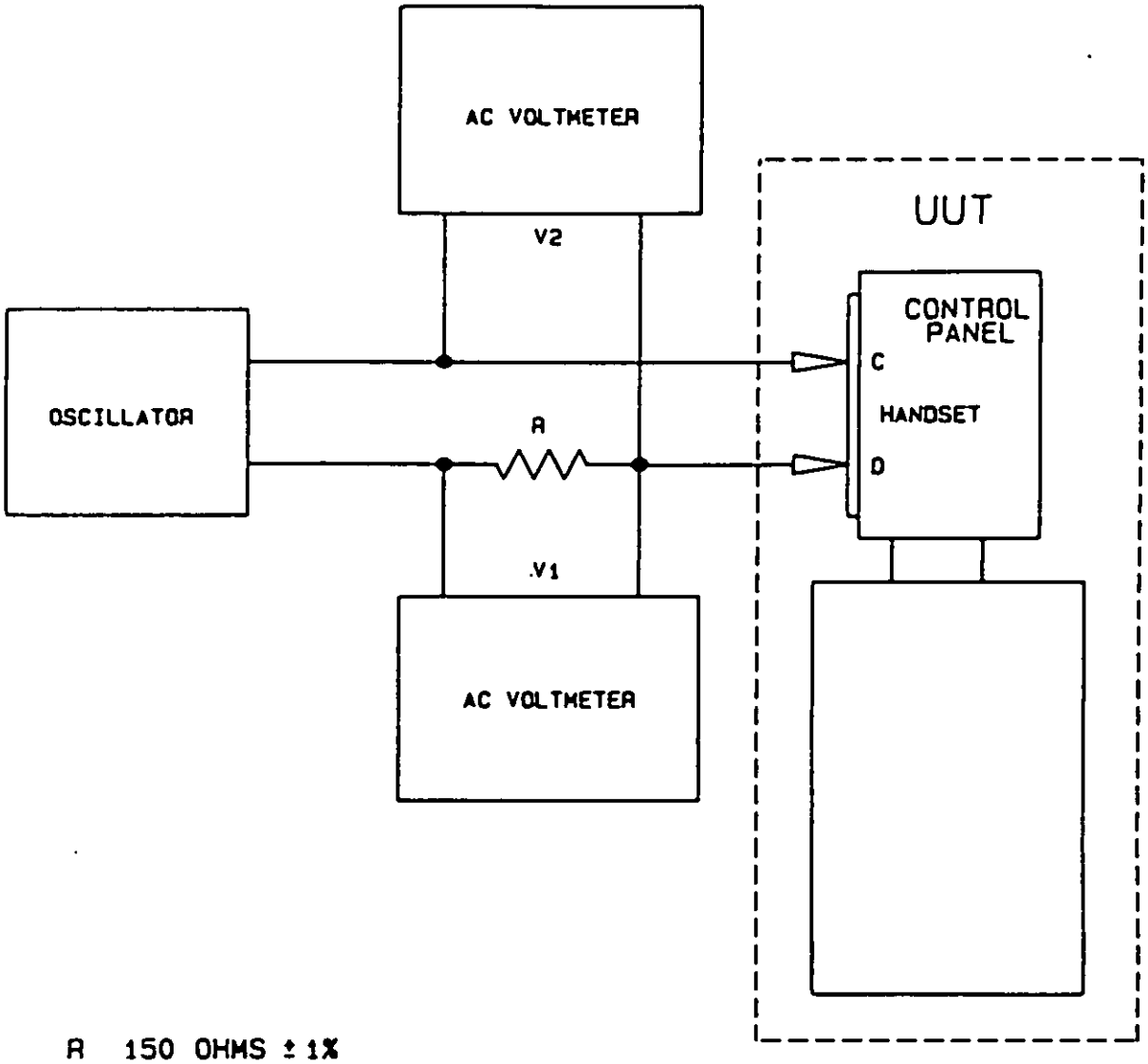
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R 600 OHMS ± 1%.

FIGURE 16. Signal detector operating range.

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R 150 OHMS ± 1%

FIGURE 17. AVOW handset input impedance.

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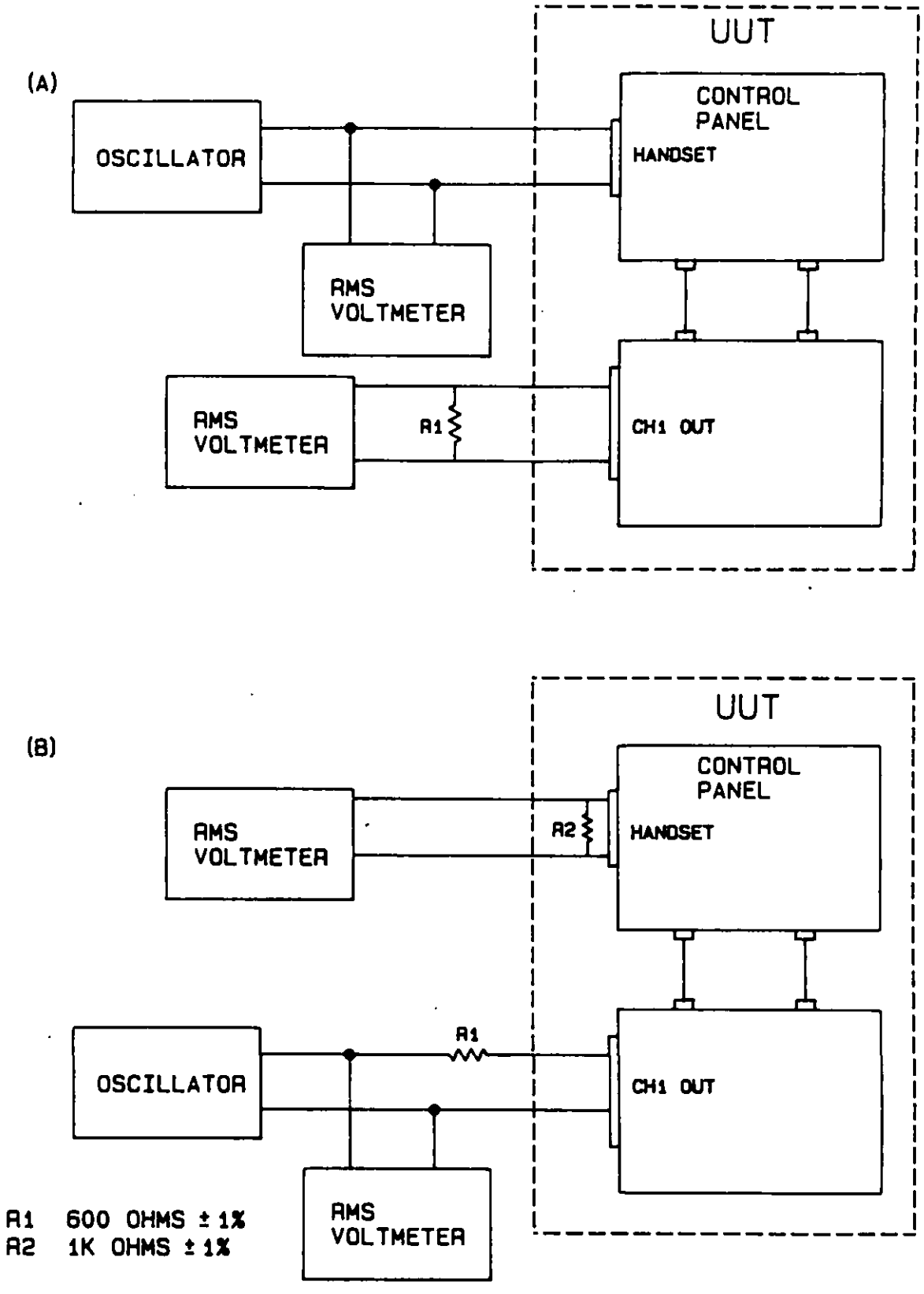


FIGURE 18. Frequency response.

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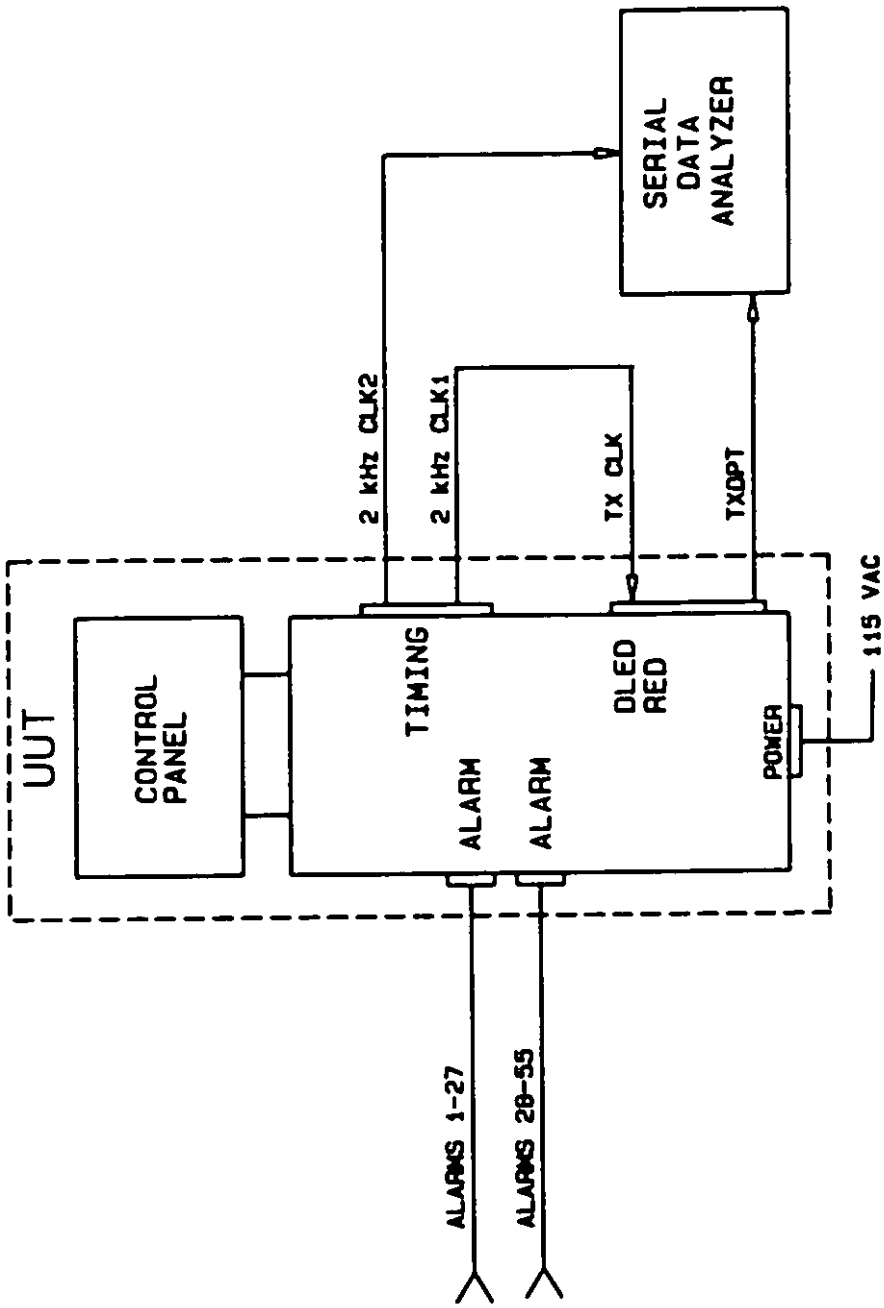


FIGURE 19. Alarm interface.

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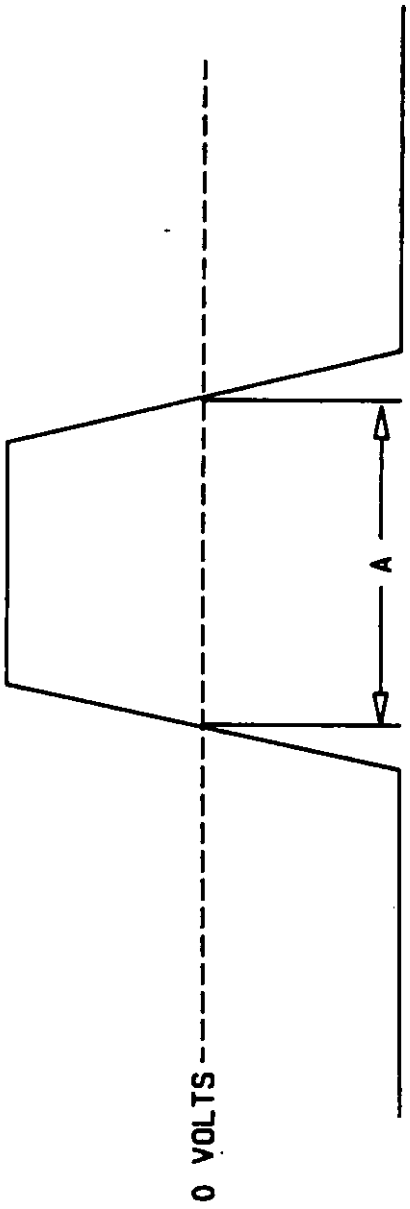


FIGURE 20. Strobe pulse width.

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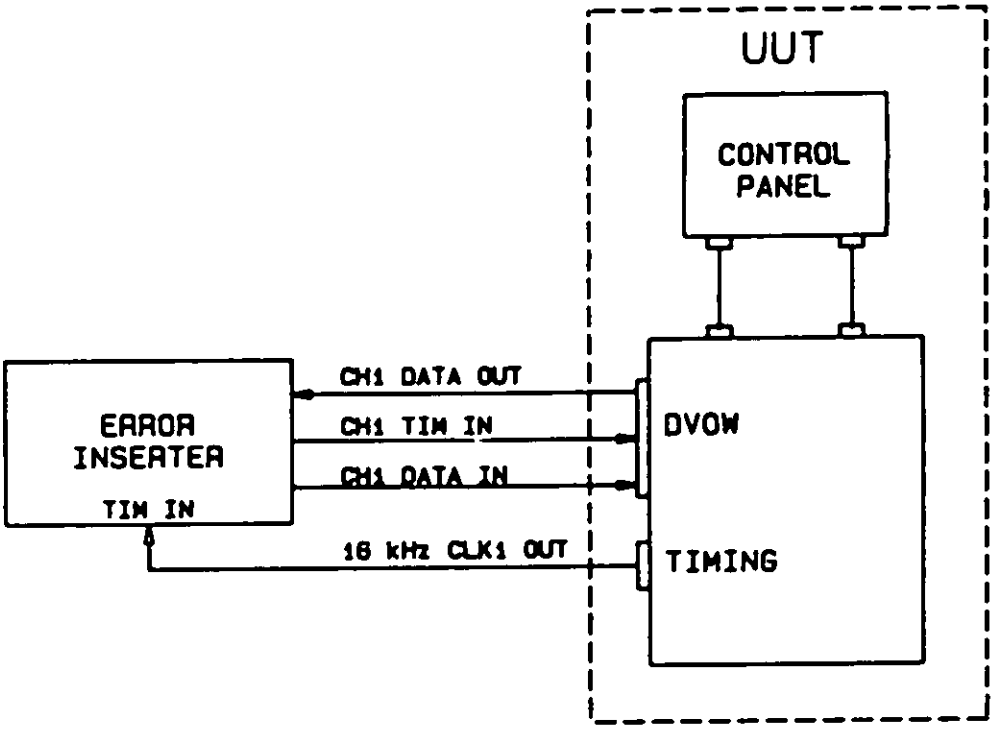


FIGURE 21. Detection with errors.

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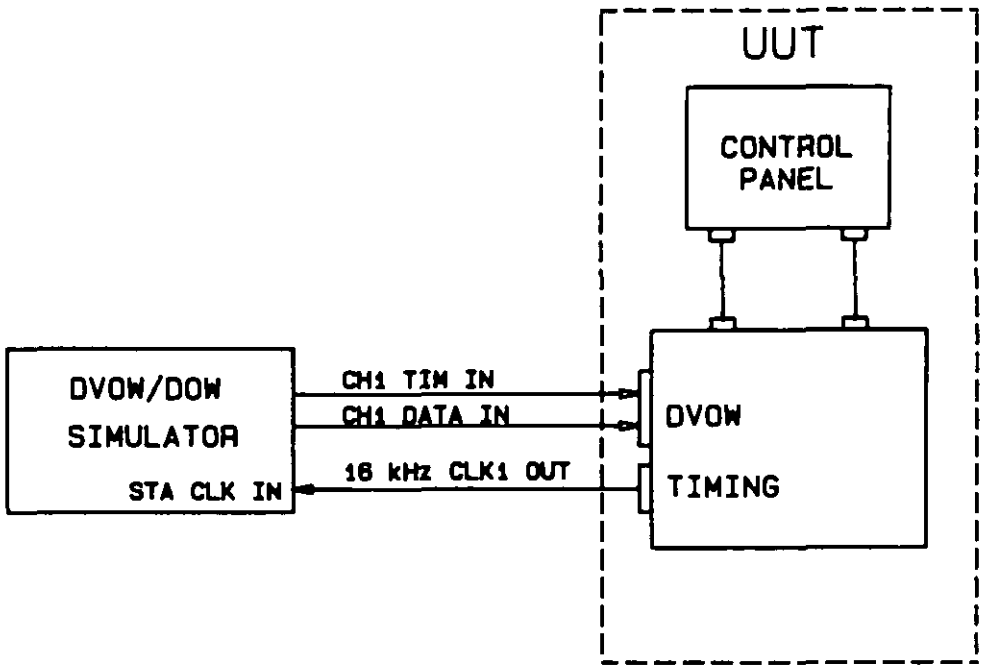


FIGURE 22. False ring detection.

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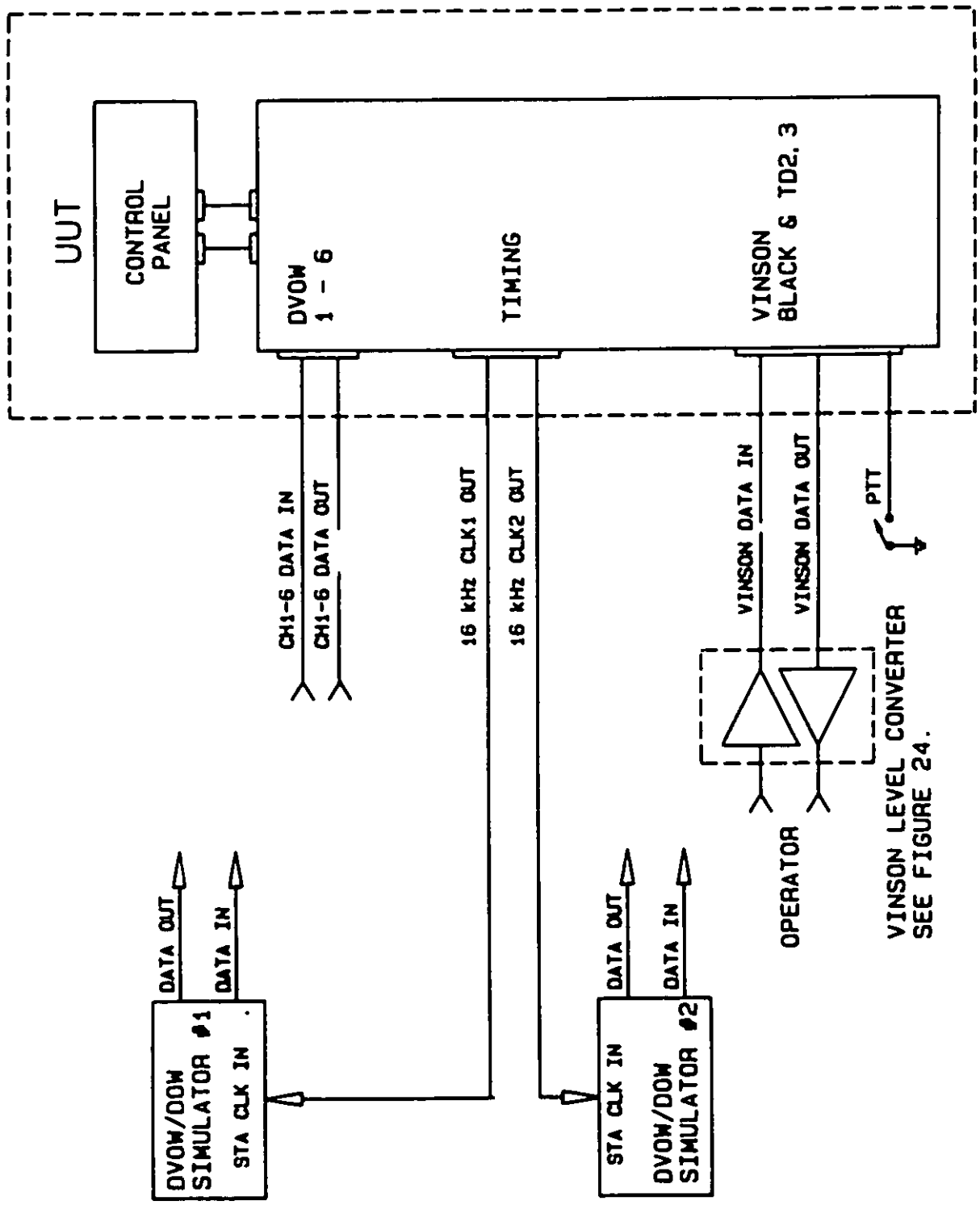


FIGURE 23. Operator access and bridging.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

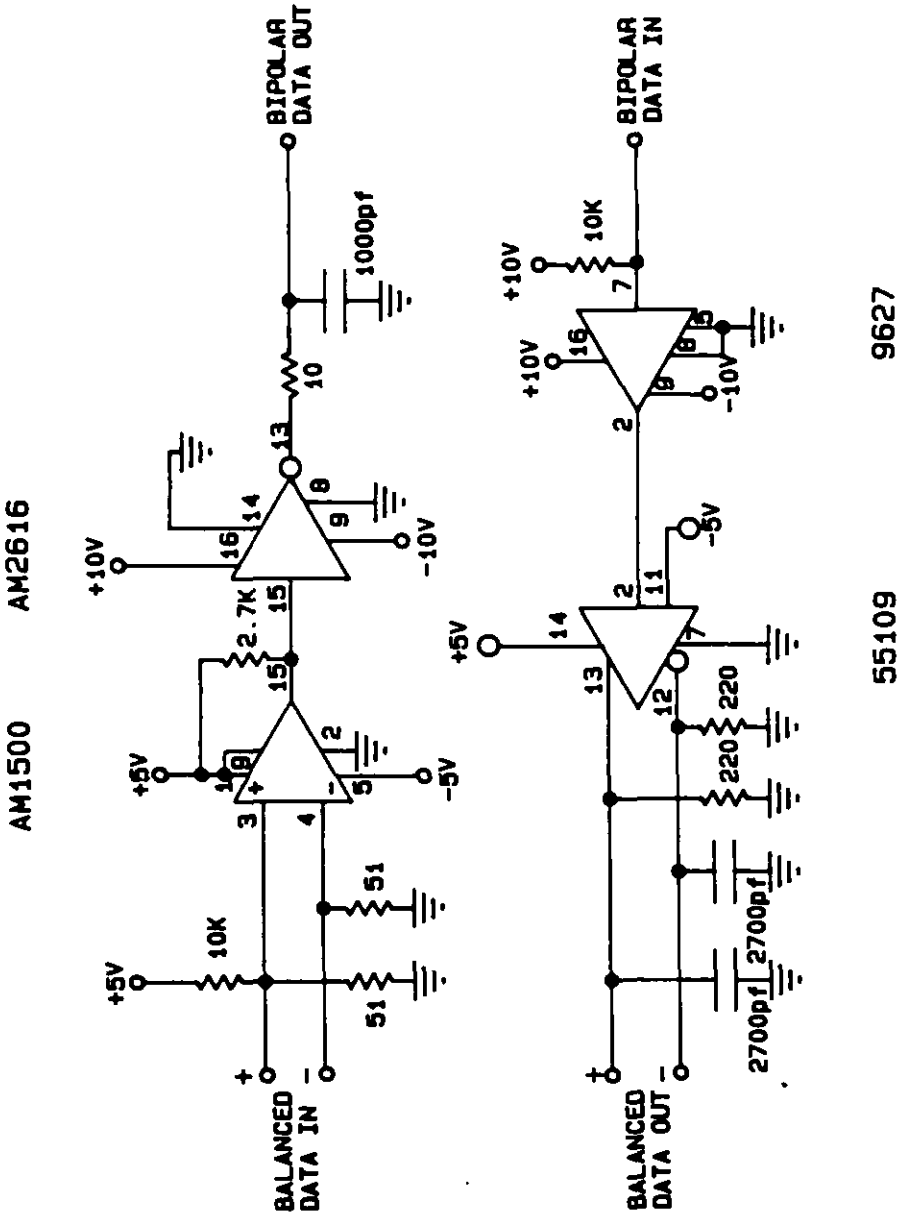


FIGURE 24. Vinson level converter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:59Z
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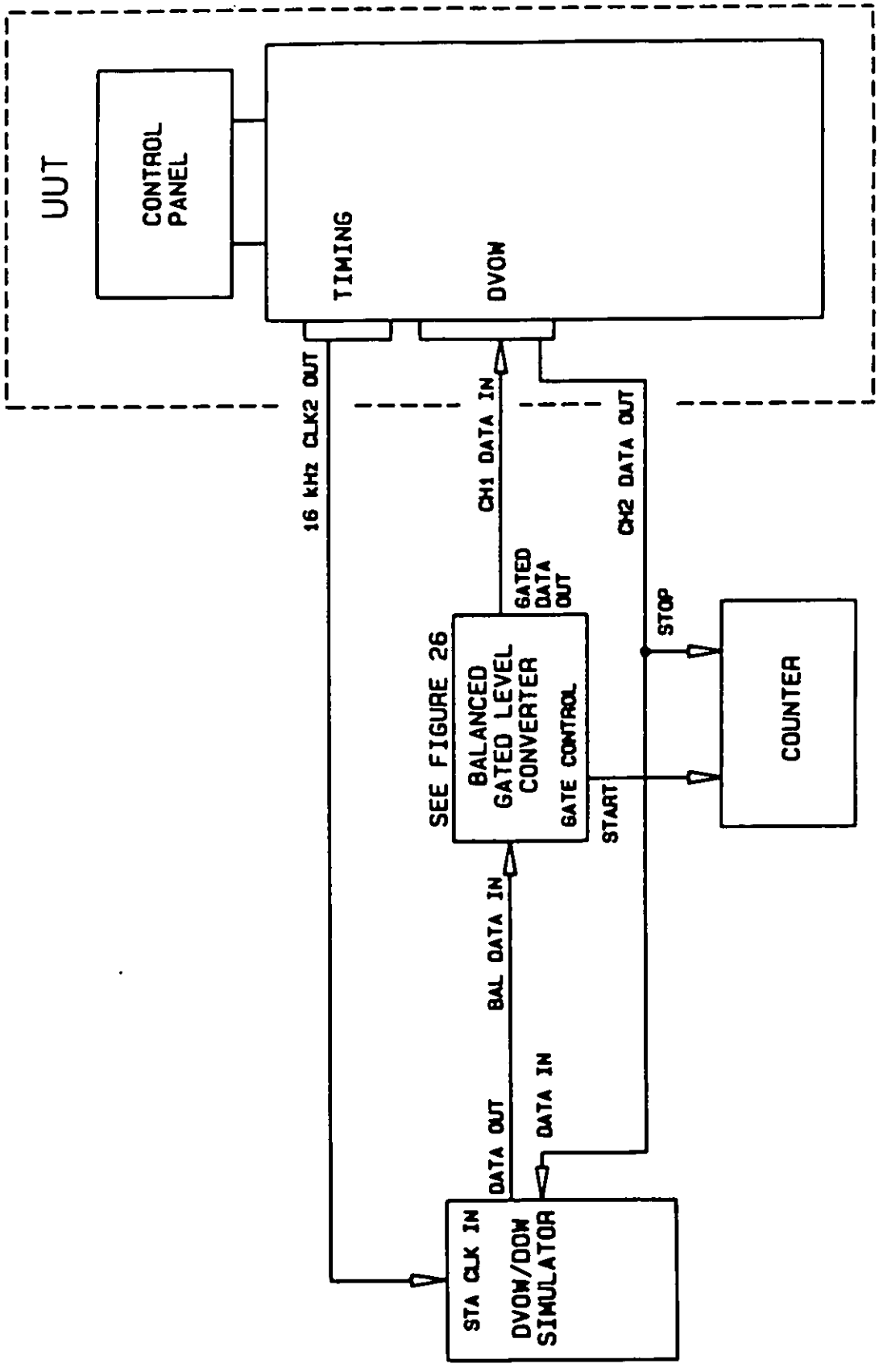
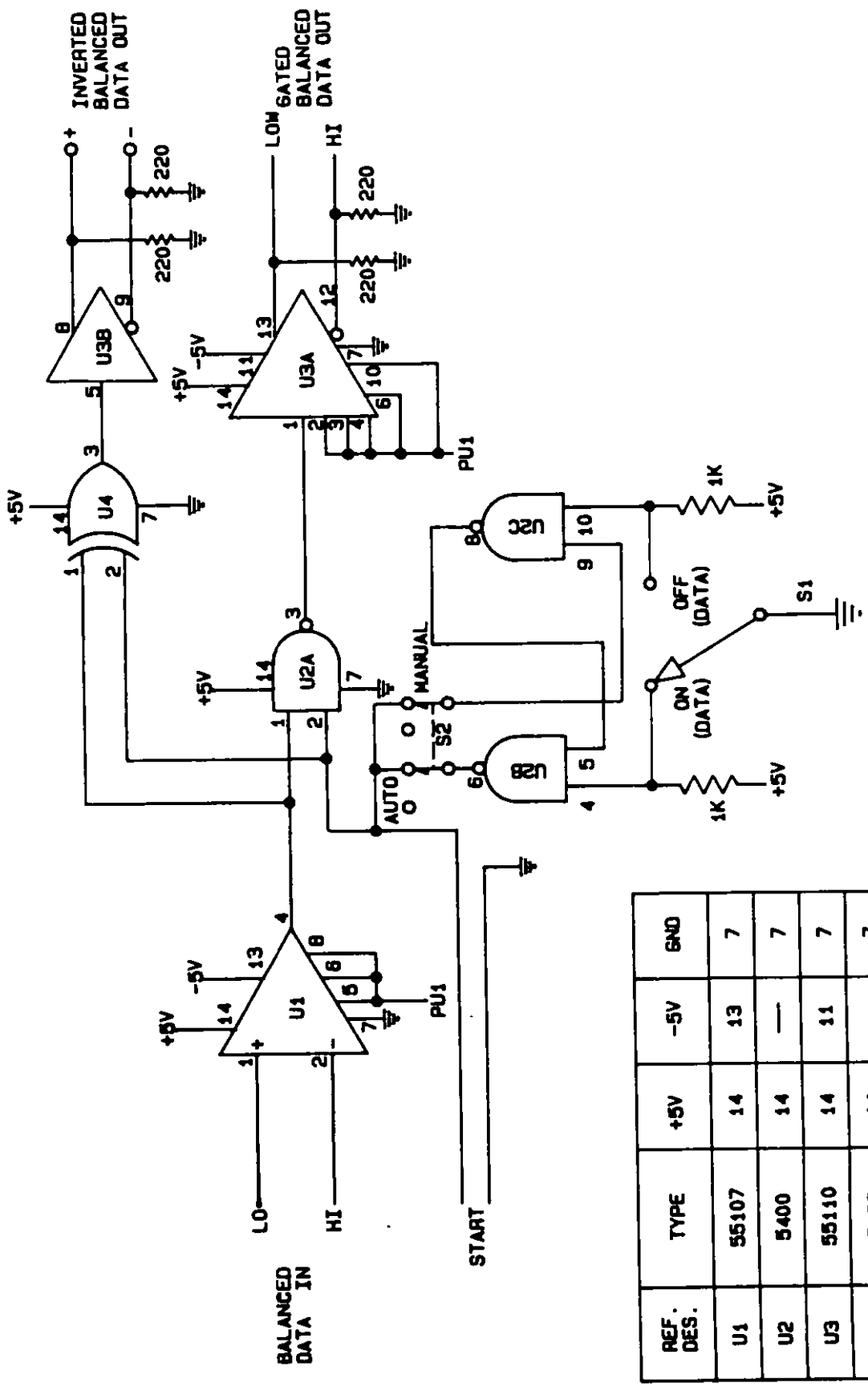


FIGURE 25. Bridge seizure and release.

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REF. DES.	TYPE	+5V	-5V	GND
U1	55107	14	13	7
U2	5400	14	—	7
U3	55110	14	11	7
U4	5486	14	—	7

FIGURE 26. Balanced gated level converter.

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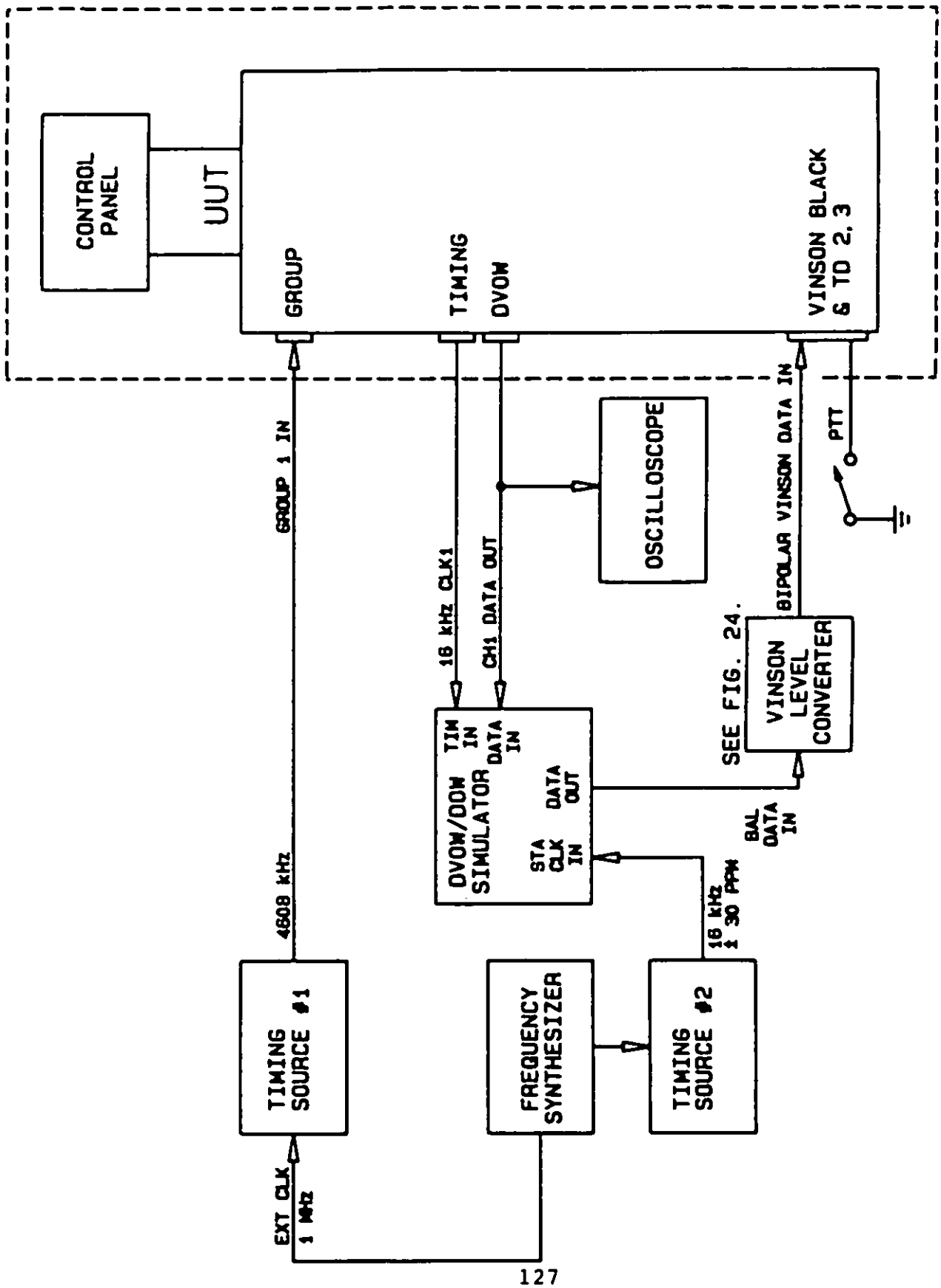


FIGURE 27. Vinson buffer operation and timing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:59Z
Check the source to verify that this is the current version before use.

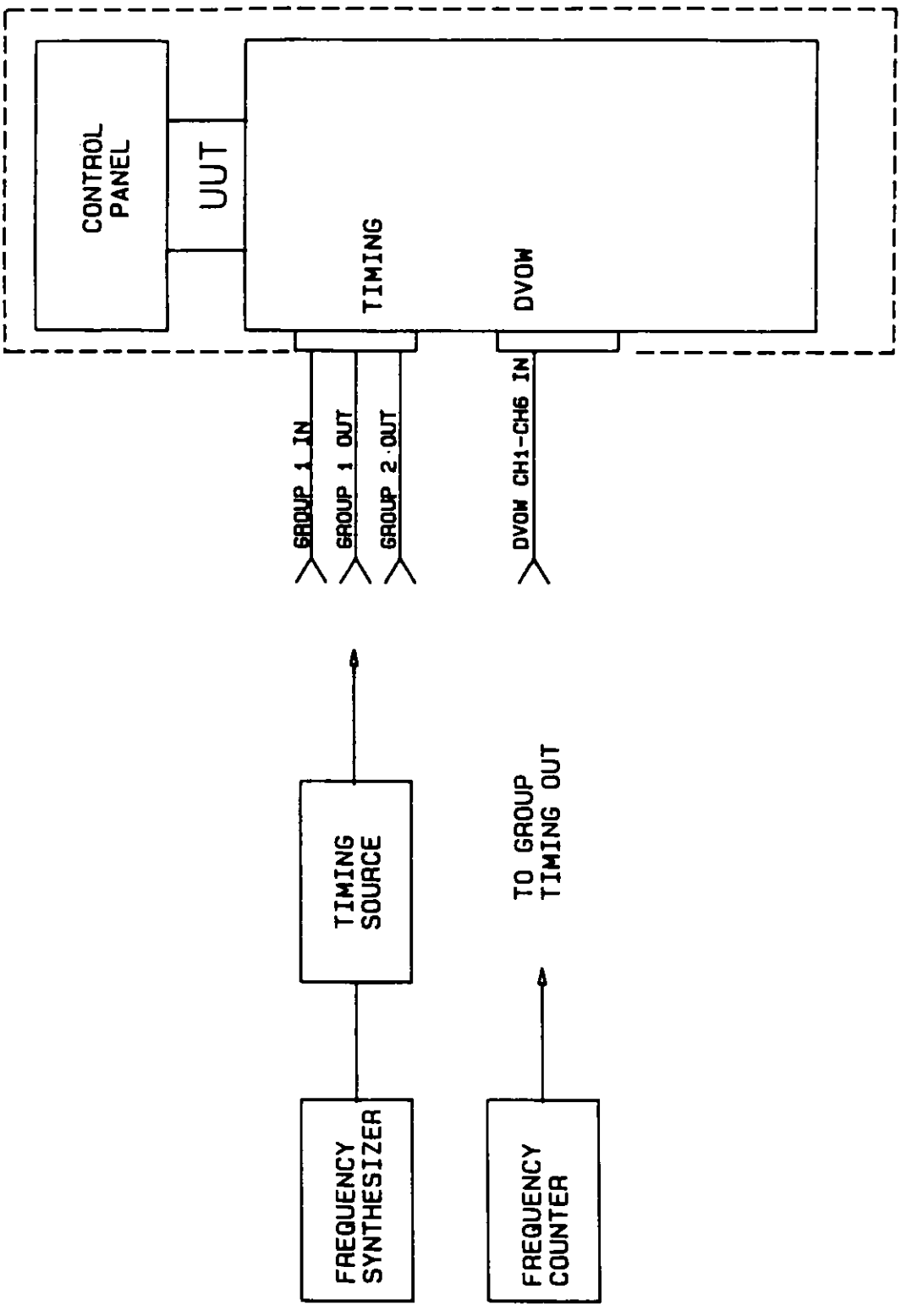


FIGURE 28. Timing source test.

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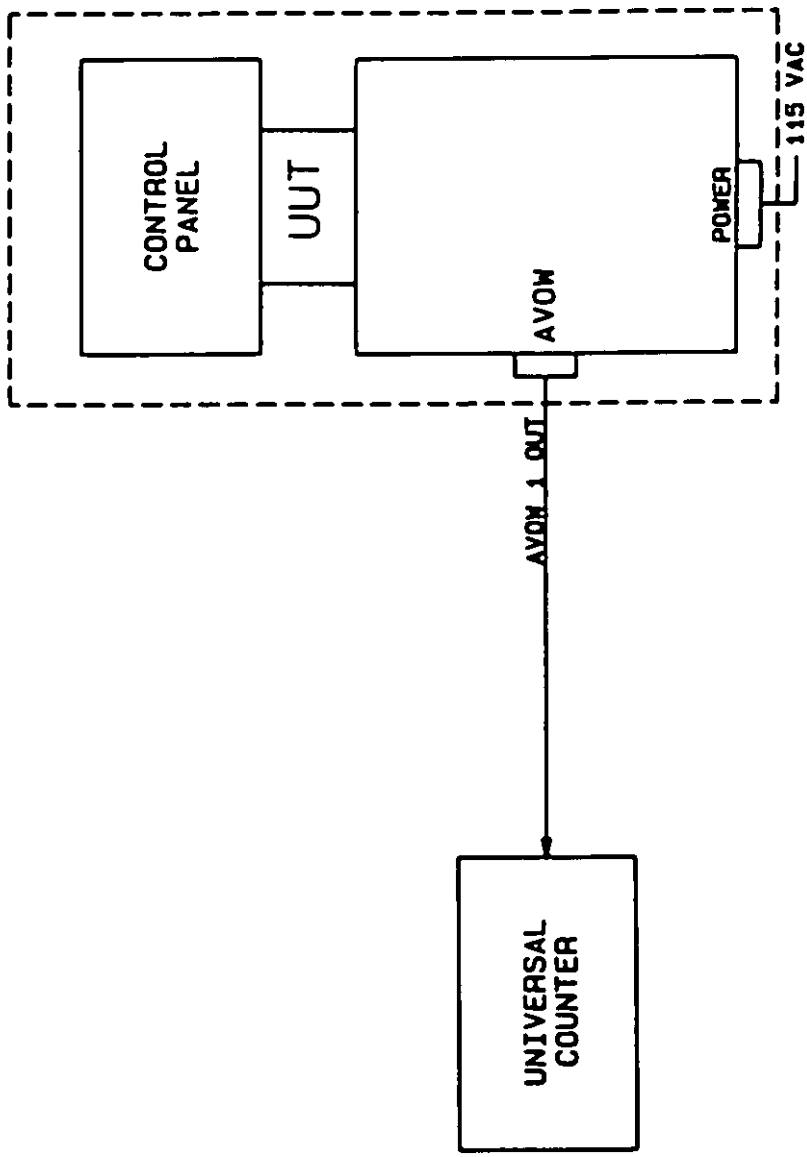


FIGURE 29. Ring generation.

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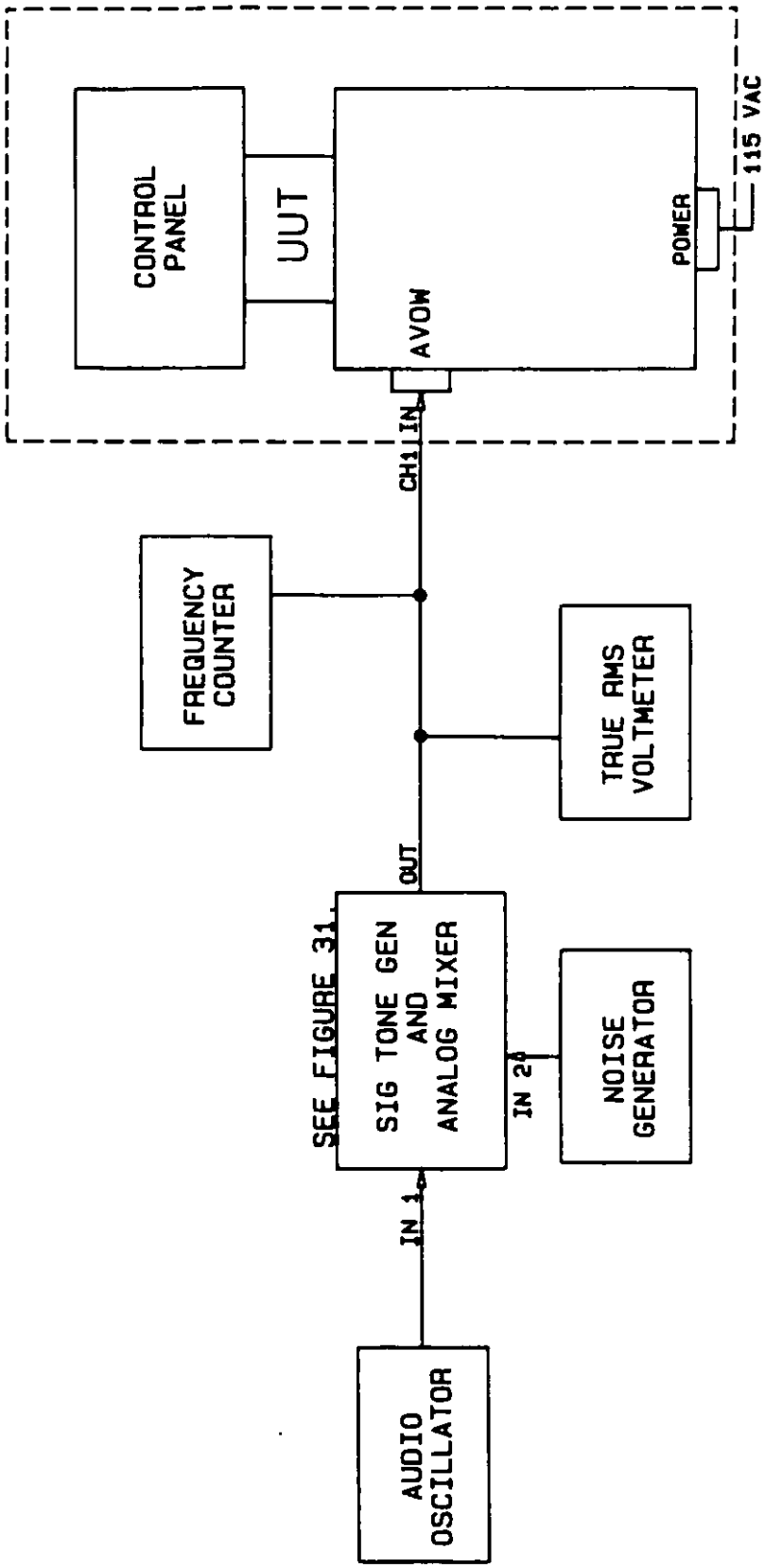
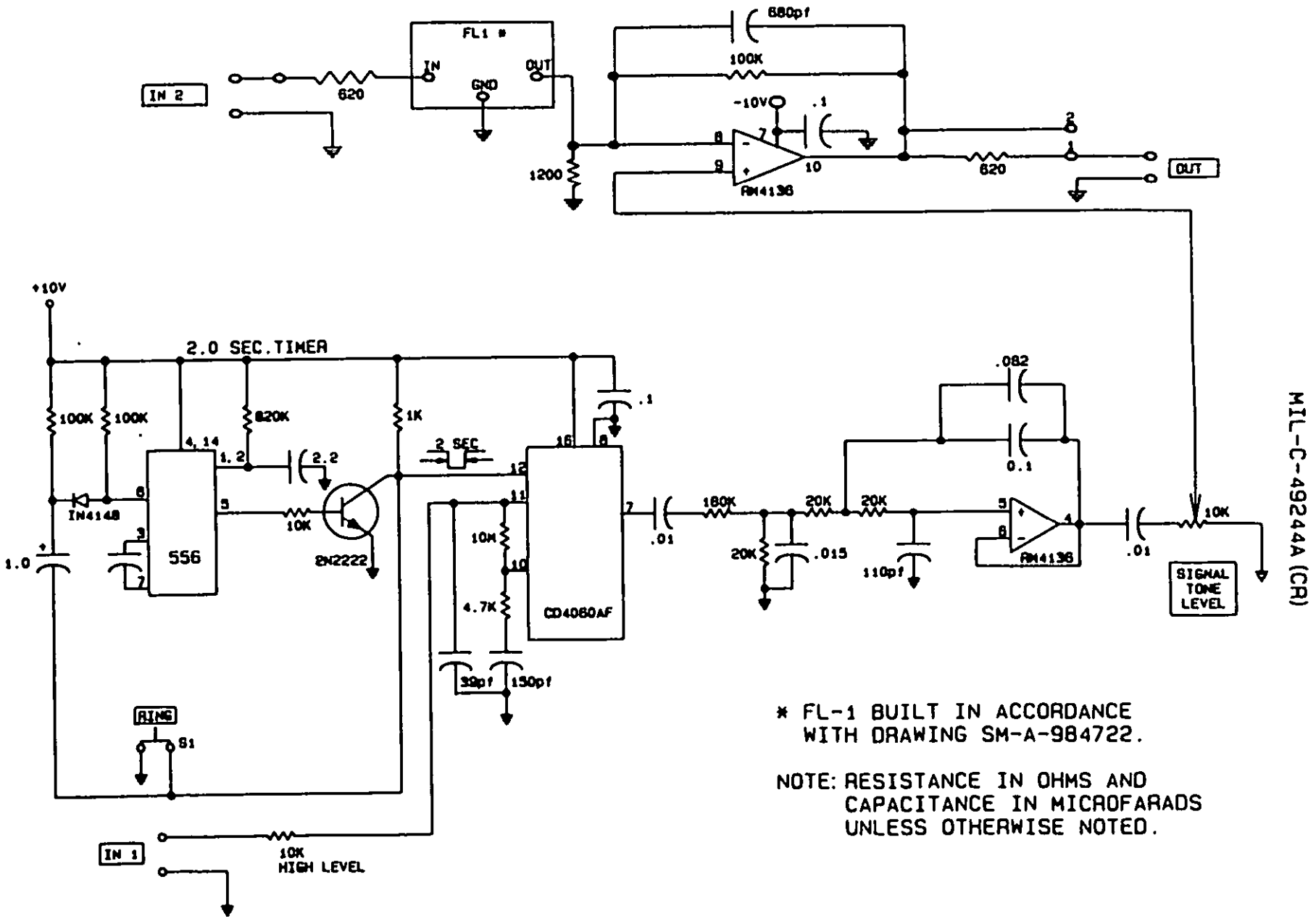


FIGURE 30. Signaling detection.

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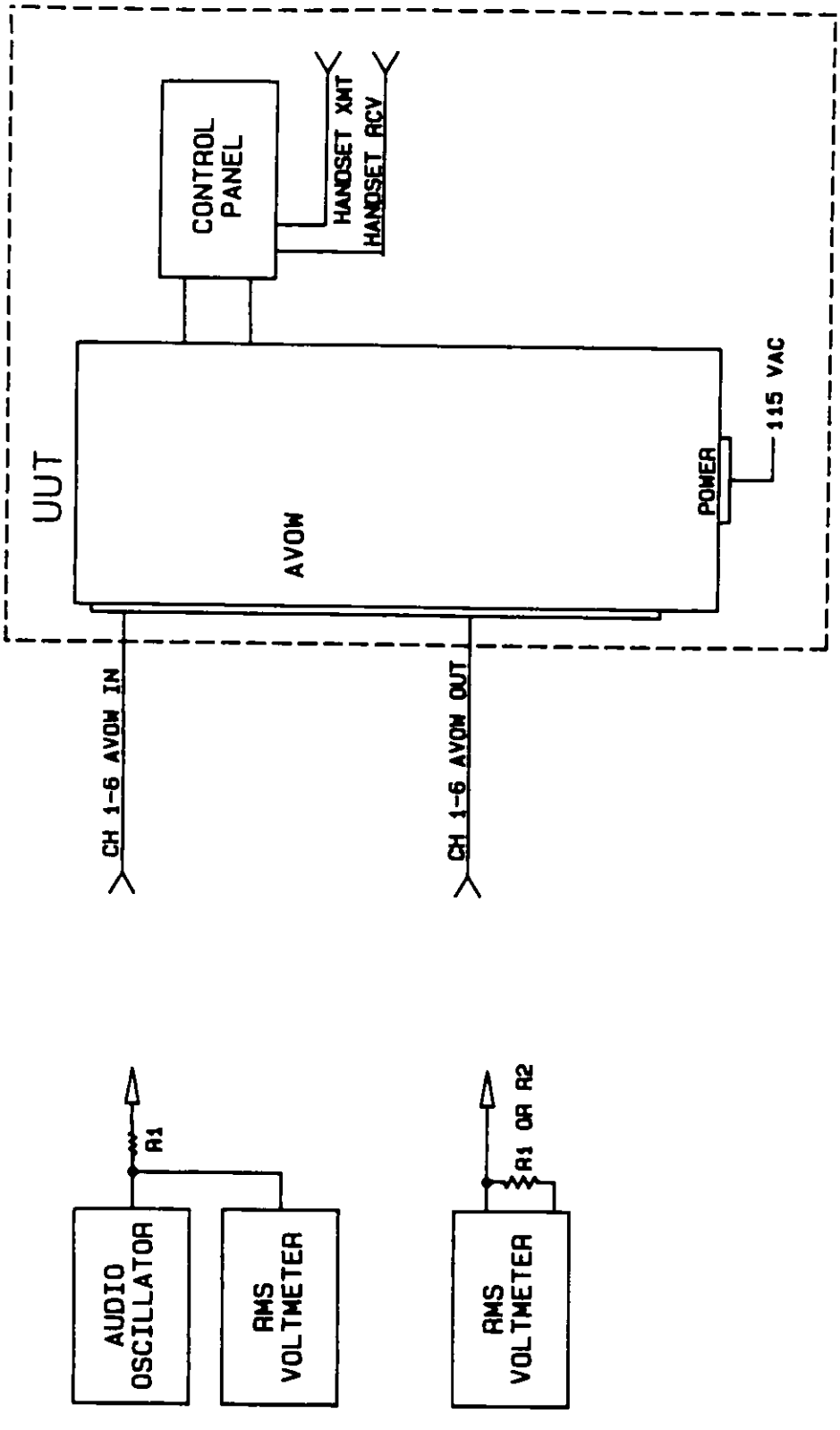


* FL-1 BUILT IN ACCORDANCE WITH DRAWING SM-A-984722.

NOTE: RESISTANCE IN OHMS AND CAPACITANCE IN MICROFARADS UNLESS OTHERWISE NOTED.

FIGURE 31. Signal tone generator and analog mixer.

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R1 600 OHMS ±1% FOR CHANNEL OUTPUTS ONLY.
R2 1000 OHMS ±1% FOR HANDSET OUTPUTS ONLY.
* REMOVE R1 FROM OSCILLATOR WHEN DRIVING HANDSET INPUT.

FIGURE 32. Operator access and bridging.

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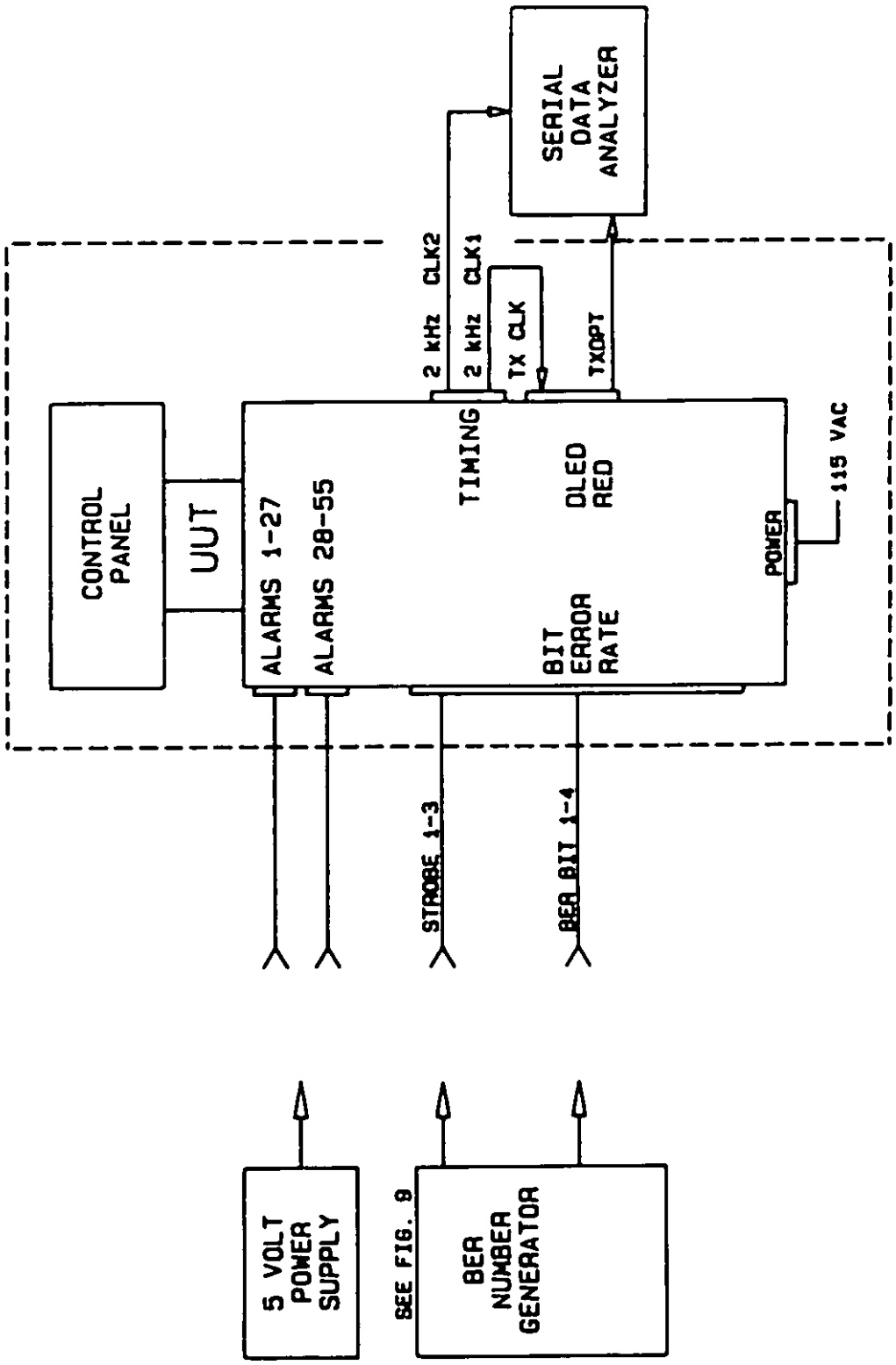


FIGURE 33. DOW performance.

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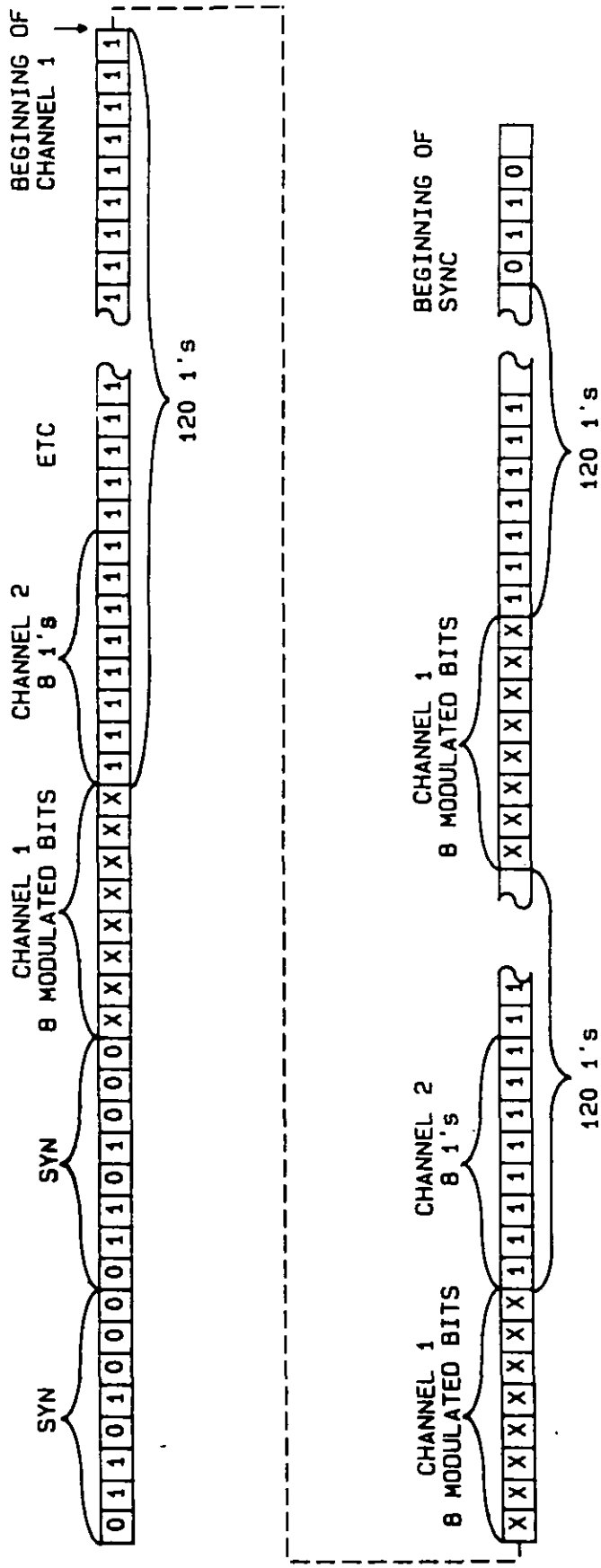


FIGURE 34. Telemetry combiner format.

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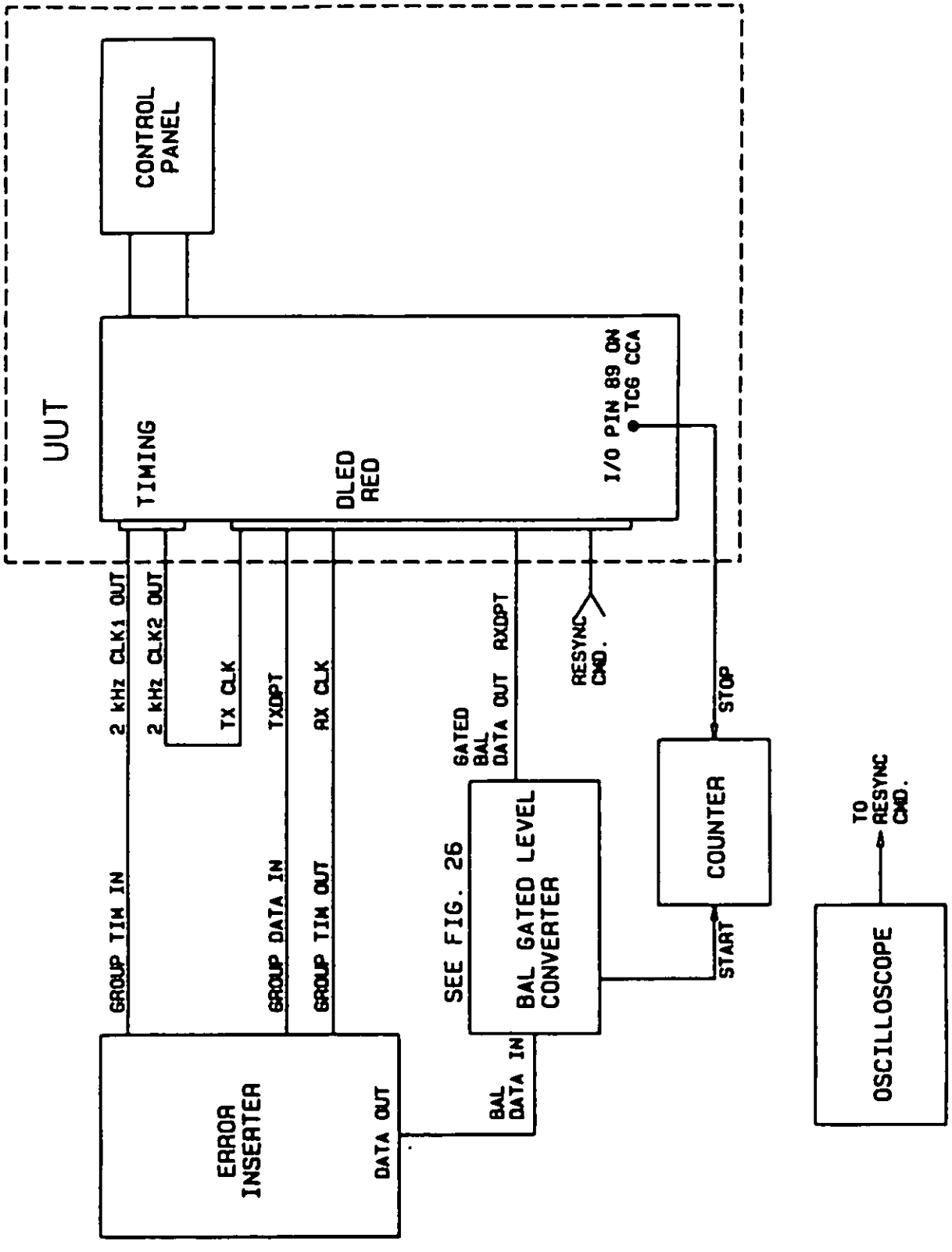


FIGURE 35. Telemetry decombiner framing.

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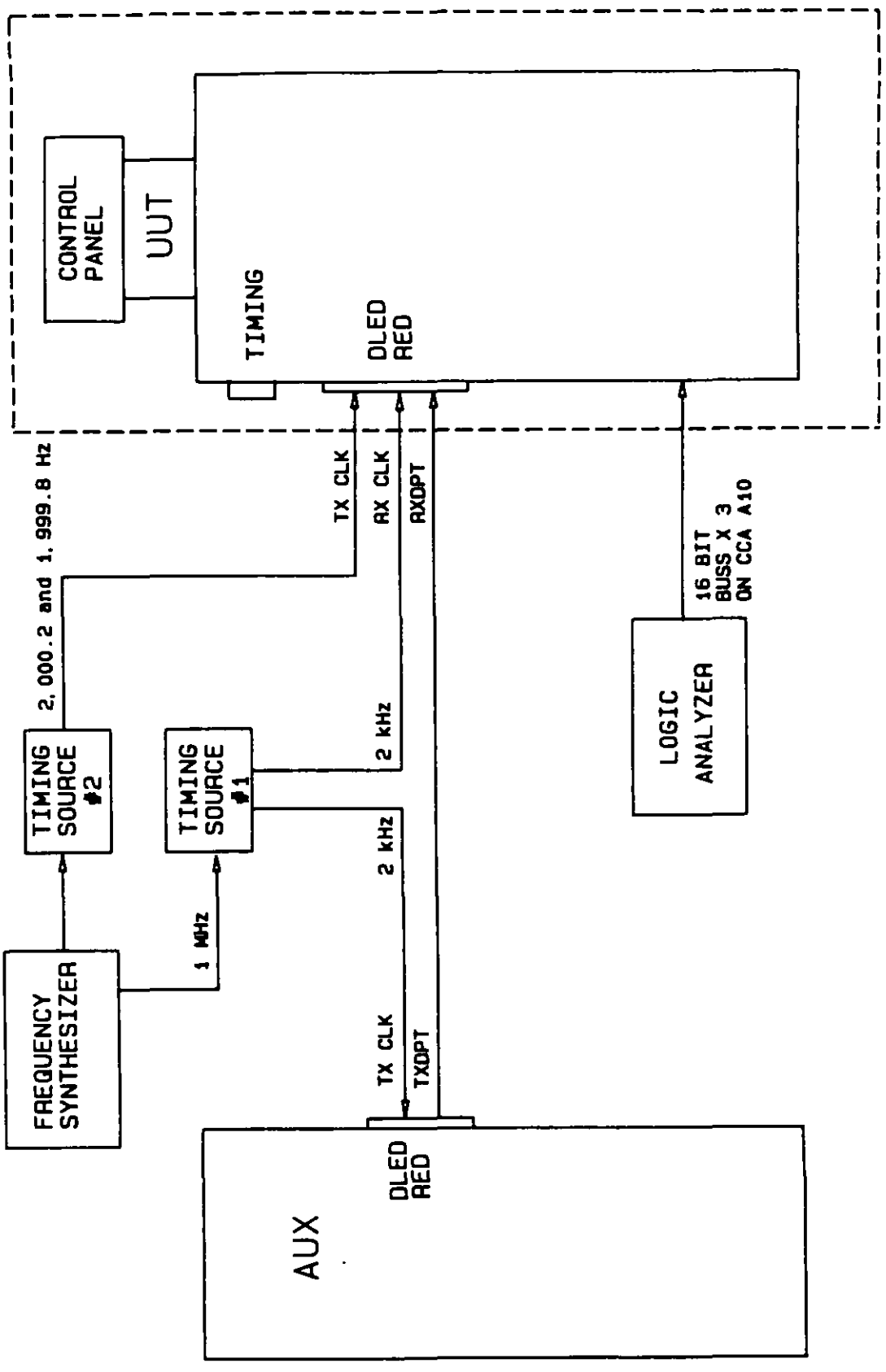


FIGURE 36. DOW BCI

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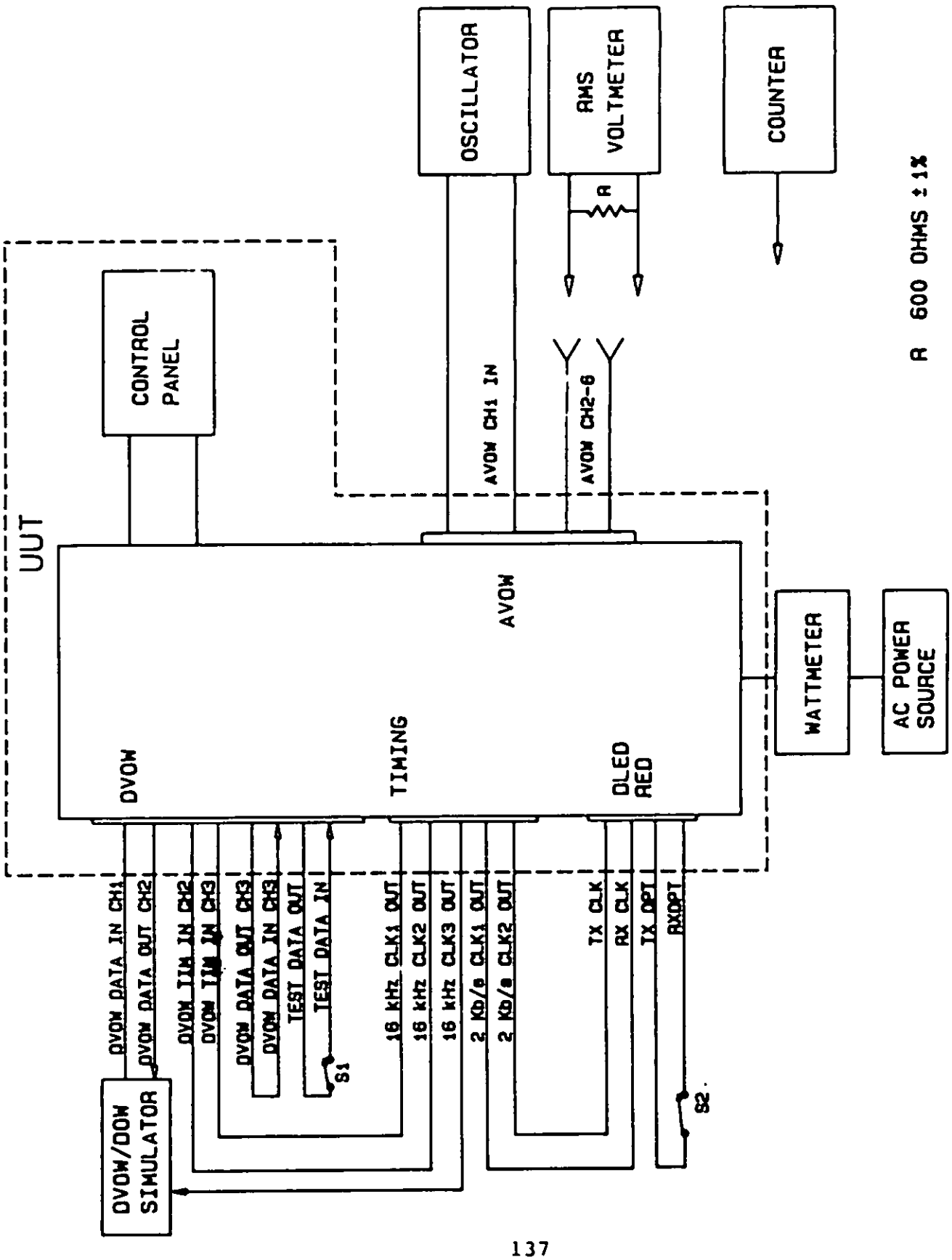


FIGURE 37. Power variation.

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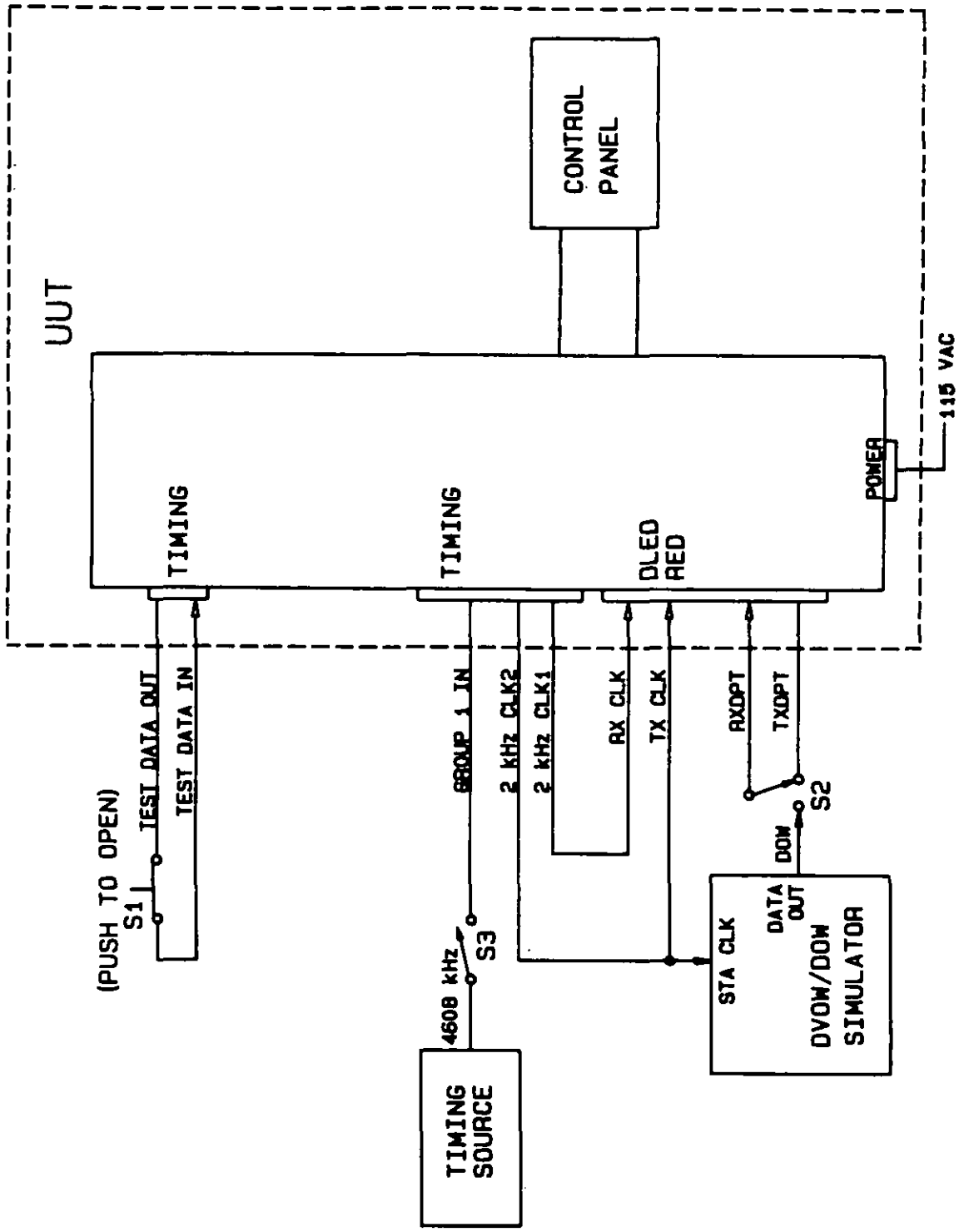


FIGURE 38. BITE and alarms.

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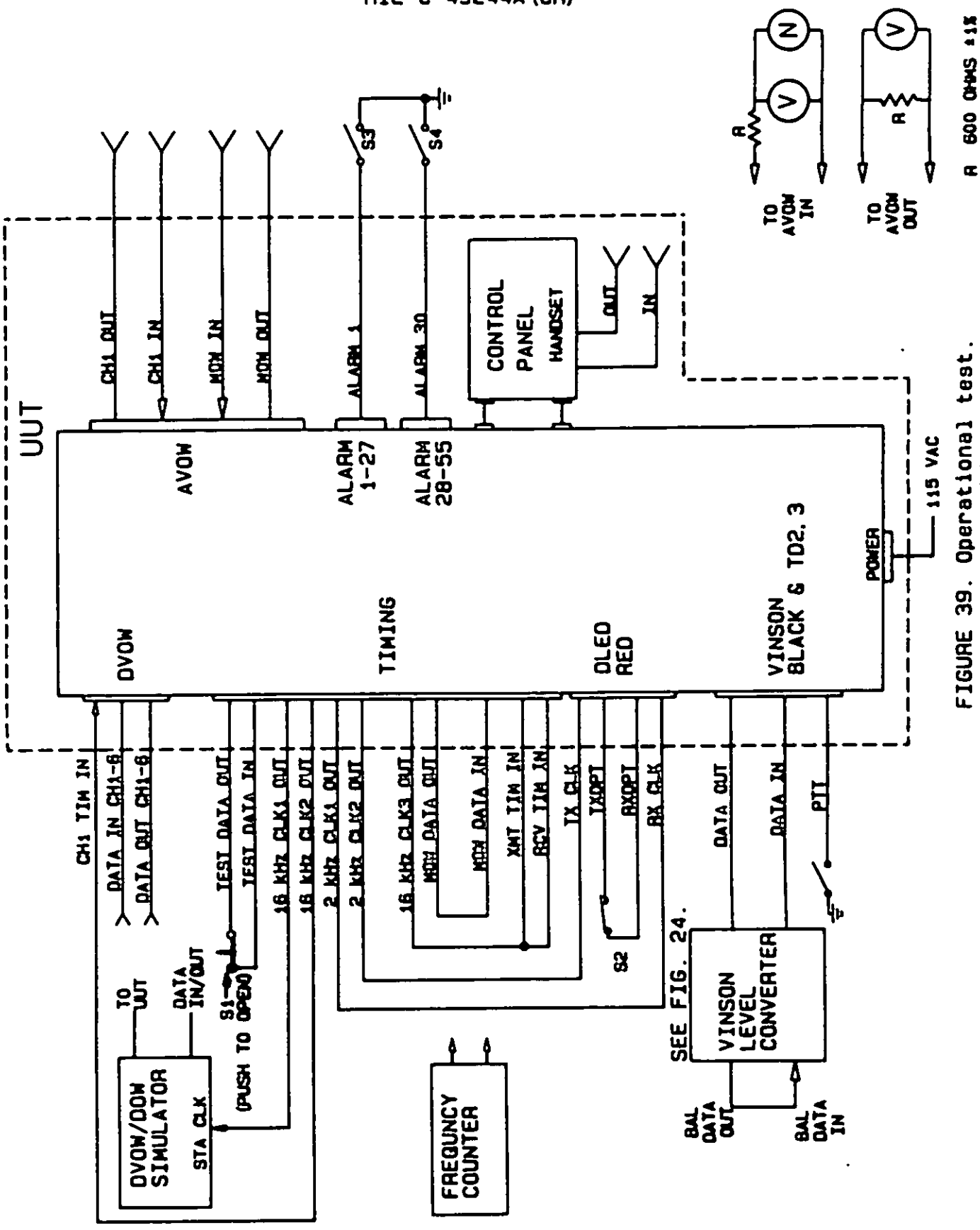


FIGURE 39. Operational test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49244A(CR)	Control, Orderwire C-10716/TRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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